GUIDANCE OF EFSA

Standard Sample Description ver. 2.0[1]

European Food Safety Authority[2, 3]

European Food Safety Authority (EFSA), Parma, Italy

ABSTRACT

Data collection is an important task of EFSA and a fundamental component of risk assessment. In order to facilitate reporting of occurrence data to EFSA from several food safety domains the Working Group on Standard Sample Description (SSD) Extension extended the previously published EFSA guidance "Standard Sample Description for Food and Feed" to cover additional data collection domains such as zoonotic agents in food and animals, antimicrobial resistance and food additives. This document provides specifications aimed at harmonising the collection of analytical data on chemical substances and microbiological agents in different matrices of non human nature (e.g. food, feed, animals, water, environmental samples and food contact materials). The revised Standard Sample Description also includes updated lists of standardised data elements, which are items describing characteristics of samples or analytical results such as country of origin, product, analytical method, limit of detection and test result. The final output of this working group is a standard model to transmit sample data from several food safety domains to EFSA.

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KEY WORDS

standard format, chemical, microbiological, XML, data, pesticides, additives

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SUMMARY

Data collection is an important task of EFSA and a fundamental component of risk assessment (Articles 22 and 23 of Regulation (EC) No 178/2002[4]). EFSA receives from different providers (Member States, European Commission, industry and other providers) an increasing volume of data in support of its risk assessments.

The Working Group on SSD Extension (WG-SSD2) was mandated by EFSA to develop a proposal to extend the previously published Guidance on "Standard Sample Description for Food and Feed" (SSD1) to include additional data collection domains such as zoonotic agents in food and animals, antimicrobial resistance and food additives and to provide a framework for the collection of harmonised analytical measurement data on chemical or microbiological contaminants in different matrices of non human nature (e.g. food, feed, animals, water, environmental samples and food contact materials).

The WG-SSD2 was requested to produce guidance documents on:

- the extended standard sample description of data on analytical measurements in food and feed samples *(Guidance on Standard Sample Description ver. 2.0)* including updated lists of standardised data elements (items describing characteristics of samples or analytical results such as country of origin, product, analytical method, limit of detection, test result), controlled terminologies and validation rules to enhance data quality;

- the updated procedures to efficiently transmit and exchange data between Member States (MSs) and EFSA (*Guidance on Data Exchange*) taking into account specific file formats for data transmission (e.g. XML, Microsoft Excel®) and specific data transmission protocols to support electronic data exchange.

The Guidance on Standard Sample Description ver. 2.0 (SSD2) specifies the data elements and the data structure of the samples and the analytical results for chemical contaminants and residues as well as microbiological contaminants, zoonotic agents and antimicrobial resistance data in food, feed, animals, environmental samples and food contact materials included in monitoring and surveillance programmes (e.g. sample description, analytical methods and analytical results). The WG-SSD2 aimed to build a description as general as possible to facilitate its application to a wide range of measurements taken for food and feed safety assessments.

The WG-SSD2 acknowledges the progress towards the harmonisation of data transmission from data providers (e.g. Member States, academia and industry) to EFSA achieved by the use of the previously published SSD1, and proposes the updated SSD2. This step should be seen in the context of the maintenance programme foreseen in the original guidance document to continue the process of adapting and improving the standard to areas not previously included in the scope of SSD1. Feedback from the implementation of SSD1 in the domains of chemical contaminants and pesticides has been incorporated into the SSD2. Introduction of the concepts of repeatable and compound data elements in the data model of SSD2 is intended to ensure that the model is more flexible than the former version (SSD1) to cater for unanticipated needs beyond the scope of the WG-SSD2, thus ensuring a stable SSD structure for the foreseeable future. Further experience in the use of the SSD in all domains will contribute to the enhancement of the SSD2 terminology catalogues over time and a defined maintenance and publication process is proposed within this guidance. The WG-SSD2 recognises that the ability of data providers to transmit data to EFSA according to the standard data model will vary. However, many data providers have already implemented SSD1 and therefore the WG-SSD2 proposes

[4] Regulation (EC) No 178/2002 of the European Parliament and of the Council, of 28 January 2002 laying down the general principles and requirements of food law, establishing the European Food Safety Authority and laying down procedures in matters of food safety. OJ L 31, 1.2.2002, p. 1-24.

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a transition period during which SSD1 and SSD2 transmissions from data providers would be accepted by EFSA. The circumstances under which this is feasible are described in this guidance.

This guidance and the SSD2 structure and catalogues should be used by data providers who have not yet implemented SSD1, when planning future developments and evolution of local, regional and national systems with the objective of harmonising data transmissions to EFSA across all data domains.

The harmonisation of data collection is recognised as a fundamental step for the development and ongoing development of the EFSA Data Warehouse.

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BACKGROUND AS PROVIDED BY EFSA

Article 33 of the Regulation (EC) 178/2002[5] of the European Parliament and of the Council states that EFSA:

- "shall search for, collect, collate, analyse and summarise relevant scientific and technical data in the fields within its mission. This shall involve in particular the collection of data relating to food consumption and the exposure of individuals to risks related to the consumption of food";
- "shall work in close cooperation with all organisations operating in the field of data collection, including those from applicant countries, third countries or international bodies".

In 2010 the Guidance on Standard Sample Description (SSD1) (EFSA, 2010a) and the Guidance on Data Exchange (GDE) (EFSA, 2010b) were published defining a standard format for describing food and feed samples and analytical results. These guidance documents describe the data model and data interchange protocol[6] for reporting the results of laboratory tests on food and feed samples. These standards are now the basis for reporting chemical occurrence monitoring data according to Regulation (EC) 2002/32[7]. Within the framework of the pesticide residues control programme according to Regulation (EC) 396/2005[8], Regulation (EU) 1274/2011[9] specifies that for 2012, 2013 and 2014 the EU coordinated multiannual control programme monitoring data must be reported to EFSA according to the SSD specifications.

Through Article 36 grant procedures EFSA has provided funding to official reporting organisations in Member States to implement the SSD within their data management systems[10]. Ireland, Belgium, Sweden, Romania, Latvia, Austria, Slovakia, Germany, Hungary and Denmark have received funding and integrated the SSD within their data reporting systems. Cyprus, Greece, Finland, Italy, the Netherlands, Portugal, Poland and Bulgaria are now starting the same project. As a consequence, the SSD is becoming the accepted European standard for describing and reporting monitoring results for chemicals and residues in food and feed. The success of the SSD makes it advisable to extend this approach to other hazards such as food additives and zoonotic agents.

In this area, the Biological Monitoring Unit (BIOMO) ran a pilot project in 2011 for the collection of Antimicrobial Resistance (AMR) data at isolate-based level (EFSA, 2012a). The existing SSD model could not be entirely adopted and consequently a preliminary ad-hoc data model was developed for the pilot. The two data models are similar, therefore as an outcome of the pilot project, the "Working group for the provision of Zoonoses data in XML and Excel format" proposed to extend the SSD to be compatible with the current draft format on AMR data at isolate-based level (EFSA, 2012b).

Further, the SSD terminology on food description should be amended in line with the structure proposed by the working group for Development of a Food Classification and Description System for exposure assessment revising food classification and description. This working group was established in late 2009 and performed its tasks for the period of approximately two years to produce a food

[5] See note 4 page 2.

[6] Electronic data interchange: it is the structured transmission of data between organizations by electronic means. It is used to transfer electronic documents or business data from one computer system to another computer system.

[7] Directive 2002/32/EC of the European Parliament and of the Council of 7 May 2002 on undesirable substances in animal feed. OJ L 140, 30.5.2002, p. 10–22

[8] Regulation (EC) No 396/2005 of the European Parliament and of the Council of 23 February 2005 on maximum residue levels of pesticides in or on food and feed of plant and animal origin and amending Council Directive 91/414/EEC. OJ L 70, 16.03.2055, p. 1-16

[9] Commission Implementing Regulation (EU) No 1274/2011 of 7 December 2011 concerning a coordinated multiannual control programme of the Union for 2012, 2013 and 2014 to ensure compliance with maximum residue levels of pesticide and to assess the consumer exposure to pesticide residues in and on food of plant and animal origin. OJ L 325, 8.12.2011, p. 24-43

[10] Electronic Transmission of Chemical Occurrence Data, see http://registerofquestions.efsa.europa.eu/roqFrontend/questionsListLoader?mandate=M-2009-0251, http://registerofquestions.efsa.europa.eu/roqFrontend/questionsListLoader?mandate=M-2010-0140

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classification and description system (FoodEx2) applicable to different risk assessment areas. The general structure of the FoodEx food classification system, the principles applied and guidance for its use are summarised in the Scientific Report of EFSA: Report on the development of a Food Classification and Description System for exposure assessment and guidance on its implementation and use published at the end of 2011 (EFSA, 2011a). This FoodEx food classification system overlaps to some extent with the SSD in terms of describing the samples and therefore an approach to resolve the overlap is now required.

A technical report on data collection was prepared by the EFSA Advisory Forum representatives for a meeting in Poland on 28-29 September 2011 (EFSA, 2010c). This document contained the following recommendation:

"A Task Force should be established to coordinate improvements in data and process integration and act as a horizontal review group on the outcomes from the various domain groups, taking into consideration all existing standards. The composition of this Task Force should ensure that all domains are represented. This group in particular should have a role in the development of a common catalogue for all data collection purposes in order to enable maximum use of data collected across the different areas of expertise. It should not be the role of the Task Force to propose new areas of work or standards."

In order to put this recommendation into operation, the Dietary and Chemical Monitoring Unit (DCM) of EFSA was tasked with forming a working group to review the SSD with the support of the BIOMO Unit and the Pesticides Unit. The review included a proposal to extend the SSD to support the reporting of biological and chemical monitoring results within the EFSA's remit not currently covered and also to integrate FoodEx2 in SSD2.

Therefore the working group was proposed to be comprised of:

- chair;
- experts in data collection from the different areas:
  - one expert from the Chemical Occurrence Expert Group;
  - two experts from the Task Force on Zoonoses Data Collection;
  - one expert from the Networking Group on Pesticide Monitoring;
- data managers:
  - one data manager for the chemical area;
  - one data manager for the microbiological area;
- experts on food classification:
  - one expert for the chemical area;
  - one expert for the microbiological area.

Other EFSA units interested in this activity were invited to join the working group activities. When necessary, sub-working groups within the networks of interest were established to prepare proposals on ad-hoc subjects (e.g. SSD catalogues for different fields). These sub-working groups were managed directly by the EFSA unit responsible for the network of interest. The proposed approach ensured the coverage of the different areas of expertise while keeping the number of experts to a minimum.

TERMS OF REFERENCE AS PROVIDED BY EFSA

The working group should produce an amended "Guidance on Standard Sample Description" making sure that the new version of the SSD will also be capable of describing sample based data for the following areas:

- Food additives;
- Isolate-based data on antimicrobial resistance;
- Sample level data on microbiological contaminants;

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- Sample level of animal/flock/herd data on zoonotic agents in animals.

In addition the SSD should be modified to fully support the new FoodEx2 food classification and description system, recently published with the contribution of the Food Classification working group.

The WG-SSD2 should also look for improvements that take into account the practical user experience gained in the pesticide residues data collection, in the chemical occurrence continuous data collection and the several article 36 projects for the implementation of electronic transmission[11].

The WG-SSD2 should evaluate and propose changes to be performed taking into account backward compatibility and the cost/benefit balance in implementing those changes.

[11] Electronic Transmission of Chemical Occurrence Data, see
http://registerofquestions.efsa.europa.eu/roqFrontend/questionsListLoader?mandate=M-2009-0251,
http://registerofquestions.efsa.europa.eu/roqFrontend/questionsListLoader?mandate=M-2010-0140,
http://registerofquestions.efsa.europa.eu/roqFrontend/questionsListLoader?mandate=M-2011-0136

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1. Introduction

The "*Standard Sample Description* ver. 2.0", ''SSD2' (or simply 'SSD' when there is no need to make any distinction with the previous version) specifies the data elements and the data structure to describe several types of samples and results coming from laboratory analytical measurements.

The SSD2, with the introduction of the biological monitoring data domain, extends the scope of the original SSD1 structure from food and feed samples to include also samples of animal origin and environmental samples for the monitoring of zoonoses, zoonotic agents and antimicrobial resistance. For this reason, in this new version, it was also necessary to revise the original title of the guidance by removing the remark 'for food and feed'.

This guidance focuses on the definition of a logical model which is independent from the file format. Therefore data providers and receivers can use different file formats - e.g. Microsoft Excel®, Comma Separated Values (CSV) and Extensible Markup Language (XML) - to submit transmissions depending on their technological constraints. The definition of the supported file formats, the actual implementation of the standard logical model and a detailed definition of the business rules will be discussed in a separate guidance of the WG-SSD2: "Guidance on Data Exchange ver. 2.0" (GDE2).

The specification of a logical model for SSD2 is composed of:

- data elements definition and structure;
- controlled terminologies;
- business rules to ensure the validity of the information supplied.

This document consists of an amended version of the previous guidance; therefore, it mainly describes the new version. The compatibility between the two versions and the mapping from SSD1 to SSD2 is presented in Appendix B.

The SSD2 maintains, as a basis, a fixed table structure as present in the SSD1. Additionally, it is enhanced with provisions for additional flexibility to accommodate data elements to be defined at a later stage. This flexibility is obtained with the introduction of repeatable and compound data elements.

The WG-SSD2 highlights that the SSD2 is compatible, from a content perspective, with the SSD1 but the introduction of repeatable and compound data elements and the implementation of the FoodEx2 prevents a complete equivalence of the two schemas.

In order to give sufficient time to all stakeholders to adapt their systems to the new structure, the WG-SSD2 suggests that both the standards (SSD2 and SSD1) should be supported for a certain period of time. Each competent data domain network should then agree on a plan and a timeframe for the implementation of the SSD2. In the case of data domains that do not yet have any Member State (MS) users of SSD1, the WG-SSD2 recommends implementation of SSD2 only. This should also be the case for MSs that have not yet started implementation of the SSD1 format for data transmissions.

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2. Analysis

The logical model of SSD2 is comprised of a combination of three main groups of terms and characteristics:

- data elements definition and structure;
- controlled terminologies;
- business rules to ensure the validity of the information supplied.

2.1. Data elements definition and structure

The data elements are referenced by a sequential alphanumeric code. A unique element name is provided; this is to be used for column names, field names and tags depending on the software programs, files or databases implementing the SSD. The unique element name is composed only of characters from the Roman alphabet and does not include spaces or any other special characters. The unique element names should always be reported case sensitively[12] to ensure compatibility with information systems especially XML standards. Therefore the unique element names should be reported in the correct case, as specified in this guidance, when the data is electronically transmitted.

In order to avoid compatibility problems with systems not capable of managing case sensitive element names, the case will not be used as a unique distinction between two different element names. The data elements are also described by a label to be used in reports, printouts or in the graphical interfaces of the software programs that will manage the SSD. A data type is associated with each data element and it defines the values that it can contain. Data types will be defined using the W3C XML schemas data types specification (W3C, online).

The SSD2 supports three types of data elements:

- *Simple data elements* can contain only one instance of an 'element value' which may be either a 'text value' or a 'numeric value' or a 'catalogue value'. These elements were the only type available in SSD1;

- *Repeatable data elements* may contain one or many instances of an 'element value' for the specified data type. The 'Action taken' (N.05) in the SSD2 is an example of a repeatable data element. Data providers may wish to indicate more than one value belonging to the catalogue ACTION e.g. 'Administrative consequences' and 'Rapid alert notification'. The detail for the syntax to report this type of element is defined in the GDE2. An example could be: '*value1$value2$...$valueN*'[13]: 'A$R'. For information Table 1 contains the meaning of the codes A and R included in the example.

Table 1: Extract from Action catalogue

Code	Name
A	Administrative consequences
R	Rapid alert notification

[12] Distinguishing upper- and lower-case letters. Often used in computer science to indicate that a distinction is made in comparison or equality of letters based on case. For example, a case-sensitive comparison will not recognize "Password" and "password" as the same, but a case-insensitive comparison would.

[13] The example of syntax is provided in order to clarify reading but it is does not define any requirement for the syntax of repeatable fields which will be under the scope of the GDE2. It was highlighted, particularly during the consultation phase, the need to use an XML subsection to describe repeatable fields.

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- *Compound data elements* may contain one or many 'simple elements' which are named 'attributes' of the 'compound element'. Each attribute can have values which can be 'text values', 'numeric values' or 'catalogue values'. In order to distinguish the different attributes inside a 'compound element' both the name of the 'attribute' and its value has to be included. In the scenario presented above, all attributes provide information at the same level and they can be provided independently. Alternatively, one attribute could be defined as the default attribute of the compound element: this attribute is referred to as 'base term'. In this case the other attributes further describe the 'base term' and refer to it providing additional information. These attributes are named 'facets', while the 'facet' values are called 'facet descriptors'. In the SSD2, a typical example of this type of element is the 'matrix analysed'. This element follows the FoodEx2 classification, where a base term from the FoodEx2 building hierarchy can be enriched with additional information through the addition of facets. The detail for the syntax to report this type of element is defined in the GDE2. An example could be: '*element1=value1$element2=value2$...$ elementN=valueN*' or '*baseTerm#facet1=value1$...$ facetN=valueN*'[14]. Some compound data elements can be defined at a later stage for collecting information relevant for a specific call for data. The 'compound data elements' introduce flexibility to the SSD model, which was not present in SSD1, in order to keep its overall structure stable for the long term. An example performed with the FoodEx2 catalogue is A01TX#F01.A057E$F02.A069J.

Table 2: Extract from FoodEx2 catalogue

Base term		**Facet**		**Facet descriptor**	
Code	**Name**	**Code**	**Name**	**Code**	**Name**
A01TX	Bovine, fresh fat tissue	F01	Food source animal	A057E	Cattle
		F02	part-nature	A069J	Fat tissue

The sequence and relationships between the different types of elements that constitute the SSD2 data structure is presented in Figure 1.

[14] The example of syntax is provided in order to clarify reading but does not define any requirement for the syntax of compound fields which will be under the scope of the GDE2. It was highlighted, particularly during the consultation phase, the need to use an XML subsection to describe compound fields.

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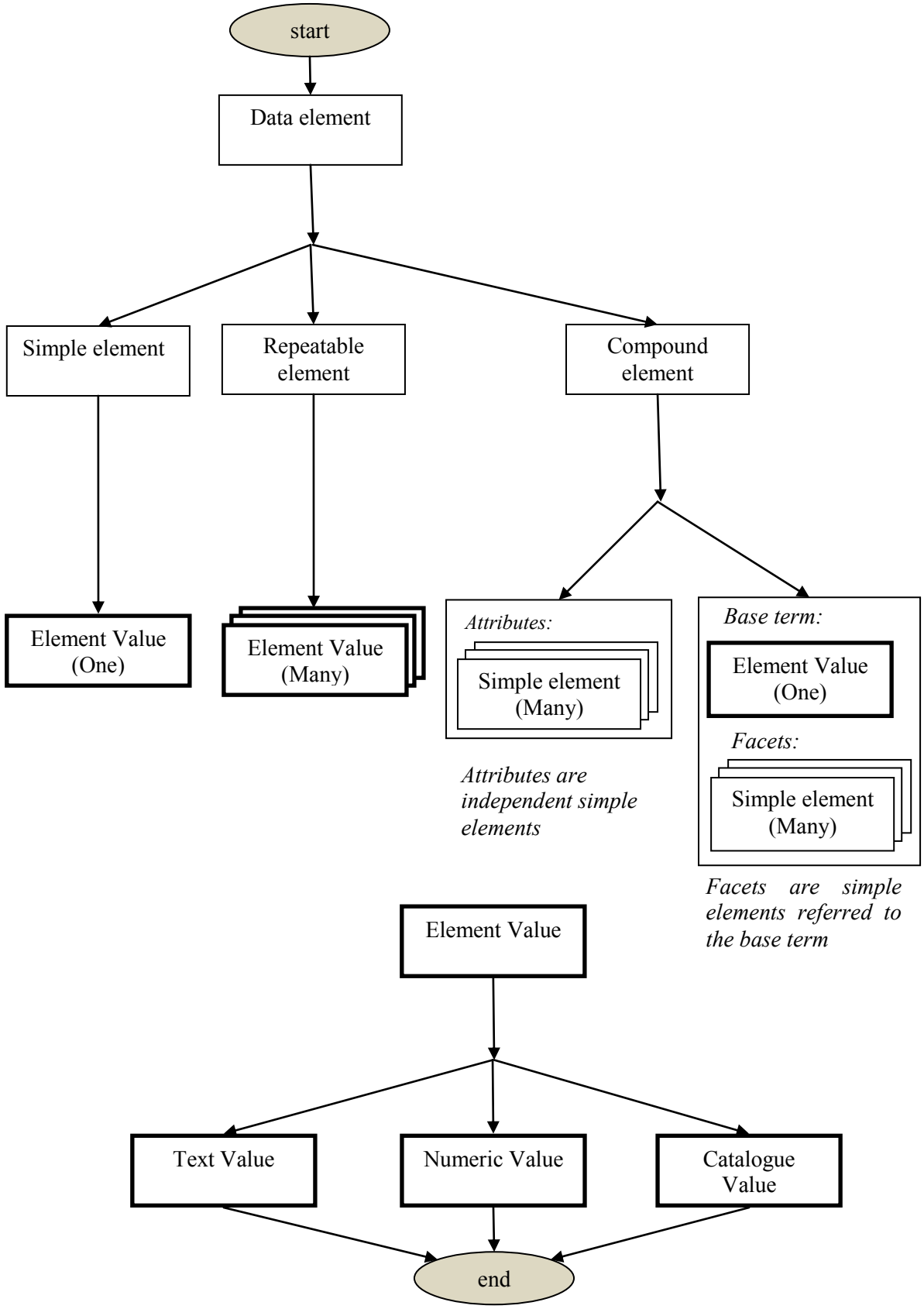

Figure 1: Data element structure in SSD2

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2.2. Controlled terminologies

The SSD includes controlled terminologies. A controlled terminology is a finite and enumerated set of terms intended to convey information unambiguously. The use of controlled terminologies facilitates the aggregation of data during analysis and ensures comparability between datasets. Controlled terminologies are language independent; only the code needs to be transmitted since the description of the code in any language can be linked to the code. This is consistent with the approach taken in SSD1 controlled terminologies and the WG-SSD2 recommends that EFSA continues to maintain the description in the controlled terminologies in English only and that optional translations for local use remain the responsibility of MS.

To ensure that the data are of sufficient quality to allow analysis at EU level, controlled terminologies have been applied extensively in the SSD. For some elements with controlled terminologies, an additional companion free text element is included. This allows the provision of relevant information when the controlled terminology is insufficient to fully characterise the item described. When this pair of fields is defined, they can be recognised since they take respectively the suffixes 'Code' and 'Text'.

2.3. Business rules

Within the SSD, the term 'business rules' is used to describe fixed rules for validation of the data provided in the data fields of a transmission. Business rules can either check the validity of a value reported in an individual data element (single data element validation) or they can cross check inter-dependent values reported in more data elements (inter-dependent data element validation). The SSD2, introducing repeatable and compound data elements, requires that the mechanism of business rule validation be extended also to this new type of element:

- *Single data element validation*: checks on specific data elements e.g. the verification whether the code reported in a field constrained to a controlled terminology is correct, or if the value reported is within a certain acceptance range. This verification will also include reference to the data domain validity of the value which is recorded in the catalogue. This will ensure that values in a controlled terminology which are invalid for a particular data domain are not transmitted in datasets for that domain;

- *Inter-dependent data element validation*: checks on two or more data elements e.g. the verification whether LOQ was supplied if the result was reported below LOQ;

- *Repeatable and compound data element validation*: checks can be performed on the whole element as well as on the individual elements of the list. E.g. for the element 'analysed matrix', it should be possible to check its consistency as whole element throughout the same sample as well as the presence/absence of required/deprecated facets or facets descriptors as requested by the specific data collection domains.

The implementation of the business rules on a field-by-field basis will be described in the GDE2. In any case, the GDE2 will address only those business rules which are applicable to all data collection domains. Certain business rules are due to a specific requirement in the relevant legislation or a particular project. These domain specific rules should be addressed in separate documents describing the data collections and they are the responsibility of the relevant networks.

EFSA, in conjunction with domain experts available in the relevant competent EFSA networks, will define the data collection specific requirements and business rules, including mandatory elements and facets for each data domain.

The WG-SSD2 highlights the importance of the validation rules to perform a consistent quality check of the data. The usage of validation rules is connected with the existence of standard terminologies. For this reason the WG-SSD2 also recommends to limit the usage of free text fields and to always prefer the use of standard terminologies.

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3. Scope of the Standard Sample Description

The SSD is targeted to support the data collection and transmission of the samples data and the results of analytical measurement of several data collections domains:

- Chemical analytical results:
  - pesticide residue concentration levels;
  - chemical contaminant concentration levels;
  - food additives concentration levels.
- Biological analytical results:
  - sample level data on microbiological contaminants;
  - isolate-based data on antimicrobial resistance;
  - animal/flock/herd level data on zoonotic agents.

The legislation texts taken into consideration for the design and specification of the SSD were:

- pesticide residues: reporting of results of the monitoring of pesticide residues in food according to
  - Regulation (EC) No 396/2005[15] of the European Parliament and of the Council and its amendments;
  - Commission implementing Regulation (EU) No 1274/2011 [16] concerning a coordinated multiannual control programme;
  - Commission Directive 2006/125/EC[17] and Commission Directive 2006/141/EC on infant and baby food[18];
  - Council Regulation (EC) No 834/2007 of 28 June 2007 on organic production and labelling of organic products and repealing Council Regulation (EEC) No 2092/91[19];
- chemical contaminants: reporting of results of the chemicals included in Commission Regulation (EC) No. 1881/2006[20] and its amendments;
- food additives: Regulation (EC) No 1333/2008 of the European Parliament and of the Council on food additives[21] and its amendments;
- biological monitoring data: Directive 2003/99/EC on monitoring of zoonoses and zoonotic agents[22] and Commission Regulation (EC) No 2073/2005 on microbiological criteria for foodstuffs[23] and its amendments.

The SSD is primarily designed for the collection of analytical results submitted to EFSA data collections. Although this model is expected to be suitable for ad-hoc studies, additional considerations, specifications and customisations are needed before using the data model outside the scope for which it was designed. Any time the SSD is applied to a new data domain, specific

[15] See note 8 page 5
[16] See note 9 page 5
[17] Commission Directive 2006/125/EC of 5 December 2006 on processed cereal-based foods and baby foods for infants and young children. OJ 339 6.12.2006, p. 16-35
[18] Commission Directive 2006/141/EC of 22 December 2006 on infant formulae and follow-on formulae and amending Directive 1999/21/EC. OJ 401 30.12.2006, p. 1-33
[19] Council Regulation (EC) No 834/2007 of 28 June 2007 on organic production and labelling of organic products and repealing Regulation (EEC) No 2092/91. OJ 189 20.7.2007, p. 1-23
[20] Commission Regulation (EC) No 1881/2006 of 19 December 2006 setting maximum levels for certain contaminants in foodstuffs. OJ 364 20.12.2006, p. 5-24
[21] Regulation (EC) No 1333/2008 of the European Parliament and of the Council of 16 December 2008 on food additives. OJ 354 31.12.2008, p. 16-33
[22] Directive 2003/99/EC of the European Parliament and of the Council of 17 November 2003 on the monitoring of zoonoses and zoonotic agents, amending Council Decision 90/424/EEC and repealing Council Directive 92/117/EEC. OJ 325 12.12.2003, p. 31-40
[23] Commission Regulation (EC) No 2073/2005 of 15 November 2005 on microbiological criteria for foodstuffs. OJ 338 22.12.2005, p. 1-26

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requirements and business rules applicable to that domain should be provided. This information should be provided, for example, in the call for data.

The SSD logical model is designed to support denormalised data transmissions and it should not be considered as a database design for *data storage* or *analysis*, since it is not optimised for these tasks.

In addition, it will be up to the relevant EFSA networks to define which file formats will be acceptable for the data transmissions in the domains under their competence.

Additional data elements are relevant for the transmission process; they are dependent on the needs of data providers and receivers and, therefore, are not considered as part of the SSD (see section 4 Context information of data transmissions).

Although the logical model of other data collection domains may be very similar to the SSD (e.g. veterinary drug residues or diseases in animals), the WG-SSD2 has validated the SSD ver. 2.0 only with examples from the domains explicitly defined in the current mandate.

The WG-SSD2 encourages the transmission of data from single results (determination or detection) obtained from one sample, sample analysed portion or isolate. In some cases these parameters could be calculated from other measures (e.g. pesticide residue definitions, dioxin TEQs).

Samples (or sample analysed portions or isolates) could have been taken from one single food/feed item, a single batch of food/feed or a defined class of food/feed (e.g. in total diet studies). Alternatively, in the field of biological samples, in addition to the previous mentioned cases, they could also originate from a single animal or a single epidemiological unit (e.g. flock/herd).

4. Context information of data transmissions

The WG-SSD2 has defined some entities which are necessary as context information for the data transmission. This contextual information mainly depends on the set up of the system of each data provider (organisation transmitting the data e.g. MS, academia and industry).

Data providers and data receivers should keep traceable accounts of transmissions and data as part of good management practice.

For submissions to EFSA, when the user will use the web interface, the context information will be automatically generated by the data collection system by selecting the correct data collection to which the file is uploaded. When using, instead, the electronic transmission system, some of this information must be explicitly mentioned in the header of the transmission file for electronic transmission of files (e.g. Web services). These concepts will be discussed in detail in the GDE2.

In the following list, the WG-SSD2 summarised the entities, which are necessary for data transmission:

A. *Sender country*: Entity describing the country of the organisation transmitting the data;

B. *Sender organisation*: Entity describing the organisation transmitting the data, the data provider organisation. The name Sender Organisation is given to stress that responsibility for the transmission is with the organisation rather than the individual user submitting the data;

C. *Receiver organisation*: Entity describing the organisation receiving the data. In the current situation it is typically EFSA, but the WG-SSD2 defined the model in such a way that it could be used by other organisations to receive data. In some countries SSD1 is already used to submit data from the Regional Competent Authorities to the National Competent Authority

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(Nicolau et al., 2012) although in the case of some data elements, the level of detail required for EFSA may be insufficient for a National Competent Authority.

D. *File transmission*: Entity linking all data submitted in a single file transmission. The entity should be described by some additional attributes such as the transmission date, the receipt date and other additional logging dates that may be needed by the transmission or receiver systems. A link to the physical original copy of the transmitted or received file should also be maintained.

E. *Data collection domain*: The data collection domain is an entity grouping together all the data collections on a specific area. The data collection domain plays an important role:

   - in the selection of domain specific controlled terminologies in the catalogues;
   - for some validation rules which could be domain dependant.

F. *Data collection*: Entity linking all files transmissions included in a single collection of data on specific areas over time. In general terms, the data receiver will define data collections on an ad-hoc basis: e.g. Heavy metal data collection in the Contaminant Area and Pesticide residues.

G. *Reference period*: Each data collection has one or more associated reporting periods which partition the data collection, for example in years e.g. Pesticide residues 2011, Pesticide residues 2012, Zoonoses data collection 2011, Zoonoses data collection 2012.

H. *Sampling event*: Represents the top level of the structure of the SSD2 for data reporting. It is represented in this data model with a dashed line, since it is defined and represented in section 6 of this document.

Figure 2 represents the entity-relationship diagrams between the entities defined above. In the diagram each line describes a relationship between two different entities. For each entity, a relationship describes the number of times ('cardinality') that each entity can appear in that relationship. The following cardinalities are possible:

- *0,1*: The entity is optional and not repeatable. The entity could be not present or it could be present once in the model;
- *1,1*: The entity is mandatory and not repeatable. It has to be present once in the data model;
- *0,n*: The entity is optional and repeatable. The entity could be not present or could be present one or more times in the model;
- *1,n*: The entity is mandatory and repeatable. The entity must be provided once or could be present several times in the model.

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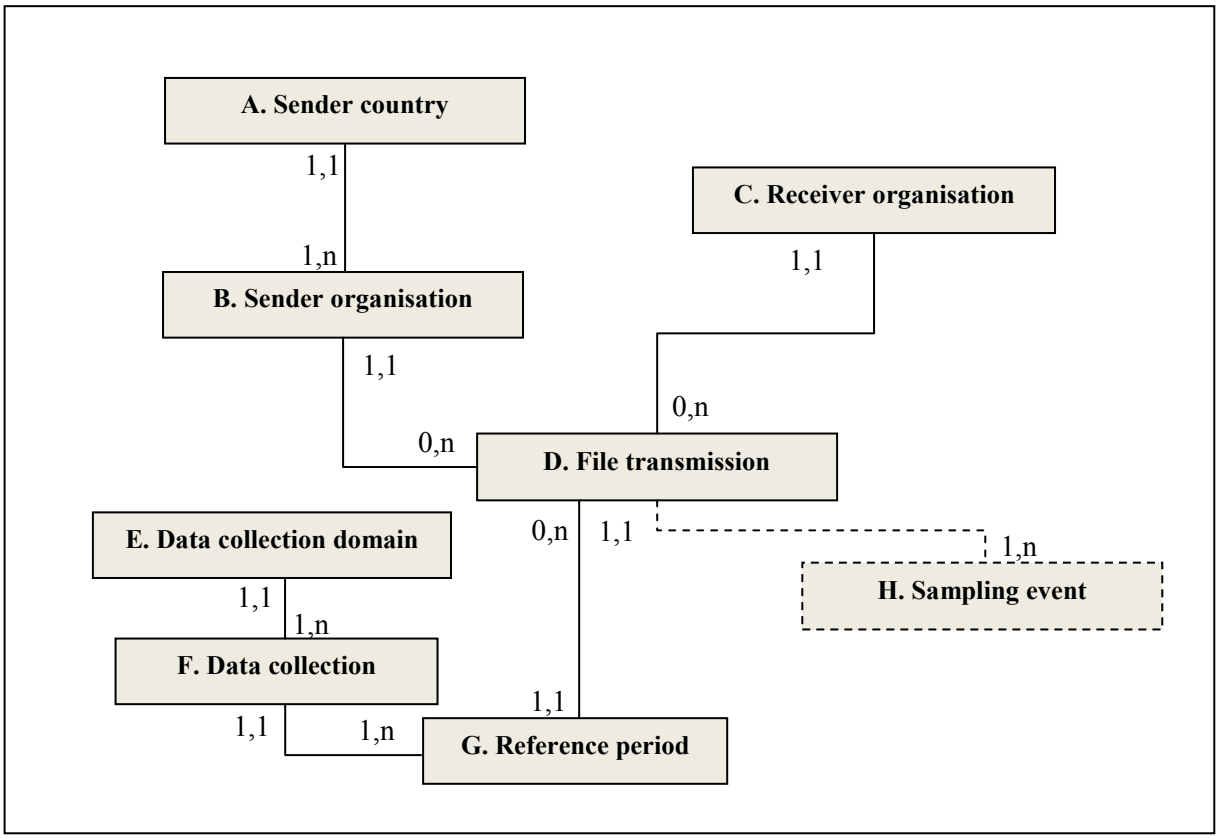

Figure 2: Structure of the main entities of the context in formation of data transmissions

5. Data structure of the Standard Sample Description

In order to support the transmission of data on samples analysed, the SSD2 needs to take into account the logical relationships between the following key entities which will be listed in different sections of the SSD in Table 4.

It should be taken into account that the terminology chosen to describe the different key entities is the best compromise the WG-SSD2 could find to harmonise divergent definitions existing in the different data collection domains and existing legislation.

A. *Local organisation*: The organisation (local/regional/National Competent Authority) that initially requested or performed the sampling and is responsible for the implementation of the sampling programme.

B. *Sampling programme*: The description of criteria, purpose, method or legislative reference used to generate a sampling event.

C. *Sampling event*: The entity representing the sampling unit extracted at a certain time from the sampled population, whose chemical or microbiological properties are the target of the sampling. In fact each sampling event is regarded as the aggregate of all samples taken at a certain time to investigate chemical or microbiological properties of the sampling unit under consideration at a certain time. Examples of 'sampling event' are an 'animal sampled at a certain time', a 'flock/herd of animals sampled at a certain time', a 'batch or lot of products sampled at certain time'. The same sampling unit sampled at different times represents different sampling events (excluding cases when for practical reasons a sampling unit may require many days to be sampled e.g. a large herd of sheep).

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D. *Sample taken*: In order to evaluate chemical or microbiological properties of the sampling event, one or more samples can be taken from the sampling unit at a certain time. For example the analysis of a sampling unit of a chicken flock can originate a series of samples such as samples from the environment (e.g. boot swabs), blood samples or chicken meat samples.

E. *Matrix sampled*: The description of the item sampled and of its relevant characteristics available before the analysis.

F. *Sample analysed:* The 'sample analysed' is usually, but not always, identical to the 'sample taken', i.e. with the sample sent to the laboratory. 'Sample analysed' describes the sample as it is analysed by the laboratory. Each 'sample taken' can be analysed as different 'samples analysed' using deliberately different experimental conditions (e.g. different time of analysis: the 'sample taken' is analysed by the laboratory at its arrival and at the end of its shelf life) or different processes applied before analysis (e.g. coffee powder and coffee as beverage for consumption). The reporting of the 'sample analysed' and the related 'matrix analysed' is only needed in these cases where the 'sample taken' is not analysed as such or the processing of the sample is well known and implicitly derivable by the 'sample taken' reported e.g. removing root and leaves of the certain commodities in the case of the pesticide residues.

G. *Matrix analysed:* Description of the 'matrix analysed' and its relevant characteristics. This will often be the same as the 'matrix sampled' but can vary as in the example of coffee powder and coffee as a beverage above (section F "Sample analysed").

H. *Sample analysed portion*: The 'sample analysed portion' (often called 'test portion') is a replicate achieved by subdividing the sample analysed. It represents the repetition of the same experimental condition defined in a 'sample analysed' to acquire data on the variability associated with such measurements. The usage of SSD element 'sample analysed portion' is only needed when an explicit requirement in a data collection domain exists to report the results of these replicas. Usually only one result of the 'sample analysed portion' should be reported which represents the best estimate for that measure. When more than one 'sample analysed portion' is taken (e.g. for aflatoxins analysis), this field is used for differentiation among 'sample analysed portions'.

I. *Isolate*: The isolate identifies, by a unique code, a culture of a biological agent, isolated from a specific 'sample taken'.

J. *Laboratory*: The laboratory in which the analysis was performed or the laboratory responsible for the results. In cases when more than one laboratory has performed analyses of the same 'sample taken', such as detection, confirmation or resistance testing, the laboratory responsible for the total results can be either the initial laboratory performing detection or the one performing confirmation or the one testing resistance.

K. *Parameter*: The specification of the parameter (analyte) determined in the matrix analysed.

L. *Analytical method*: The laboratory method of analysis and diagnostic method used to generate the result.

M. Result: The result of the laboratory tests, as a quantitative or qualitative outcome value. Information relating to the result also includes the accreditation procedure, limits of detection and quantification and the result value for the analytical method.

N. *Evaluation:* Assessment of the result, evaluating its compliance with a defined limit.

It should be noted that, for their definition, 'sampling event', 'sample taken', 'sample analysed' and 'sample analysed portion' are considered always present. But there are many cases where these key entities may be not relevant or overlapping. To simplify the reporting in these cases, a mechanism for

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the automatic completion of the relevant SSD elements was established and this is specified in the next section of this document (section 6).

The diagram in Figure 3 represents the relationship between the represented entities. The meaning of the cardinalities reported in the figure is described in section 4. The data structure reported in Figure 3 is implemented in the SSD in *denormalised* format. The pros and cons of the denormalised approach are summarised below.

Pros:

- Simplified generation of the data files since no nested elements are included;
- Similarity with a spread sheet table, into/from which the denormalised structure can be easily imported/exported.

Cons:

- Possible errors due to incorrect repetition of the values in the upper level entities (e.g. Sample taken identifier);
- The file size is larger because the same information is repeated.

The denormalised structure was agreed because the simplicity of data file generation was judged to be of higher priority. In order to mitigate the negative aspects of a denormalised structure the WG-SSD2 suggests:

- A strict management of the codes assigned to the different entities represented in the SSD model of Figure 2 which should be, in most cases, database managed;
- The definition of specific business rules to verify automatically the compliance between the entries reported and codes assigned to the different entities;
- The use of compressed archives (such as 'zip files') and a limit for the number of records that can be transmitted in a single file. Additional details on this aspect can be found in the 'GDE2'.

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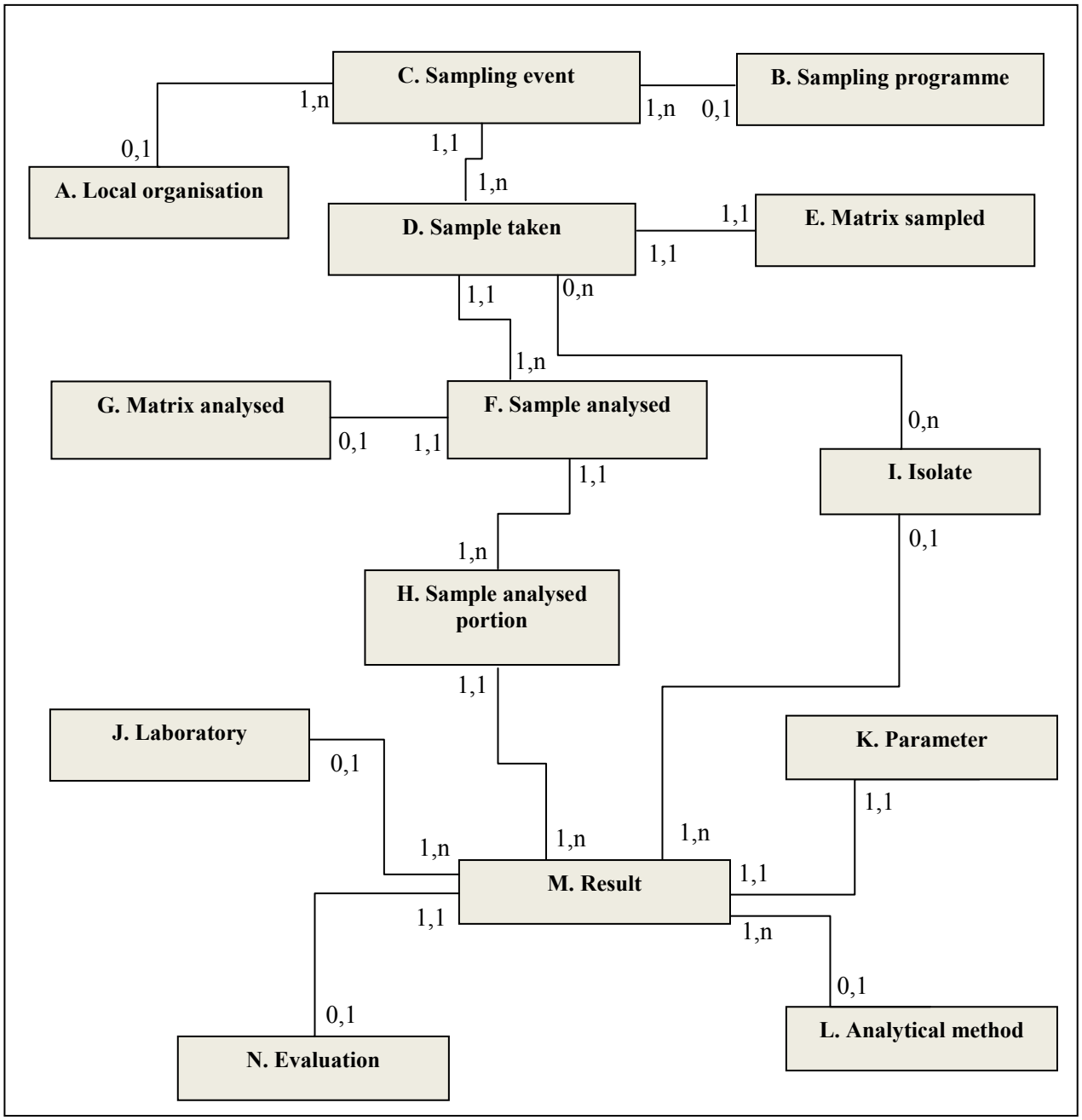

Figure 3: Structure of the main Standard Description entities

As pointed out in the definition above, the WG-SSD2 believes that the identification of the right mapping of the data against the entities 'sampling event', 'sample taken', 'sample analysed' and 'sample analysed portion' is not always unambiguous, due to heterogeneity of definition in the different legislations. Therefore the WG-SSD2 created a guidance table (Table 3) to help the data providers in the selection of the appropriate level for the data collection domains described in this guidance.

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Table 3: Guidance on mapping the existing different reporting levels in data collection against the SSD sampling event, sample taken, sample analysed, sample analysed portion and isolate entities

Legislation	***Data collection***	***Sampling event***	***Sample taken***	***Sample analysed***	***Sample portion analysed***	***Isolate***
Zoonoses	Food/ feed	Single or batch	Sample. In case the sample comprises of several subsamples, each subsample is reported in different row.	In most cases the same as the *Sample taken*. In case the *Sample taken* is analysed in different times (e.g. at the arrival to the laboratory and then at the end of shelf life), two or more *Samples analysed* are reported in individual rows.	In most cases not used. In case replicates are obtained by subdividing the *Sample analysed*, two or more *Samples portion analysed* are reported in individual rows.	It is used to identify a culture of a biological agent used for further analyses and to group all results from this culture.
	Animals	Animal/flock/herd /holding or slaughtered animal batch	Sample In case the sample comprises of several subsamples, each subsample is reported in different row.	In most cases the same as the *Sample taken*. In case the *Sample taken* is analysed in different time, two or more *Samples analysed* are reported in individual rows.	In most cases not used. In case replicates are obtained by subdividing the *Sample analysed*, two or more *Samples portion analysed* are reported in individual rows.	It is used to identify a culture of a biological agent used for further analyses and to group all results from this culture.
	Shelf-life studies e.g. *Listeria monocytogenes* baseline survey 2010 - 2011 and other food pathogens	Single or batch	Sample (1 to 5) *(meat sample, cheese sample, etc..)*	For each sample taken, 2 samples analysed (at the arrival - at the end of shelf life)	Not used	Not used
Pesticides	Pesticides	Lot	Laboratory sample -> only description of matrix taken should be provided	Analytical sample (composition as defined in legislation) therefore description of matrix analysed should not be provided since it is provided in the legislation	Not used	Not used

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Legislation	*Data collection*	*Sampling event*	*Sample taken*	*Sample analysed*	*Sample portion analysed*	*Isolate*
Contaminants	Aflatoxins	Lot	Laboratory sample	Analytical sample (same matrix and identification as laboratory sample, therefore it should not be provided)	Analytical portion: Several analytical portion should be reported	Not used
	Furan (and process contaminants)	Lot	Laboratory sample	Analytical sample(s) (Several: matrix description different identification - processing/storage etc.)	Not used	Not used
	Other contaminants	Lot	Laboratory sample	Analytical sample: Usually the same as the laboratory sample; in this case it should not be reported. In case analytical sample identification is different from the sample taken identification, it should be reported.	Not used	Not used
Food additives	Additives	Lot	Laboratory sample	Analytical sample(s) (Several: Composition different - separation etc.).	Not used	Not used
Food contact materials	Food contact materials composition	Lot	Laboratory sample, plastic sampled	Analytical sample(s) (Identification is the same as in laboratory sample therefore should not be provided)	Not used	Not used
	Food contact materials migration	Lot	Laboratory sample, plastic sampled	Food simulant: the matrix analysed will be the food simulant, for total migration param= 'total migration'.	Not used	Not used

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6. Data elements

List of elements

Table 4 contains the list of the data elements for the SSD. In this table, the meaning of the columns is as follows:

- *Element code*: An alphanumeric code providing a unique identifier for the data element. The element code is made of the section identifier code plus a progressive number.
- *Section code*: The section code identifies the entity of the SSD data model, represented in Figure 3.
- *Section*: The section describes the key entity of the SSD data model, represented in Figure 3.
- *Element name*: Unique element name is provided; this is to be used for column names, field names and tags depending on the software programs, files or databases implementing the SSD. The unique element name is composed only of characters from the Roman alphabet and does not include spaces or any other special characters. The unique element names should be considered case sensitive[24] to ensure compatibility with information systems especially XML standards.
- *Element label*: The data elements are described also by a label to be used in reports, print outs or in the graphical interfaces of the software programs that will manage the SSD.
- *Type*: A data type is associated to each data element and it defines the values that it can contain. Data types will be defined using the W3C XML schemas data type specification (W3C, online).
- *S/R/C*: Single, repeatable or compound data element. It can contain 'S' (*Single*) if the data element can be reported only once (generic structure: '*value1')*, 'R' (*Repeatable*) if one or more values can be reported within the data element. 'C' (*Compound*) is used for those data elements that are made from one optional base term plus facets or from many attributes. Additional information is available in section 2.1.
- *M*: Mandatory elements are flagged in this column with the value 'M'. This column contains the value mandatory when it applies for all data domains within the mandate of the WG-SSD2. Additional mandatory elements can be defined in specific data domains or in specific data collections. Some data elements can be made mandatory also by the business rules in special circumstances, only when some values are present in related data elements. Additional information on this aspect is provided in the business rules section in the GDE2.
- *Controlled terminology:* Provides the acronym of the catalogue that can be used to populate the data element. A catalogue is a finite and enumerated set of terms intended to convey information unambiguously.
- *Description:* Provides a short description on what the data element should contain.

[24] Distinguishing upper- and lower-case letters. Often used in computer science to indicate a distinction is made in comparison or equality of letters based on case. For example, a case-sensitive comparison will not recognize "Password" and "password" as the same, but a case-insensitive comparison would.

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Table 4: List of the data elements for the SSD

Element Code	*Section Code*	*Section*	*Element Name*	*Element Label*	*Type*[(a)]	*S/R/C*	*M*	*Controlled terminology*	*Description*
A.01	A	Local organisation	localOrgId	Local organisation identification code	xs:string (100)	S			Unique identification of the local or regional or national organisation (Competent Authority or company affiliate) requesting the analysis.
A.02	A	Local organisation	localOrgCountry	Local organisation country	xs:string (2)	S		COUNTRY	Country where the local organisation is placed. (ISO 3166-1-alpha-2).
A.03	A	Local organisation	localOrgInfo	Local organisation additional information	*CompoundType*[(b)]	C			Additional specific information and comments on the local organisation depending on specific requirements of the different data collection domains.
B.01	B	Sampling programme	progId	Sampling programme identification code	xs:string (100)	S			Unique identification code of the programme or project for which the sampling unit was taken.
B.02	B	Sampling programme	progLegalRef	Programme legal reference	xs:string (5)	S		LEGREF	Reference to the legislation for the programme defined by programme code. Reference to the legislation on what to sample, how to evaluate the sample etc.
B.03	B	Sampling programme	sampStrategy	Sampling strategy	xs:string (5)	S		SAMPSTR	Sampling strategy describe how the sample was selected (ref. EUROSTAT - Typology of sampling strategy performed in the programme or project identified by programme code (e.g. objective and selective sampling)).
B.04	B	Sampling programme	progType	Programme type	xs:string (5)	S		PRGTYP	Indicate the type of programme for which the samples have been collected (National, EU programme, Total diet study, Control and eradication programme).
B.05	B	Sampling programme	sampMethod	Sampling method	xs:string (5)	S		SAMPMD	Reference to the method for sampling (e.g. EU legislation).
B.06	B	Sampling programme	sampler	Sampler	xs:string (5)	S		SAMPLR	Define which organisation (private or public) is performing the sample.

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Element Code	*Section Code*	*Section*	*Element Name*	*Element Label*	*Type*[(a)]	*S/R/C*	*M*	*Controlled terminology*	*Description*
B.07	B	Sampling programme	sampPoint	Sampling point	xs:string (5)	S		SAMPNT	Point, in the food chain, where the sample was taken. (See Doc. ESTAT/F5/ES/155 "Data dictionary of activities of the establishments").
B.08	B	Sampling programme	progInfo	Additional sampling program information	*CompoundType*[(b)]	C			Additional specific information and comments on the sampling programme depending on specific requirements of the different data collection domains such as if the programme is used for the verification of the *Salmonella* reduction target, number of animal under the control program, total number of samples tested, etc.
C.01	C	Sampling event	sampEventId	Sampling event identification code	xs:string (100)	S			Unique identification of the sampling event. The entity representing the sampling unit extracted at certain time from the sampled population, whose chemical or microbiological properties are the target of the sampling.
C.02	C	Sampling event	sampUnitType	Sampling unit type	xs:string (5)	S		SAMPUNTYP	Define the type of sampling unit taken in this event: a batch, an animal, a flock, a herd, etc.
C.03	C	Sampling event	sampUnitSize	Sampling unit size	xs:double	S			It contains the size/amount of the sampling unit.
C.04	C	Sampling event	sampUnitSizeUnit	Sampling unit size unit	xs:string (5)	S		UNIT	It contains the Unit in which the sampling unit size is expressed.
C.05	C	Sampling event	sampUnitIds	Other sampling unit identifications	*CompoundType*[(b)]	C			Additional identification codes for the sampling unit, at a more detailed level than the sampling event ID e.g. herd code or animal ear tag number.
C.06	C	Sampling event	sampEventInfo	Additional sampling event information	*CompoundType*[(b)]	C			Additional information on the sampling event depending on specific requirements of the different data collection domains such as status of the holding, the vaccination status, the date and country of slaughtering, etc.
D.01	D	Sample taken	sampId	Sample taken identification code	xs:string (100)	S	M		Identification code of the sample taken.

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Element Code	*Section Code*	*Section*	*Element Name*	*Element Label*	*Type*[(a)]	*S/R/C*	*M*	*Controlled terminology*	*Description*
D.02	D	Sample taken	repCountry	Reporting country	xs:string (2)	S		COUNTRY	The country the reported data refer to (ISO 3166-1-alpha-2).
D.03	D	Sample taken	sampCountry	Country of sampling	xs:string (2)	S	M	COUNTRY	Country where the sample was taken for laboratory testing (ISO 3166-1-alpha-2).
D.04	D	Sample taken	sampArea	Area of sampling	xs:string (5)	S		NUTS	Area where the sample was collected (Nomenclature of territorial units for statistics - NUTS - coding system valid only for EEA and Switzerland).
D.05	D	Sample taken	repYear	Reporting year	xs:integer (4)	S			The year the reported data refer to.
D.06	D	Sample taken	sampY	Year of sampling	xs:integer (4)	S	M		Year of sampling. In case the sampling has been performed over a period of time the start date (as year) of sampling should be reported.
D.07	D	Sample taken	sampM	Month of sampling	xs:integer (2)	S			Month of sampling. In case the sampling has been performed over a period of time the start date (as month) of sampling should be reported.
D.08	D	Sample taken	sampD	Day of sampling	xs:integer (2)	S			Day of sampling. In case the sampling has been performed over a period of time the start date (as day) of sampling should be reported.
D.09	D	Sample taken	sampSize	Sample taken size	xs:double	S			Total size/amount of the sample.
D.10	D	Sample taken	sampSizeUnit	Sample taken size unit	xs:string (5)	S		UNIT	Unit in which the size/amount of the sample is expressed.
D.11	D	Sample taken	sampInfo	Additional Sample taken information	*CompoundType*[(b)]	C			Additional information on the sampling taken depending on specific requirements of the different data collection domains (e.g. day of arrival in the lab).
E.01	E	Matrix sampled	sampMatType	Type of matrix	xs:string (5)	S	M	MTXTYP	Type of sample taken (e.g. food, food stimulants, animal, feed, environment; food contact material), identifying the sub-domain of the matrix catalogue to be used.
E.02	E	Matrix sampled	sampMatCode	Coded description of the matrix of the sample taken	*CompoundType*[(b)]	C	M	MTX	Description of the sample taken characteristics using the FoodEx2 catalogue.

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Element Code	*Section Code*	*Section*	*Element Name*	*Element Label*	*Type*[(a)]	*S/R/C*	*M*	*Controlled terminology*	*Description*
E.03	E	Matrix sampled	sampMatText	Text description of the matrix of the sample taken	xs:string (250)	S			Description of the sample taken characteristics using free text.
E.04	E	Matrix sampled	origCountry	Country of origin of the sample taken	xs:string (2)	S		COUNTRY	Country of origin of the sample taken (ISO 3166-1-alpha-2 country code).
E.05	E	Matrix sampled	origArea	Area of origin of the sample taken	xs:string (5)	S		NUTS	Area of origin of the sample taken (Nomenclature of territorial units for statistics - NUTS - coding system valid only for EEA and Switzerland).
E.06	E	Matrix sampled	origFishAreaCode	Area of origin for fisheries or aquaculture activities code of the sample taken	xs:string (10)	S		FAREA	Fisheries or aquaculture area specifying the origin of the sample (FAO Fisheries areas).
E.07	E	Matrix sampled	origFishAreaText	Area of origin for fisheries or aquaculture activities text of the sample taken	xs:string (250)	S			Fisheries or aquaculture area specified in free text.
E.08	E	Matrix sampled	procCountry	Country of processing of the sample taken	xs:string (2)	S		COUNTRY	Country where the food was processed (ISO 3166-1-alpha-2).
E.09	E	Matrix sampled	procArea	Area of processing of the sample taken	xs:string (5)	S		NUTS	Area of product processing (Nomenclature of territorial units for statistics - NUTS - coding system valid only for EEA and Switzerland).
E.10	E	Matrix sampled	sampMatInfo	Additional information on the matrix sampled	*CompoundType*[(b)]	C			Additional specific information and comments on the matrix sampled, depending on specific requirements of the different data collection domains.
F.01	F	Sample analysed	sampAnId	Sample analysed identification code	xs:string (100)	S			Identification code of the analysed sample, by default the same as the sampId.
F.02	F	Sample analysed	sampAnRefTime	Sample analysis reference time	xs:string (5)	S		REFTM	Define the time at which the sample was analysed e.g. ‘analysed at arrival to the laboratory’, ‘analysed at the end of shelf-life’

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Element Code	*Section Code*	*Section*	*Element Name*	*Element Label*	*Type*[(a)]	*S/R/C*	*M*	*Controlled terminology*	*Description*
									(according to European legislation on microbiological criteria Reg. 2073/2005).
F.03	F	Sample analysed	analysisY	Year of analysis	xs:integer (4)	S	M		Year when the analysis was completed.
F.04	F	Sample analysed	analysisM	Month of analysis	xs: integer (2)	S			Month when the analysis was completed.
F.05	F	Sample analysed	analysisD	Day of analysis	xs: integer (2)	S			Day when the analysis was completed.
F.06	F	Sample analysed	sampAnInfo	Additional information on the sample analysed	*CompoundType*[(b)]	C			Additional specific information and comments on the sample analysed depending on specific requirements of the different data collection domains.
G.01	G	Matrix analysed	anMatCode	Coded description of the analysed matrix	*CompoundType*[(b)]	C		MTX	Encoding of the matrix analysed characteristics using the FoodEx2 catalogue. By default this element has the same value as "sampMatCode".
G.02	G	Matrix analysed	anMatText	Text description of the matrix analysed	xs:string (250)	S			Description of the matrix analysed characteristics using free text.
G.03	G	Matrix analysed	anMatInfo	Additional information on the analysed matrix	*CompoundType*[(b)]	C			Additional specific information and comments on the matrix analysed depending on specific requirements of the different data collection domains.
H.01	H	Sample analysed portion	anPortSeq	Sample analysed portion sequence	xs:string (100)	S			Sequence number (e.g. 1, 2, 3) reflecting the sample analysed portion actually under analysis. The default value is 1.
H.02	H	Sample analysed portion	anPortSize	Sample analysed portion size	xs:double	S			Size / amount of the sample analysed portion, i.e. amount of sample weight for analysis (weight of test portion).
H.03	H	Sample analysed portion	anPortSizeUnit	Sample analysed portion size unit	xs:string (5)	S		UNIT	Unit in which the size of the sample analysed portion is expressed.
H.04	H	Sample analysed portion	anPortInfo	Additional information on the sample	*CompoundType*[(b)]	C			Additional information and comments on the sample analysed portion depending on specific requirements of the different data

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Element Code	*Section Code*	*Section*	*Element Name*	*Element Label*	*Type*[(a)]	*S/R/C*	*M*	*Controlled terminology*	*Description*
				analysed portion					collection domains.
I.01	I	Isolate	isolId	Isolate identification	xs:string (100)	S			Identification code used to group an isolate identification with antimicrobial susceptibility tests performed on the same isolate.
I.02	I	Isolate	isolParamCode	Coded description of the isolate	CompoundType[(b)]	C		PARAM	Encoding of the isolate parameter code according to the PARAM catalogue. It is used to report the speciation or serotyping of the isolate.
I.03	I	Isolate	isolParamText	Text description of the isolate	xs:string (250)	S			Description of the isolate parameter (e.g. speciation/ serotyping) using free text.
I.04	I	Isolate	isolInfo	Additional information on the isolate	*CompoundType*[(b)]	C			Additional specific information and comments on the isolate depending on specific requirements of the different data collection domains.
J.01	J	Laboratory	labId	Laboratory identification code	xs:string (50)	S			Identification code of the laboratory (National laboratory code if available). This code should be nationally unique and consistent through all data domain transmissions.
J.02	J	Laboratory	labAccred	Laboratory accreditation	xs:string (1)	S		LABACC	The accreditation status of the laboratory and its reference procedure.
J.03	J	Laboratory	labCountry	Laboratory country	xs:string (2)	S		COUNTRY	Country where the laboratory is located (ISO 3166-1-alpha-2).
J.04	J	Laboratory	labInfo	Additional information on the laboratory	*CompoundType*[(b)] **Error! Bookmark not defined.**	C			Additional specific information and comments on the laboratory (e.g. total number of isolates available in the laboratory) depending on specific requirements of the different data collection domains.
K.01	K	Parameter	paramType	Type of parameter	xs:string (5)	S	M	PARAMTYP	Define if the parameter reported is an individual residue/ analyte, a summed residue definition or part of a summed residue definition.
K.02	K	Parameter	paramCode	Coded description of the parameter	*CompoundType*[(b)]	C	M	PARAM	Encoding of the parameter/ analyte according to the PARAM catalogue.

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Element Code	*Section Code*	*Section*	*Element Name*	*Element Label*	*Type*[a]	*S/R/C*	*M*	*Controlled terminology*	*Description*
K.03	K	Parameter	paramText	Parameter text	xs:string (250)	S			Description of the parameter/ analyte using free text.
L.01	L	Analytical method	anMethRefId	Analytical method identification	xs:string (50)	S			Identifier for the method used in the laboratory.
L.02	L	Analytical method	anMethRef Code	Analytical method reference code	xs:string (5)	S		ANLYREFMD	When validated methods are used, the official reference code should be provided.
L.03	L	Analytical method	anMethType	Analytical method type	xs:string (5)	S		ANLYTYP	Type of analytical method used.
L.04	L	Analytical method	anMethCode	Analytical method code	*CompoundType*[b]	C		ANLYMD	Encoding of the method or instrument used from the ANLYMD catalogue.
L.05	L	Analytical method	anMethText	Analytical method text	xs:string (250)	S			Description of the method or instrument using free text, particularly if 'other' was reported for 'Analytical method code'.
L.06	L	Analytical method	anMethInfo	Additional information on the analytical method	*CompoundType*[b] **Error! Bookmark not defined.**	C			Additional specific information and comments on the analytical method depending on specific requirements of the different data collection domains such as disk concentration and diameter for antimicrobial resistance diffusion method, method sensitivity and method specificity, migration time, migration temperature, etc....
M.01	M	Result	resId	Result identification code	xs:string (100)	S	M		Identification code of an analytical result (a row of the data table) in the transmitted file. The result identification code must be maintained at organisation level and it will be used in further updated/deletion operation from the senders.
M.02	M	Result	accredProc	Accreditation procedure for the analytical method	xs:string (5)	S		MDACC	The accreditation status of the analytical method used and its reference procedure.
M.03	M	Result	resUnit	Result unit	xs:string (5)	S		UNIT	Unit of measurement for the values reported in 'Result LOD', 'Result LOQ', 'ResLLWR', 'ResULWR', 'CC alpha', 'CC beta', 'Result value', 'Result value uncertainty standard deviation, 'Result value uncertainty', 'Limit for the result evaluation' and 'Limit for the

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Element Code	*Section Code*	*Section*	*Element Name*	*Element Label*	*Type*[(a)]	*S/R/C*	*M*	*Controlled terminology*	*Description*
									result evaluation (High limit)'.
M.04	M	Result	resLOD	Result LOD	xs:double	S			Limit of detection expressed in the unit specified by the element 'Result unit'.
M.05	M	Result	resLOQ	Result LOQ	xs:double	S			Limit of quantification expressed in the unit specified by the element 'Result unit'.
M.06	M	Result	resLLWR	Result lower limit of the working range	xs:double	S			Lower limit of the working range expressed in the unit specified by the element 'Result unit'.
M.07	M	Result	resULWR	Result upper limit of the working range	xs:double	S			Upper limit of the working range expressed in the unit specified by the element 'Result unit'.
M.08	M	Result	CCalpha	CC alpha	xs:double	S			CC alpha value (decision limit) expressed in the unit specified by the element 'Result unit'.
M.09	M	Result	CCbeta	CC beta	xs:double	S			CC beta value (detection capability) expressed in the unit specified by the element 'Result unit'.
M.10	M	Result	resVal	Result value	xs:double	S			The result of the analytical measure expressed in the unit specified by the element 'Result unit'.
M.11	M	Result	resValRec	Result value recovery rate	xs:double	S			Recovery value associated with the concentration measurement expressed as a percentage (%). i.e. report 100 for 100 %.
M.12	M	Result	resValRecCorr	Result value corrected for recovery	xs:string (1)	S		YESNO	Define if the result value has been corrected for recovery.
M.13	M	Result	exprResPerc	Expression of result percentage	*CompoundType*[(b)]	C			This compound field can be used to report the percentage of a measured specific matrix component (e.g. fat, alcohol, moisture) used as reference to express the analytical result (e.g. on fat basis, on alcohol basis and on dry weight basis).
M.14	M	Result	exprResType	Expression of result type	xs:string (5)	S		EXPRRES	Code to describe how the result has been expressed: whole weight, fat weight, dry

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Element Code	*Section Code*	*Section*	*Element Name*	*Element Label*	*Type*[(a)]	*S/R/C*	*M*	*Controlled terminology*	*Description*
									weight, etc.
M.15	M	Result	resQualValue	Result qualitative value	xs:string (3)	S		POSNEG	This field should be completed only if the result value is qualitative e.g. positive/ present or negative/ absent. In this case the element 'Result value' should be left blank.
M.16	M	Result	resType	Type of result	xs:string (3)	S		VALTYP	Indicate the type of result, whether it could be quantified/determined or not.
M.17	M	Result	resValUncert	Result value uncertainty	xs:double	S			Indicate the expanded uncertainty value (usually 95% confidence interval) associated with the measurement expressed in the unit reported in the field 'Result unit'.
M.18	M	Result	resValUncertSD	Result value uncertainty Standard deviation	xs:double	S			Standard deviation for the uncertainty of measurement.
M.19	M	Result	resRefId	Result reference identification	xs:string (100)	S			When the result is a complex structure, the identification code to the external structure is stored here (e.g. molecular typing images).
M.20	M	Result	resInfo	Additional information on the result	*CompoundType*[(b)]	C			Additional specific information and comments on the result section depending on specific requirements of the different data collection domains.
N.01	N	Evaluation	evalLowLimit	Limit for the result evaluation	xs:double	S			Report the reference or legal limit, limit or cut-off value for the parameter/analyte for the relevant matrix or the lower level of three-class evaluation limit analyte. It is expressed in the unit specified by the element "Result unit".
N.02	N	Evaluation	evalHighLimit	Limit for the result evaluation (High limit)	xs:double	S			Report the higher legal limit of the analyte for the three-class evaluation limit analyte. It is expressed in the unit specified by the element "Result unit".

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Element Code	*Section Code*	*Section*	*Element Name*	*Element Label*	*Type*[(a)]	*S/R/C*	*M*	*Controlled terminology*	*Description*
N.03	N	Evaluation	evalLimitType	Type of limit for the result evaluation	xs:string (5)	S		LMTTYP	Type of legal limit used to evaluate the result. ML, MRPL, MRL, action limit, cut-off value etc.
N.04	N	Evaluation	evalCode	Evaluation of the result	xs:string (5)	S		RESEVAL	Evaluation of the result. If the result exceeds a limit specified above or contains the evaluation on Sampling Event, Sample Taken, or Sample Analysed as indicated by evalLowLimit (N.01).
N.05	N	Evaluation	actTakenCode	Action Taken	*xs:string(1)*	R		ACTION	Describe any follow-up actions taken as a result higher than the legal limit.
N.06	N	Evaluation	evalInfo	Additional information on the evaluation	*CompoundType*[(b)]	C			Additional specific information and comments on the evaluation section depending on specific requirements of the different data collection domains.

(a): W3C, online. XML Schema Part 2: Datatypes Second Edition. W3C recommendation 28 October 2004. Available online:,http://www.w3.org/TR/xmlschema-2/

(b): Compound elements or repeatable elements have been defined as an XML custom type compoundType, so that its definition can be defined in the GDE2. It is suggested that for preliminary usage of the data structure, the compoundType is assumed xs:string (4000). This preliminary length provided should allow sufficient space to report the information for a compound element as free text.

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The sections and elements further described in the next sections may use controlled terminologies as already listed in Table 4. The lists of all values belonging to the controlled terminology are reported in an appendix document to this guidance: Appendix A List of controlled terminologies as Microsoft Excel® workbook 'StandardSampleDescription.xls'. The definition of the compound fields with all the applicable facets is also available in the same document.

Data elements definitions

This section contains the lists of standardised data, describing characteristics of samples or analytical results such as country of origin, product, analytical method, limit of detection, result, etc. Most of the data have to be submitted in a standardised way (controlled terminologies and validation rules) to enhance data quality.

A Local organisation section

This section is designed to indicate that the sender organisation received the data in the transmission from a local/ regional/ national organisation or a local/ regional/ national competent authority or an affiliate for a commercial organisation, which implemented the sampling programme. This section is important for commercial organisations where the data may be submitted by the headquarters but be collected by a variety of local entities. If the data collection is decentralised it is possible that duplicated samples could be transmitted to EFSA or other data receivers. In this case the information about the local organisation can help in the detection of duplicate samples. This section is characterised by the elements reported below.

A.01 Local organisation identification code (localOrgId)

This element defines the identification code for the local or regional or national organisation (Competent Authority or company affiliate) requesting the analysis.

A.02 Local organisation country (localOrgCountry)

This element defines the country where the local organisation is located. This field will follow the ISO 3166-1-alpha-2 country code list available in the COUNTRY catalogue.

A.03 Local organisation additional information (localOrgInfo)

This compound field refers to additional specific information and comments on the 'local organisation' depending on specific requirements of the different data collection domains.

B Sampling programme section

This section contains all the data elements necessary to describe the criteria, purpose, method and legislative reference of the sampling programme under which the sampling event occurred.

B.01 Sampling programme identification code (progId)

This data element should contain the laboratory, sender organisation or local organisation's unique identification code for the sampling programme or project for which the sample described by the SSD2 was taken. Each code will identify a specific statistical sample investigated for a programme or project. Further description on the reason for sampling can be provided in the element 'Programme legal reference' (see below element B.02).

B.02 Programme legal reference (progLegalRef)

This data element allows the selection from a specific controlled terminology (LEGREF catalogue), listing relevant European legislation regarding the reason for which the sampling was performed. This field should be used every time relevant legislation is available, otherwise when not applicable

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(sampling performed under a specific national legislation or other reason) the field should not be reported.

B.03 Sampling strategy (sampStrategy)

Sampling strategy describes how the sample was selected from the population being monitored or surveyed. A list was defined adapting the Eurostat definition to the naming convention adopted in this document (SAMPSTR catalogue) (EUROSTAT, 2010).

Code	Description	Definition
ST10A	Objective sampling	Strategy based on the selection of a random sample from a population on which the data are reported. Random sample is a sample which is taken under statistical consideration to provide representative data.
ST20A	Selective sampling	Strategy based on the selection of a random sample from a subpopulation (or more frequently from subpopulations) of a population on which the data are reported. The subpopulations may or may not be determined on a risk basis. The sampling from each subpopulation may not be proportional: the sample size is proportionally bigger for instance in subpopulations considered at high risk.
ST30A	Suspect sampling	Selection of an individual product or establishment in order to confirm or reject a suspicion of non-conformity. It's not a random sampling, therefore there is no sample extracted from the population.
ST40A	Convenient sampling	Strategy based on the selection of a sample for which units are selected only on the basis of feasibility or ease of data collection. It's a not random sampling. The data reported refer themselves to units selected according to this strategy. Special instance of selective sampling where no randomisation is performed in extracting the sample but units are selected only on the basis of feasibility or ease of data collection. The subpopulations may or may not be determined on a risk basis. The sampling from each subpopulation may not be proportional: the sample size is proportionally bigger for instance in subpopulations considered at high risk.
ST50A	Census	When the totality of a population or sub-population, on which the data are reported, is controlled.
ST90A	Other	
STXXA	Not specified	

It should be considered that Eurostat's definitions refer to reporting of aggregated data. As the SSD2 data model refers to single sampling units (according to the data model), instead of aggregated data as required by the sampling strategies of Eurostat, the 'Sampling strategy' (B.03) has to reflect the requirements of the programme specified in 'Sampling programme identification code' (B.01). In this context, the 'Sampling strategy' (B.03) relies on the proper identification of all the results from the same sample of the population described through the 'Sampling programme identification code' (B.01). If the element 'Sampling programme identification code' (B.01) is absent (and cannot be derived or implied from elsewhere in the data reported) the 'Sampling strategy' (B.03) loses its context and its meaning.

Some examples:

Example 1: a sample of milk tested for the presence of melamine belongs to a programme on 'Controls on <u>all</u> milk imported from China'; the sampling strategy to assign to this sample is 'Census'. The population of reference is all lots (consignments) of milk imported from China and there should

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be a 'Sampling programme identification code' (B.01) associated with these samples, all of which should have a 'Sampling strategy' (B.03) of 'Census'.

Example 2: a commodity included in the 'EU coordinated programme' (e.g. Orange) sampled randomly from retail outlets in a MS and tested for the determination of pesticides specified in the EU coordinated programme; the sampling strategy to assign to this sample is 'Objective sampling'. The population of reference is all lots of oranges available to the consumer in the MS. There should be a 'Sampling programme identification code' (B.01) associated with these samples, all of which should have a 'Sampling strategy' (B.03) of 'Objective'.

Example 3: a sample of orange tested for the determination of a pesticide residue belongs to a programme on 'random controls on oranges imported from any non-EU Countries'. The sampling strategy to assign to this sample is 'Selective sampling'. The population of reference is the set of all sampling units (lots, consignments) of oranges imported from extra-EU Countries, sampled for convenience at EU border post controls. There should be a 'Sampling programme identification code' (B.01) associated with these samples, all of which should have a 'Sampling strategy' (B.03) of 'Selective'.

Example 4: a sample of orange tested for the presence of a pesticide residue available from a precise wholesaler, as a consequence of previous non-conformity results. The sampling strategy to assign to this sample is 'Suspect sampling'.

Example 5: A pair of boot swabs from a slaughter flock of *Gallus gallus* is tested prior to slaughter for the presence of *Salmonella*. As all flocks within the *Salmonella* control programme are analysed for *Salmonella*, the sampling strategy is 'Census'.

B.04 Programme type (progType)

The programme type should be reported using a controlled terminology (PRGTYP catalogue) to indicate the type of control programme or other type of source to which the sample belongs. It is important to determine whether the programme was designed to assess consumer exposure at the EU level or at national level as the items sampled may differ depending on the dietary habits of the population under study.

B.05 Sampling method (sampMethod)

This element defines the way the samples have been taken from the original population using the controlled terminology from SAMPMD catalogue.

E.g. If the sample is analysed individually or it is analysed after the pooling of samples.

Sampling methods may refer to procedures in EU legislation, e.g. for pesticide monitoring "N009A" (According to Directive 2002/63/EC[25]).

B.06 Sampler (sampler)

This element defines the organisation (Competent authority, industry, private organisations, and officials[26]), public or private, which performed the sampling.

[25] Commission Directive 2002/63/EC of 11 July 2002 establishing Community methods of sampling for the official control of pesticide residues in and on products of plant and animal origin and repealing Directive 79/700/EEC

[26], The term refers to "staff performing official controls",as defined in article 6 of Regulation No 882/2004 of the European Parliament and of the Council of 29 April 2004 on official controls performed to ensure the verification of compliance with feed and food law, animal health and animal welfare rules. Official Journal L 165, 30.4.2004, p. 1-141

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B.07 Sampling point (sampPoint)

This element defines the point of sampling in the food chain (e.g. stage in the production chain) where the sample was taken.

The controlled terminology (SAMPNT catalogue) to be used in the data element is based on the data dictionary of activities (EUROSTAT, 2007). The list details the activities of establishments at different points in the food chain. The activities are described at different levels of detail, and data providers are requested to report at the most detailed level available. This list of activities is intended to indicate the type of establishment from which the sample was taken.

B.08 Additional sampling programme information (progInfo)

This compound field refers to data collection specific information on the sampling programme information depending on specific requirements of the different data collection domains, such as if the programme is used for the verification of the *Salmonella* reduction target, number of animal under the control program, total number of samples tested.

C Sampling event section

The entity represents the sampling unit extracted at a certain time from the sampled population, whose chemical or microbiological properties are the target of the sampling.

In fact each sampling event is regarded as an aggregate of all samples taken at a certain time to investigate chemical or microbiological properties of the sampling unit under consideration at a certain time.

Examples of 'sampling event' are one or more samples from an animal sampled at a certain time (as represented in Table 5. Example showing the usage of the sampling event to group different samples taken from the same animal), a flock/herd of animals sampled at a certain time, a batch of products sampled at a certain time. The same sampling unit sampled at different times represents different sampling events (excluding cases when for practical reasons a sampling unit may require many days to be sampled e.g. a large herd of sheep).

C.01 Sampling event identification code (sampEventId)

The sample event identification code represents a unique identifier of the sampling event, at data provider level for the sampling unit selected for sampling in a place at a certain time. The entity representing the sampling unit extracted at certain time from the sampled population, whose chemical or microbiological properties are the target of the sampling.

C.02 Sampling unit type (sampUnitType)

This entity indicates what type of sampling unit was selected for sampling. Examples of sampling unit type are a batch, an animal, a flock, a herd etc.

C.03 Sampling unit size (sampUnitSize) and C.04 Sampling unit size unit (sampUnitSizeUnit)

The size of the sampling unit and its unit of measurement can be reported using these two fields. These two fields can be used to indicate a sampling unit (lot) of 10 kilograms as well as a sampling unit (herd) of 10 animals (e.g. in this case 'sample unit size unit' should be set to 'unit').

C.05 Other sampling unit identifications (sampUnitIds)

This compound field can be used to report additional identification codes for the sampling unit, providing, for example, additional existing identification numbers for the sampling unit. Examples of

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these identification codes are 'herd code' or 'animal ear tag number' or 'animal identification number', 'batch number', 'flock identification number'. Multiple identification numbers for the same sampling unit can be reported.

This compound field may contain different sampling unit identifier codes (e.g. 'animalId' - animal identification number, 'flockId' - flock identification number), but each of these elements cannot be repeated more than once.

All the elements that can be listed in compound fields are available in Table 17 Preliminary attributes of the compound elements.

The detail for the syntax to report this type of elements is defined in the GDE2. An example could be: animalId='AT3245'$flockid='AT239849'

An example of usage of this field to report additional sampling unit information is provided in Table 5. This field can be also used to group different sampling events belonging to a higher-level sampling unit as represented in Table 6. This example, in fact, presents how different sampling events, relating to animals in the same herd can be still grouped via the 'herdId' in the compound field 'sampUnitIds'.

C.06 Additional sampling event information (sampEventInfo)

This compound field can be used to report additional information on the sampling event depending on specific requirements of the different data collections, e.g. status of holding regarding infection, the vaccination status, the country of birth and/or the country of slaughtering.

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Table 5: Example showing the usage of the sampling event to group different samples taken from the same animal

sampEventId	*sampUnitType*	*sampUnitIds*	*sampUnitSize*	*sampUnitSizeUnit*	*SampId*	*sampMatText*	*anPortSeq*	*resId*	*paramText*	*resQualValue*	*resType*
IT1876	animal	AnimalId=IT-189283728$SlaughterhouseId=IT-567648	1	G005A	IT_20120907_0001	Cattle blood	1	IT_20120907_0001_01/1.1	Salmonella	POS	BIN
IT1876	animal	AnimalId=IT-189283728$SlaughterhouseId=IT-567648	1	G005A	IT_20120907_0002	Cattle urine	1	IT_20120907_0002_01/1.1	Salmonella	NEG	BIN
IT1876	animal	AnimalId=IT-189283728$SlaughterhouseId=IT-567648	1	G005A	IT_20120907_0003	Cattle muscle	1	IT_20120907_0003_01/1.1	Salmonella	POS	BIN

Table 6: Example showing the usage of sampling unit identifiers to group samples taken from different animals from the same herd

sampEventId	*sampUnitType*	*sampUnitIds*	*sampUnitsize*	*sampUnitSizeUnit*	*SampId*	*sampMatText*	*resId*	*paramText*	*resQualValue*	*resType*
AT16389_1	animal	herdId=AT-16389	1	G005A	AT_20120910_01	Cattle blood	AT_20120910_01_01/01.1	Salmonella	NEG	BIN
AT16389_2	animal	herdId=AT-16389	1	G005A	AT_20120910_02	Cattle blood	AT_20120910_02_01/01.1	Salmonella	NEG	BIN
AT16389_3	animal	herdId=AT-16389	1	G005A	AT_20120910_03	Cattle blood	AT_20120910_03_01/01.1	Salmonella	NEG	BIN
AT16389_4	animal	herdId=AT-16389	1	G005A	AT_20120910_04	Cattle blood	AT_20120910_04_01/01.1	Salmonella	NEG	BIN
AT16389_5	animal	herdId=AT-16389	1	G005A	AT_20120910_05	Cattle blood	AT_20120910_05_01/01.1	Salmonella	NEG	BIN
AT16389_6	animal	herdId=AT-16389	1	G005A	AT_20120910_06	Cattle blood	AT_20120910_06_01/01.1	Salmonella	NEG	BIN
AT16389_7	animal	herdId=AT-16389	1	G005A	AT_20120910_07	Cattle blood	AT_20120910_07_01/01.1	Salmonella	NEG	BIN
AT16389_8	animal	herdId=AT-16389	1	G005A	AT_20120910_08	Cattle blood	AT_20120910_08_01/01.1	Salmonella	NEG	BIN
AT16389_9	animal	herdId=AT-16389	1	G005A	AT_20120910_09	Cattle blood	AT_20120910_09_01/01.1	Salmonella	NEG	BIN
AT16389_1	animal	herdId=AT-16389	1	G005A	AT_20120911_01	Cattle muscle	AT_20120911_01_01/01.1	Salmonella	NEG	BIN
AT16389_2	animal	herdId=AT-16389	1	G005A	AT_20120911_02	Cattle muscle	AT_20120911_02_01/01.1	Salmonella	NEG	BIN
AT16389_3	animal	herdId=AT-16389	1	G005A	AT_20120911_03	Cattle muscle	AT_20120911_03_01/01.1	Salmonella	NEG	BIN
AT16389_4	animal	herdId=AT-16389	1	G005A	AT_20120911_04	Cattle muscle	AT_20120911_04_01/01.1	Salmonella	NEG	BIN
AT16389_5	animal	herdId=AT-16389	1	G005A	AT_20120911_05	Cattle muscle	AT_20120911_05_01/01.1	Salmonella	NEG	BIN
AT16389_6	animal	herdId=AT-16389	1	G005A	AT_20120911_06	Cattle muscle	AT_20120911_06_01/01.1	Salmonella	NEG	BIN
AT16389_7	animal	herdId=AT-16389	1	G005A	AT_20120911_07	Cattle muscle	AT_20120911_07_01/01.1	Salmonella	NEG	BIN
AT16389_8	animal	herdId=AT-16389	1	G005A	AT_20120911_08	Cattle muscle	AT_20120911_08_01/01.1	Salmonella	NEG	BIN
AT16389_9	animal	herdId=AT-16389	1	G005A	AT_20120911_09	Cattle muscle	AT_20120911_09_01/01.1	Salmonella	NEG	BIN

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D Sample taken section

The elements belonging to this section describe information related to the sample event. In order to evaluate chemical or microbiological properties of the sampling event, one or more samples can be taken from the sampling unit at a certain time. With the term 'sample', SSD indicates the sample as taken by the sampling officer. Samples can be of different types such as environmental samples (e.g. swabs), animal samples (e.g. urine, blood, tissue or organs), food samples or feed samples. It should be stressed that entities such as an animal, a flock, a herd, a batch of food are not samples, but they are considered as sampling units (refer to the sampling event section in this guidance (Section C) for further information).

D.01 Sample taken identification code (sampId)

The sample taken by the sampling officer shall be identified by a unique 'sample identification' code.

There is no obligation on the data providers regarding the format of the sample identification number but data providers must ensure that the sample taken identification code is unique at data provider level.

To provide guidance to those organisations that still have to implement a policy for this identifier in their database, the WG-SSD2 suggests the generation of a unique sample code at country level as presented in Table 7 Possible methods suggested to build the sample code from available information. This method combines some information that should be available in the dataset.

Table 7: Possible methods suggested to build the sample code from available information

Country code	***Acronym of local organisation***	***Sampling***			***Data provider's sample number***	***Sample taken identification***
		Year	Month	Day		
BE	FASFC	2012	10	01	0004	BE-FASFC-2012-10-01-0004
BE	FASFC	2012			00000004	BE-FASFC-00000004

D.02 Reporting country (repCountry)

This element contains the country for which the data is reported. In specific cases this element can be different from the country of the organisation reporting the data or from the country in which the sample was taken. Such use should be specified by the EFSA data domain network.

D.03 Country of sampling (sampCountry)

This element contains the country in which the sample was taken for laboratory testing. Countries should be encoded using the standard ISO-3166-1-alpha-2 coding system. An extract is reported in the COUNTRY catalogue. As a result, only MSs or EEA country codes should be used.

In addition to the ISO standard codes, the codes EU, AA, XC, XD and XX have been added according to the provisions of the ISO-3166-1-alpha-2 for user-assigned code elements. When the country is unknown these options must be used in the provided order, being as specific as possible:

EU - Unspecified country that is part of European Union (EU);

AA - Unspecified country that is part of the European Economic Area (EEA) including EU;

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XC - Unspecified third country non EEA;

XD - Country not domestic, import[27];

XX - Unknown (i.e. nothing is known about the country or referring to international waters).

The EEA is made up of the 27[28] EU MSs and the three EEA EFTA States (Iceland, Liechtenstein and Norway).

D.04 Area of sampling (sampArea)

The area of sampling provides more detailed geographical information on locations according to the definitions described in the section above (D.03 Country of sampling). Use the Nomenclature of Territorial Units for Statistics (NUTS) code as described in NUTS catalogue. This coding system only covers regions within countries in EEA.

D.05 Reporting Year (repYear)

This element contains the year for which the data are reported. Data collections are organised often in periods of one year. This element should match the 'reference period' information available in the data collection context information. In order to streamline the data collection process and reduce the probability of the generation of duplicates, the data collection should clearly state the date to which the reporting year refers to e.g. Date Sampled or Date Analysed or other approaches. The definition of the reporting year must be specified by each EFSA data domain network.

D.06 Year of sampling (sampY), D.07 Month of sampling (sampM) and D.08 Day of sampling (sampD)

Sampling date divided into the year, month and day elements. It is mandatory to report the year while reporting the month or the day is optional. It is possible to report the day only if the month is also reported.

In case the sampling has been performed over a period of time the start date of sampling should be reported.

D.09 Sample taken size (sampSize) and D.10 Sample taken size unit (sampSizeUnit)

The pair of elements can be used to report the size and the unit of measure of the 'sample taken'.

D.11 Additional Sample taken information (sampInfo)

This element contains additional information on the sampling taken depending on specific requirements of the different data collection domains (e.g. day of arrival in the laboratory).

E Matrix sampled section

The elements belonging to this section describe information related to the matrix sampled, the area of origin of the sample and the country of processing and all the information related to the matrix sampled.

[27] The country code 'XD - Country non domestic, import' is provided mainly for managing historical data where the mainly distinguish for the origin of the matrix was 'domestic' or 'import'. Data providers should seek a more detailed definition of the country of origin when collecting new data.

[28] As available on the EEA web site http://www.efta.int/eea/eea-agreement.aspx, consulted on 05/09/2013, Croatia is not yet an EEA member although it is part of EU.

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E.01 Type of matrix (sampMatType)

This field defines the type of matrix that is analysed. At the time this guidance was prepared the type of matrices available in the MTXTYP catalogue are:

- Food;
- Food simulants;
- Feed;
- Animal;
- Environmental samples;
- Food contact material.

The type of matrix performs an additional selection on the food classification system, restricting further the possible terms that a data provider can report to a specific data collection (linked to a data collection domain) to a more specific sub-domain[29]. For this element the choice 'unknown' should be avoided since it would make it impossible to restrict the classification system to the proper sub-domain.

E.02 Coded description of the matrix of the sample taken (sampMatCode) and E.03 Text description of the matrix of the sample taken (sampMatText)

The sampMatCode (E.02) element contains the FoodEx2 code of the sample taken. The field allows the reporting of a compound code according to FoodEx2 catalogue (EFSA, 2011b).

The structure of the code is made up of a term taken from the applicable domain (*base term*), selected by the data collection domain linked to the data collection and by sub-domain referenced by the data element matrix type. This "base term" has then to be followed by the combination of facet descriptors, which can be either pre-assigned in the FoodEx2 system (*implicit facets*) or defined by the data provider (*explicit facets*).

The data provider, after describing the matrix with the most detailed level of information available using FoodEx2, should also report the full textual sample description in the E.03 field (free text field). This will provide a double check for the codes reported in case of data quality problems. Finally, an additional comment field is also available to provide additional information regarding the matrix (E.10 sampMatInfo element of the system).

There currently exists legislation with specific classifications and compliance criteria based on these, such as Regulation (EC) No 1333/2008[30] on food additives or Regulation (EC) No 396/2005[31] on pesticides.

The pesticide classification terms (i.e. MATRIX terms in SSD1) are all included as such in FoodEx2 as Raw Primary Commodities (RPC) and hold the pre-assigned code from the matrix catalogue. In addition, the food items derived from a single RPC are also linked to the pesticide MATRIX code. Thus the pesticide MATRIX code can be easily derived from the FoodEx2 code provided. There exists some mapping issues for composite foods where the matrix code cannot be unambiguously defined. For these items, the FoodEx2 code will link to the MATRIX code 'Not in list'.

By contrast, food additive classification, containing a majority of processed and composite foods, cannot be derived easily and unambiguously from FoodEx2 coding. For the additives domain, a new facet (legislative additive code) has been added to FoodEx2. As far as possible, the correct descriptor for this facet is already provided as an implicit facet of the FoodEx2 code. Otherwise, the users of the system shall add it to the FoodEx2 code as an explicit facet, when applicable.

29 Domains and sub-domains are defined in the section of controlled terminologies
30 See note 21 page 13
31 See note 8 page 5

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It should be noted that together with 'source' and 'part-nature' facets, the facet 'process technology' also covers an important role in FoodEx2 since it is of fundamental importance for the reporting of food additive and pesticide residues data. Whilst the 'source' and 'part-nature' facets are always implicitly completed for each 'core' and 'extended' term of the building hierarchy, the facet 'process technology' is implicit only for those terms where it can be unambiguously derived by the term name or scope-notes.

Due to its importance in data reporting and analysis, the facet 'process technology' may be made mandatory in some domains (i.e. pesticide residues). For this reason the system allows the reporting of the term 'unprocessed'. The reporting of this term is allowed only in those domains where the facet 'process technology' is mandatory. In other domains where this facet is not mandatory, the term 'unprocessed' should not be used. It should be noted that the term 'unprocessed' is domain dependent and therefore cannot be indicated as an implicit facet by the system.

Users should be aware that the reporting of generic facet descriptors such as 'unprocessed' makes the coding domain specific, therefore reportable only to one specific domain. Use of more specific facets descriptors is encouraged as this makes their usage possible within multiple data domains.

E.04 Country of origin of the sample taken (origCountry) and E.08 Country of processing of the sample taken (procCountry)

The country of origin of the sample taken (E.04) should be considered the place where the main commodity was grown, raised etc. The country of processing (E.08) is the location where the processed commodity was manufactured. The element E.08 should be used for processed commodities only. Countries should be encoded using the standard ISO-3166-1-alpha-2 coding system. An extract is reported in the COUNTRY catalogue. The same guidance as for country of sampling applies.

E.05 Area of origin of the sample taken (origArea) and E.09 Area of processing of the sample taken (procArea)

The area of origin (E.05) and the area of processing (E.09) provide more detailed geographical information on locations according to the definitions described in the section E.04 and E.08.

Use the Nomenclature of Territorial Units for Statistics (NUTS) code as described in NUTS catalogue. This coding system only covers regions within countries in EEA.

E.06 Area of origin for fisheries or aquaculture activities code of the sample taken (origFishAreaCode) and E.07 Area of origin for fisheries or aquaculture activities text of the sample taken (origFishAreaText)

Fishing areas are coded using the FAO fishing area coding system, prefixed with the letter 'M'. Additional codes have been added in case details on the part of the ocean are unknown or if the fishing area is unknown. More detail on the fishing place (e.g. ICES codes, name of river or lake, place of catch) can be reported in the element 'Area of origin for fisheries or aquaculture activities code of the sample taken (origFishAreaText)'.

E.10 Additional information on the matrix sampled (sampMatInfo)

This compound field can be used to report additional information and comments on the matrix sampled depending on the specific requirements of the different data collection domains.

F Sample analysed section

The elements belonging to this section describe information related to the sample analysed. Each 'sample analysed identification code' identifies the matrix, which is actually analysed in the laboratory. It takes into account that the sample can be analysed in different conditions e.g. after separation in different parts, with and without processing or after different storage periods. A number of cases where this can occur are detailed below. In order to simplify reporting, the SSD allows that

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the 'sample analysed identification code' and the 'matrix analysed' will often be the same as the 'sample taken identification code' and 'matrix sampled', respectively. With this assumption the 'sample analysed identification code' and 'matrix analysed' should only be reported where they vary from the sample taken information and can, in some circumstances, be left blank.

In some cases, (e.g. furan) it may be requested to perform the analysis of the 'sample taken' in different forms. The same 'sample' of coffee powder can be analysed as 'coffee powder' (food as purchased) and also as 'coffee beverage' (food as consumed). In this example, the same 'sample taken' by the sampling officer will generate two different 'samples analysed'. For certain microbiological analyses, it may be requested to perform the analysis twice, once at the arrival in the laboratory and once at the end of the shelf life. Also in this case, in order to properly link the related information, it is required to report only one sample but separate the two analyses performed into two different 'samples analysed'.

Table 8: Sample taken instant coffee analysed for furan as purchase and as consumed

SampId	*sampMatText*	*sampAnId*	*anMatText*	*paramText*	*resLOD*	*resLOQ*	*resVal*	*resType*
NL_20100105_0001	Instant coffee	NL_20100105_0001_01	Instant coffee, with water, as consumed	Furan	1	2	8	VAL
NL_20100105_0001	Instant coffee	NL_20100105_0001_02	Instant coffee, powder, as purchased	Furan	1	2	5	VAL

The same principle can be applied to complex 'sample taken' where the different homogeneous parts are analysed separately. An example can be the analysis of candies, where the hard exterior part can be analysed separately from the soft filling. This separation is necessary also since different 'maximum limits' (ML) may apply for evaluation.

Table 9: Sample taken for candies where hard and filling were separated before analysis

SampId	*sampMatText*	*sampAnId*	*anMatText*	*paramText*	*resLOD*	*resLOQ*	*resVal*	*resType*
NL_20100105_0002	Candies	NL_20100105_0002_01	Candies, hard	Aspartame	1	2		LOD
NL_20100105_0002	Candies	NL_20100105_0002_02	Candies, filling	Aspartame	1	2		LOD

Finally, it is necessary to distinguish the 'sample analysed' from the sample taken in the analysis of the food contact materials. Here, the same plastic can be analysed for composition and for migration. In the first case, the determination of Bisphenol A (BPA) in plastic, the 'matrix sampled' will be the same as the 'matrix analysed'. In the second case, the determination of BPA migration, the 'matrix sampled' is 'polycarbonate dishware' while the 'matrix analysed' is 'oil' as a 'food simulant'.

Table 10: Sample taken plastic, analysed for composition and migration

SampId	*sampMatText*	*sampAnId*	*anMatText*	*paramText*	*resLOD*	*resLOQ*	*resVal*	*resType*
NL_20100105_0003	Polycarbonate dishware	NL_20100105_0003_01	Polycarbonate dishware	BPA	1	2	8	VAL
NL_20100105_0003	Polycarbonate dishware	NL_20100105_0003_02	Oil as food simulant	BPA	1	2	5	VAL

Contrary to these examples there are cases where the matrix of 'sample analysed' follows by legislation from the matrix of 'sample taken' (e.g. in the pesticide data domain: Rhubarb must be analysed as "Stalks after removal of roots and leaves"). In such cases the EFSA data domain network may decide to report 'sample taken' only.

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F.01 Sample analysed identification code (sampAnId)

This element contains an alphanumeric identification code of the sample analysed and the code is mandatory for each 'sample analysed'. Since the 'analysed sample section' (section F) is not needed in all cases, to simplify the population of the data model, this element will be assumed to have the same value as the 'sample taken identification code' (element D.01 "sampId") unless otherwise populated.

Where multiple results corresponding to different parameters are reported for the same 'sample analysed' the 'unique sample identification number' must be maintained for that 'sample analysed' in all transmissions.

F.02 Sample analysed reference time (sampAnRefTime)

This element provides a qualitative reference for the time at which the sample was analysed. The reporting of this information is important for some microbiological analyses. A catalogue "REFTM" containing values such as 'Analysed at the arrival to laboratory', 'Analysed at the end of shelf-life' is assigned to this element.

F.03 Year of analysis (analysisY), F.04 Month of analysis (analysisM) and F.05 Day of analysis (analysisD)

Analysis date divided into the year, month and day elements. It is mandatory to report the year while reporting the month or the day is optional. It is possible to report the day only if the month is also reported.

F.06 Additional information on the sample analysed (sampAnInfo)

This compound field can be used to report additional information and comments on the sample analysed or specific data collection tags. If the analysis has been performed over a period of time the completion date of analysis should be stated in this field.

G Matrix analysed section

The elements belonging to this section describe information related to the 'matrix analysed' and its relevant characteristics. This section allows the inclusion of an additional encoded and textual description of the 'matrix analysed', in the cases where the matrix analysed is different from the matrix sampled. Please also refer to Section F 'Sample analysed' (above) for examples of instances where 'Sample taken' and 'Sample analysed' can differ.

G.01 Coded description of the matrix analysed (anMatCode) and G.02 Text description of the matrix analysed (anMatText)

These two fields report the sample analysed characteristics encoded using the FoodEx2 catalogue (G.01 compound field) and described as free text (G.02). This element will be assumed to have the same value as "sampMatCode" (E.02) and "sampMatText" (E.03), respectively, unless otherwise populated.

G.03 Additional information on the analysed matrix (anMatInfo)

This compound field can be used to report additional specific information and comments on the analysed matrix.

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H Sample analysed portion section

The elements belonging to this section describe information related to the 'sample analysed portion' (often called 'test portion') as portion of the 'sample analysed'. These are sometimes referred to as replicate samples since they are identical in every way and each is representative of the 'sample taken' and/or 'sample analysed'. The reporting of multiple 'sample analysed portions' is required in some cases by the legislation e.g. aflatoxins in dried fruits where three 'sample analysed portions' must be analysed (Commission Regulation (EC) No. 401/2006 and its amendments). The concept of 'sample analysed portions' should not be used if the 'matrix analysed' is not identical in all the 'sample analysed portions'; in case this constraint is not valid, this analysis must be reported as a different 'sample analysed'.

The 'sample taken' could be analysed for the same parameter more than once to perform a counter-analysis or to confirm a positive sample. In these cases, the only result to be reported is the one for which the evaluation is performed and the 'sample analysed portion' should not be used.

H.01 Sample analysed portion sequence (anPortSeq)

The 'sample analysed portion sequence' is a sequence number (1, 2, 3, n) listing the different portions of the 'sample analysed'. In cases where the 'sample analysed portion' is not explicitly reported, it will be assumed to be coincident with the 'sample analysed' and equal to 1.

Table 11 shows a correct usage of the 'sample analysed portion' element.

Table 11: Example of a correct usage of sample analysed portion

SampId	*sampMatText*	*sampAnId*	*sampAnMatText*	*anPortSeq*	*paramText*	*resLOQ*	*resVal*	*resType*
PL_2009_001	Peanuts	PL_2009_001	Peanuts	1	Aflatoxin G2	2		LOQ
PL_2009_001	Peanuts	PL_2009_001	Peanuts	1	Aflatoxin G1	2	7	VAL
PL_2009_001	Peanuts	PL_2009_001	Peanuts	1	Aflatoxin B2	2	10	VAL
PL_2009_001	Peanuts	PL_2009_001	Peanuts	1	Aflatoxin B1	2	5	VAL
PL_2009_001	Peanuts	PL_2009_001	Peanuts	2	Aflatoxin B2	2	4	VAL
PL_2009_001	Peanuts	PL_2009_001	Peanuts	2	Aflatoxin G2	2		LOQ
PL_2009_001	Peanuts	PL_2009_001	Peanuts	2	Aflatoxin B1	2	5	VAL
PL_2009_001	Peanuts	PL_2009_001	Peanuts	2	Aflatoxin G1	2	3	VAL
PL_2009_001	Peanuts	PL_2009_001	Peanuts	3	Aflatoxin G2	2		LOQ
PL_2009_001	Peanuts	PL_2009_001	Peanuts	3	Aflatoxin G1	2		LOQ
PL_2009_001	Peanuts	PL_2009_001	Peanuts	3	Aflatoxin B1	2		LOQ
PL_2009_001	Peanuts	PL_2009_001	Peanuts	3	Aflatoxin B2	2		LOQ

Table 12 reports an example of wrong usage of 'sample analysed portion' in aflatoxins B1 and B2 reporting. The same sample coded as 'BE-FASFC-01' has been analysed three times for the parameters aflatoxin B1 and aflatoxin B2 but the matrices analysed are not identical. The 'anMatText' column reports three different matrices: 'Maize bran', 'Maize grain' and 'Maize grain, boiled'. This is in conflict with the stated constraint that 'sample analysed portions' should not be used if the matrix analysed is not identical in all the 'sample analysed portions'.

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Table 12: Example of incorrect reporting of 'sample analysed portion'

sampAnId	anMatText	anPortSeq	paramText	resId	resVal
BE-FASFC-01	Maize bran	1	Aflatoxin B1	BE-FASFC-01/01/01	3.5
BE-FASFC-01	Maize bran	1	Aflatoxin B2	BE-FASFC-01/01/02	7.4
BE-FASFC-01	Maize grain	2	Aflatoxin B1	BE-FASFC-01/02/01	3.9
BE-FASFC-01	Maize grain	2	Aflatoxin B2	BE-FASFC-01/02/02	6.4
BE-FASFC-01	Maize grain, boiled	3	Aflatoxin B1	BE-FASFC-01/03/01	3.2
BE-FASFC-01	Maize grain, boiled	3	Aflatoxin B2	BE-FASFC-01/03/02	7.1

Also not accepted by the model is the reporting of results for the same parameter in the same 'sample analysed' without reporting different 'sample analysed portions' (as shown in Table 13 all the fields of column 'anPortSeq' are unfilled).

Table 13: Example of incorrect reporting due to repeated parameter in the same 'sample analysed'

sampAnId	anMatText	anPortSeq	paramText	resId	resVal
BE-FASFC-01	Maize bran		Aflatoxin B1	BE-FASFC-01/01/01	3.5
BE-FASFC-01	Maize bran		Aflatoxin B2	BE-FASFC-01/01/02	7.4
BE-FASFC-01	Maize bran		Aflatoxin B1	BE-FASFC-01/02/01	3.9
BE-FASFC-01	Maize bran		Aflatoxin B2	BE-FASFC-01/02/02	6.4
BE-FASFC-01	Maize bran		Aflatoxin B1	BE-FASFC-01/03/01	3.2
BE-FASFC-01	Maize bran		Aflatoxin B2	BE-FASFC-01/03/02	7.1

H.02 Sample analysed portion size (anPortSize) and H.03 Sample analysed portion size unit (anPortSizeUnit)

These elements are provided to report the size/ amount and the unit in which the sample analysed portion size is expressed.

H.04 Additional information on the sample analysed portion (anPortInfo)

This compound field can be used to report additional information and comments on the sample analysed portion depending on the specific requirements of the different data collection domains.

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I Isolate section

The isolate identifies, by a unique code, a culture of a biological agent, isolated from a specific 'sample taken'. Isolate section is specifically used to group the results of the susceptibility testing of isolates to different antimicrobial substances.

Whilst isolates always derive from a sample taken, and this is the recommended reporting in the model, the data model supports also those cases where they are not always explicitly linked to the 'sample taken' from which they were isolated.

I.01 Isolate identification (isolId)

This field contains the identification code used to group an isolate identification with antimicrobial susceptibility test results performed on the same isolate. The isolate code identifies the isolate (bacterial strain) derived from a microbial detection. It should be unique for each data provider. In case data providers are collecting isolates from different laboratories, they should ensure that different isolates are transmitted with different isolate codes.

I.02 Coded description of the isolate (isolParamCode) and I.03 Text description of the isolate (isolParamText)

Data element 'isolParamCode' (I.02) contains the code, according to the 'Parameter catalogue', of the isolate. This data element is especially important in cases when the link between the isolate and data of the sample are missing (origin, laboratory, etc.). Data element 'isolParamText' (I.03) contains the description of the isolate parameter (e.g. speciation/ serotyping) using free text. A complete example of reporting of AMR tests is described in the Table 14 "Isolate identification example".

I.04 Additional information on the isolate (isolInfo)

This compound field can be used to report additional information and comments on the isolate depending on the specific requirements of the different data collection domains.

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Table 14: Isolate identification example

SampId	*sampMatText*	*sampAnId*	*isolId*	*isolParamText*	*resId*	*paramText*	*resVal*	*resQualValue*	*resType*
IT_20110317_46	chicken meat	IT_20110317_46_1			IT_20110317_46_1_1	*Salmonella*		POS	BIN
IT_20110317_46	chicken meat	IT_20110317_46_1	IT_ISOL_59	*Salmonella* Enteritidis	IT_20110317_46_1_2	*Salmonella* Enteritidis		POS	BIN
IT_20110317_46	chicken meat	IT_20110317_46_1	IT_ISOL_59	*Salmonella* Enteritidis	IT_20110317_46_1_3	Tetracycline	16		VAL
IT_20110317_46	chicken meat	IT_20110317_46_1	IT_ISOL_59	*Salmonella* Enteritidis	IT_20110317_46_1_4	Ampicillin	32		VAL
IT_20110317_46	chicken meat	IT_20110317_46_1			IT_20110317_46_1_5	*Campylobacter*		NEG	BIN
IT_20110317_46	chicken meat	IT_20110317_46_1	IT_ISOL_60	*Salmonella* Dublin	IT_20110317_46_1_6	*Salmonella* Dublin		POS	BIN
IT_20110317_46	chicken meat	IT_20110317_46_1	IT_ISOL_60	*Salmonella* Dublin	IT_20110317_46_1_7	Tetracycline	16		VAL
IT_20110317_46	chicken meat	IT_20110317_46_1	IT_ISOL_60	*Salmonella* Dublin	IT_20110317_46_1_8	Ampicillin	32		VAL

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J Laboratory section

The elements belonging to this section describe information related to the laboratory performing the analysis and responsible for the results. In cases when more than one laboratory has performed analyses of the same 'sample taken', such as detection, confirmation or resistance testing, the laboratory responsible for the total results can be either the initial laboratory performing detection or the one performing confirmation or the one testing resistance.

J.01 Laboratory identification code (labId)

The identification code of the laboratory providing the results should be reported here. If a national laboratory coding system exists, this code should be reported. This code should be reported consistently for all transmissions of data. When further information is requested separately, for instance participation in proficiency tests, the same unique code should be reported in these cases.

J.02 Laboratory accreditation (labAccred)

This element indicates whether accreditation of the laboratories performing the analysis has been achieved. In accordance with Art 12 of Regulation 882/2004, laboratories designated for official controls must be accredited to ISO/IEC17025, or avail of the derogation in Art 18 of Regulation 2076/2005.

J.03 Laboratory country (labCountry)

This element defines the country where the laboratory performing the testing is located. This field will follow the ISO 3166-1-alpha-2 country code list available in the 'COUNTRY' catalogue.

J.04 Additional information on the laboratory (labInfo)

This compound field can be used to report additional information and comments on the laboratory (e.g. total number of isolates available in the laboratory) depending on the specific requirements of the different data collection domains.

K Parameter section

The elements belonging to this section describe the parameters (analyte) for which the samples have been analysed.

K.01 Type of parameter (paramType), K.02 Coded description of parameter code (paramCode) and K.03 Parameter text (paramText)

In order to facilitate the reporting of parameters according to complex parameter definitions and ensure that the assessment of these complex parameter definitions in a certain sample is accurate, the 'Type of Parameter' (K.01) data element must be completed to indicate whether a parameter reported is summed according to a parameter sum or other type of complex parameter definition, a part of a parameter sum or residue definition or an individual parameter or residue, making the calculation of summed parameters more transparent. This is necessary because in some cases it is difficult to reproduce, on the receiver side, which individual parameters are included in summed parameters (such as residues definition, dioxin TEQ). Specific codes are used for those parameters which are defined in legislation.

The compound field 'Parameter code' (K.02) can be used to report the coding system currently in use in the different data collection domains to describe parameters under analysis. Data collection specific guidelines should be consulted to determine the correct section of the coding system to be used. In case the parameter is not included in the PARAM catalogue, the code 'Not in list' should be reported and the name of the parameter should be specified in the 'Parameter Text' element (K.03).

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Presently the features of a compound field is not utilised for 'Parameter Code' (K.02). The field is defined as a compound field to ensure flexibility to future demands.

L Analytical method section

The elements belonging to this section describe information related to the analytical methods used in the laboratory performing the sample analysis.

L.01 Analytical method identification (anMethRefId)

This data element identifies the method used. The identifier used should enable the laboratory to uniquely identify the actual method applied to perform the analysis.

L.02 Analytical method reference code (anMethRefCode)

This data element shall contain, when a standard (Official) method is used, the reference to the standard method. When a non-standard method is used the data provider shall choose the source of the reference for the method (e.g. "in house method", "published method (scientific paper/guidelines)", "alternative methods validated against standards"). This field shall be populated with values from the ANLYREFMD catalogue.

L.03 Analytical method type (anMethType)

This data element identifies the type of analytical method applied to perform the analysis (e.g. AT03A is Phagetyping). This field shall be populated with values from the ANLYTYP catalogue.

L.04 Analytical method code (anMethCode) and L.05 Analytical method text (anMethText)

The compound field Analytical method code (L.04) can be used to report the code describing the type of method applied (sometimes represented by the main instrument in chemical determination used in the method). If the method is not in the list or it is necessary to provide more details with respect to the method available in the list, the associated free text field Analytical method text (L.05) should be used. The free text field must be used if 'Classification not possible' was reported for 'Analytical Method Code' (e.g. F026A is LC-MS Liquid Chromatography Mass Spectrometry). The catalogue for the element L.04, named ANYMD in SSD1 and ANLYTYP in SSD2 is principally designed for chemical analyses. Some entries were added for microbiological analysis based on the experience of the ongoing data collections. BIOMO unit started a working group [M-2012-0028] in February 2013 to complete the list for microbiological methods of analysis. The result of this WG will be integrated in the ANLYMD catalogue in the 2014 major release.

When standard (Official) methods are used, the official reference code should be provided (e.g. ISO 6579:2002 or ISO 6579) in L.02 'Analytical method reference code' (or in L.05 'Analytical method text' if unavailable in the ANLYREFMD catalogue).

L.06 Additional information on the analytical method (anMethInfo)

This compound data element can be used to report additional data collection specific information related to the analytical method depending on the specific requirements of the different data collection domains (such as disk concentration and disk diameter for antimicrobial resistance diffusion method, method sensitivity, method specificity, contact time and contact temperature for migration tests regarding food packaging materials).

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M Result section

The elements belonging to this section describe information related to the result of the laboratory tests, as a quantitative or qualitative outcome value. Information relating to the result includes the accreditation procedure, limits of detection and quantification and result value for the analytical method.

M.01 Result identification code (resId)

This element should contain the unique identification of an analytical result (a row of data) in the transmitted file. This identification code is mandatory, as it will be used as reference for operation of deleting or updating an individual result if this procedure is supported by the data collection protocol. In addition, this is the element that is referenced in the acknowledgment file and in the detailed error report to describe where a specific error is located in the file transmitted by the data provider.

M.02 Accreditation procedure for the analytical method (accredProc)

This element contains the appropriate option to state the quality assurance system applied in the analytical procedure in the laboratory, as described in Article 12 of Regulation (EC) No 882/2004 on "Official controls performed to ensure the verification of compliance with feed and food law, animal health and welfare rules" data providers should state which quality assurance procedures (e.g. ISO/IEC17025, other third party quality assessment, internal validation) were used. It is essential that any data which is submitted to EFSA is of sufficient quality that it is fit-for-purpose. Multiple options are not possible; the data provider must choose the option which is most appropriate. This field shall be populated with values from the 'MDACC catalogue'.

M.03 Result unit (resUnit)

This data element indicates the unit of measurement for the values reported in 'Result LOD' (M.04), 'Result LOQ' (M.05), 'Result Value' (M.10), 'Result lower limit of the working range' (M.06), 'Result upper limit of the working range' (M.07), 'Result value uncertainty' (M.17), 'Result value uncertainty standard deviation' (M.18), 'CC alpha' (M.08), 'CC beta' (M.09), 'Limit for the result evaluation (N.01)' or 'Limit for the result evaluation (High limit) (N.02)'. This field shall be populated with values from the 'UNIT catalogue'.

This should be consistent for all elements reported in a single result row. This field is by default not mandatory, but it becomes mandatory if at least one of the fields listed above is provided.

M.04 Result LOD (resLOD)

The limit of detection (LOD)[32] is the lowest concentration level that can be determined to be statistically different from a blank (Keith et al., 1983). Usually a confidence level of 95% or 99% is used. This value must be expressed in the same units as reported in the data element 'Result unit' (M.03).

[32] Many definitions of LOD and LOQ have been suggested throughout the years and in different analytical areas. For recent international definitions see: e.g. Report of the thirtieth session of the Codex Committee on methods of analysis and sampling. (Codex Alimentarius Commission, FAO, WHO, 2009); Method validation and quality control procedures for pesticide residue analysis in food and feed (Document N° SANCO/10684/2009), http://ec.europa.eu/food/plant/protection/resources/qualcontrol_en.pdf; Commission Directive 2009/90/EC laying down, pursuant to Directive 2000/60/EC of the European Parliament and of the Council, technical specifications for chemical analysis and monitoring of water status (OJ L 201, 31.07.2009, p.36-38); IUPAC Compendium of Chemical Terminology (IUPAC, 2013).

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M.05 Result LOQ (resLOQ)

The limit of quantification (LOQ)[33] is the level above which quantitative results may be obtained with a specified degree of confidence (Keith et al., 1983). This value must be expressed in the same units as reported in the data element 'Result unit'.

In the specific case of complex (residue) definitions, i.e. in cases when expression of result covers more than one substance (usually 'sum of … (selected compounds / substances / residues)'), there are many possibilities of how to express and report a result of analysis when all components are less than LOQ/LOD (<LOQ/<LOD). Among these possibilities is to use lowest LOQ/LOD, use of highest LOQ/LOD, use of sum of all LOQs/LODs contributions, use of 'zero' value and also use no expression of LOQ/LOD value for such cases. The rules on how LOQ and/or non-quantified values for parameter sums should be calculated have to be stated by the respective EFSA networks for each domain. The same approach has to be taken also for the following fields resLLWR, resULWR, LOD, CC alpha, CC beta.

M.06 Result lower limit of the working range (resLLWR) and M.07 Result upper limit of the working range (resULWR)

The range of concentration or mass that can be adequately determined by an analytical method. These values must be expressed in the same unit as reported in the data element 'Result unit' (M.03).

M.08 CC alpha (CCalpha)

Decision limit (CCα)[34] means the limit at, and above which, it can be concluded with an error probability of α that a sample is non-compliant.

If no permitted limit has been established for a substance, the decision limit is the lowest concentration level at which a method can discriminate with a statistical certainty of 1 - α that the particular analyte is present.

If a permitted limit has been established for a substance, the decision limit is the concentration above which it can be decided with a statistical certainty of 1 - α that the permitted limit has been truly exceeded.

Alpha (α) error means the probability that the tested sample is compliant, even though a non-compliant measurement has been obtained (false non-compliant decision).

'CC alpha' must be expressed in the same unit as reported in the data element 'Result unit' (M.03).

M.09 CC beta (CCbeta)

Detection capability (CCβ)[35] means the smallest content of the substance that may be detected, identified and/or quantified in a sample with an error probability of β.

In the case of substances for which no permitted limit has been established, the detection capability is the lowest concentration at which a method is able to detect truly contaminated samples with a statistical certainty of 1 - β.

In the case of substances with an established permitted limit, this means that the detection capability is the concentration at which the method is able to detect permitted limit concentrations with a statistical certainty of 1 - β.

[33] Seee note 34 page 56.
[34] Commission Decision of 12 August 2002 implementing Council Directive 96/23/EC concerning the performance of analytical methods and the interpretation of results. OJ L 221, 17.08.2002, p. 8-36
[35] See note 36

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Beta (β) error means the probability that the tested sample is truly non-compliant, even though a compliant measurement has been obtained (false compliant decision).

'CC beta' must be expressed in the same unit as reported in the data element 'Result unit' (M.03).

M.10 Result value (resVal), M.15 Result qualitative value (resQualValue), M.16 Type of result (resType),

The data elements 'Result value' (M.10), 'Result qualitative value' (M.15) and 'Type of result' (M.16) are used to describe different type of results of an analysis.

'Result value' (M.10) stores the numeric value reported for a directly measured or calculated quantity. If the result is numeric, the data element 'Result value' (M.10) must be completed and the 'Type of result' (M.16) should be set to 'VAL' (numerical value) for results at or above the 'LOQ/CC alfa/LLWR'. In this case the result of the analytical measure must be reported in the same unit as reported by the data element 'Result unit' (M.03). The 'Result value' (M.10) should not be corrected for uncertainty.

In cases when the 'Result value' cannot be determined, the result value could be left empty but the 'Type of result' must indicate which type of limit was used (e.g. 'LOD', 'LOQ', 'CCalfa', 'LLWR', 'ULWR'). In cases when the measured result value is below 'LOQ' (or lower than 'CC alpha' or 'LLWR'), but higher then 'LOD', the data providers are invited to report the measured result value, although at the same time stating 'Type of result' equal to 'LOQ' (or 'CCA' or 'LLWR'). Likewise for results above the 'ULWR' (in this case stating 'Type of result' equal to 'ULWR').

'ResVal' (M.10) has to be absent when 'resType' (M.16) is 'LOD'.

In some cases the analytical method may not have a LOQ defined; in these cases, if the analytical method is taking into account a limit of reporting, this information should be reported in the LOQ data element.

If the result of the analysis is qualitative i.e. "positive/present" or "negative/absent" then the data element 'Result qualitative value' (M.15) should be populated and the 'Type of result' (M.16) should be set to 'BIN' (Qualitative value).

M.11 Result value recovery rate (resValRec) and M.12 Result value corrected for recovery (resValRecCorr)

Recovery value is associated with the concentration measurement expressed as a percentage (%).

The value reported in the element 'result value recovery rate' (M.11) may or may not be used by the data provider to correct the result value reported. This depends from the specific requirements of the data collections.

If not otherwise specified, the result expressed is considered not corrected for recovery. If the data provider reports a result corrected for recovery, the data element 'Result value corrected for recovery' (M.12) must be set to 'yes'. It should be noted that for specific data collections (e.g. pesticide residues) the result reported should not be corrected for recovery.

In any case, the data provider should report in the 'resVal' (M.10), the best estimated value for the parameter analysed and no additional correction of this value should be performed on the data receiver side.

The recovery for the result must be provided as percentage (e.g. the value 120 indicates 120% recovery, the value 1.20 indicates 1.20%). In order to avoid mistakes in reporting, value below 10 in

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the 'result value recovery rate' (M.11) will be rejected during the business rules validation since this level of recovery would not be realistic and would tend to indicate an error in reporting.

M.13 Expression of result percentage (exprResPerc)

This compound field can be used to report the percentage of a measured specific matrix component (e.g. fat, alcohol, moisture) used as reference to express the analytical result (e.g. on fat basis, on alcohol basis and on dry weight basis). Therefore the reporting of these percentages would be, for example, as follows: "moistPerc=13", "alcoholPerc=40", "fatPerc=60". The detail for the syntax to report more components is defined in the GDE2. An example could be: 'moistPerc=13$alcoholPerc=40'.

Even if the result is expressed on the whole weight, the fat percentage may be useful to know for some contaminants.

This item describing the result relates to the percentage found in the original sample on analysis and should not be confused with the facets 'Fat-content' and 'Alcohol-content', which should be used as descriptors within FoodEx2 to describe the sample as labelled.

M.14 Expression of result type (exprResType)

Analytical results should be usually expressed on a 'whole' weight basis. Only when the legislation explicitly requires the expression of results on another basis (e.g. dry weight or fat basis), the information should be reported in this manner. If not specified, the result is considered therefore expressed in 'whole weight'.

If the 'Result value' (M.10) is not expressed on a whole weight basis, the type of expression of result basis should be indicated using the element 'Expression of result type' (M.14) (e.g. Fat, Moisture, Alcohol) and the numeric percentage should be indicated in the element 'Expression of result percentage' (see M.13).

This item describing the result relate to the percentage found in the original sample on analysis and should not be confused with the facets 'Fat-content' and 'Alcohol-content', which should be used as descriptors within FoodEx2 to describe the sample as labelled.

M.17 Result value uncertainty (resValUncert) and M.18 Result value uncertainty standard deviation (resValUncertSD)

The uncertainty associated with the measure must be expressed using the data elements 'Result value uncertainty standard deviation' (M.18) and 'Result value uncertainty' (M.17). The uncertainty and the uncertainty standard deviation should always be provided in the same units as the 'result value'.

'Result value uncertainty' (M.17) indicates the expanded uncertainty (usually 95% confidence interval) value associated with the measurement expressed in the unit reported in the field 'Result unit' (M.03).

'Result value uncertainty standard deviation' (M.18) indicates the Standard deviation for the uncertainty of measurement.

M.19 Result reference identification (resRefId)

In some cases, depending of the type of analysis, the result of the analytical measure may not be a simple number or a qualitative value, but a more complex structure. In case of molecular typing of isolates it is necessary to submit an image. This type of information may be stored in separate data models and described *by ad-hoc specification documents*. The linkage between the complex structure returned by the analytical method and the other information stored in the SSD2 is obtained by an identification number, stored in the complex structure and provided in the 'result reference identification' (M.19) of the SSD2 data model.

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M.20 Additional information on the result (resInfo)

This compound field can be used to report additional information, comments and tags for each analytical result depending on the specific requirements of the different data collection domains.

N Evaluation section

The elements in this section report the evaluation (assessment) of an analytical result: the reference or the legal limits of the analyte for the relevant matrix, the type of the limit, the evaluation and the actions taken.

N.01 Limit for the result evaluation (evalLowLimit)

This element is used to report the reference or legal limit for the residue/chemical/microbiological parameter for the relevant matrix or the lower level of three-class evaluation limits (as defined in Reg. 2073/2005 as amended, regarding microbiological criteria for food). The 'Result value' should be reported according to the legal limit definition and in the same unit as the legal limit. The legal limit reported should be the one applicable at the time of compliance assessment. Where the official EU legal limit has not been used to evaluate the result, then the legal limit which was used should be provided.

When the limit is placed on a calculated parameter result derived from more than one analytical result (e.g. pesticide residue definitions), this data element should be used only for the calculated parameter and not for the individual analyses.

N.02 Limit for the result evaluation (High limit) (evalHighLimit)

This element is used to report the higher level of three-class evaluation limits (as defined in Reg. 2073/2005 as amended, regarding microbiological criteria for food).

N.03 Type of limit for the result evaluation (evalLimitType)

Report the type of legal limit used to assess the result value. If a legal limit is provided in N.02 then the type of legal limit used should be also provided in this field (N.03).

When the limit is placed on a calculated parameter result derived from more than one analytical result (e.g. pesticide residue definitions), this data element should be used only for the calculated parameter and not for the individual analyses.

Examples are reported in the element 'Evaluation of the result' (N.04).

N.04 Evaluation of the result (evalCode)

This element should contain the evaluation of the measured parameter in the analysed matrix done by the laboratory or the risk manager. Additional evaluations of the 'sample analysed', the 'sample taken', the 'sampling event' are also reported in this element.

If the laboratory considers the result has failed based on internal quality control procedures, the result should not be transmitted. The result should be compared with the legislative or other limit applicable at the time of sampling and the correct code from the RESEVAL controlled terminology selected to indicate whether the result was compliant with the limit or not.

When 'sample analysed portions' are analysed and the limit applies cumulatively to the entire sample, the 'evaluation of the result' should be reported for each sample analysed portion for the parameter for which the limit exists. An example is reported in Table 15 Reporting sample evaluation for histamine - alternative way of evaluation (without additional row of evaluation).

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When the limit is placed on a calculated parameter derived from more than one analytical result (e.g. pesticide residue definitions), 'resEval' data element for the individual analyses should report 'not evaluated'. An example is reported in Table 16 Evaluation for calculated parameter (e.g. residue definitions). In this case the level for the evaluation for the pesticide residue definition is 'result' level.

Examples of compliance assessment at different levels are:

- Sampling event assessment (compliant/not compliant/not assessed);
- Sample taken assessment (compliant/not compliant/not assessed);
- Sample analysed assessment (compliant/not compliant/not assessed).

N.05 Action taken (actTakenCode)

This compound field can be used to report information on any follow-up actions taken in the case of a result higher than the legal limit. More than one action from the terminology in ACTION catalogue may be reported in this compound element. The detail for the syntax to report this type of elements is defined in the GDE2. An example could be: '*value1$value2$...$valueN*'.

N.06 Additional information on the evaluation (evalInfo)

This element is a compound element that contains additional specific information and comment for the evaluation section depending on the specific requirements of the different data collection domains.

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Table 15: Reporting sample evaluation for histamine - alternative way of evaluation (no additional row)

SampId	sampMatText	anPortId	Param	resId	resVal	resType	evalLow Limit	evalHigh Limit	evalLimitType	evalCode	evalInfo
BE-FASFC-17	Canned Tuna	1	Histamine	BE-FASFC-17/01/01	50	VAL	200	400	ML	Satisfactory	sampEventAsses =Non Compliant
BE-FASFC-17	Canned Tuna	2	Histamine	BE-FASFC-17/02/01	70	VAL	200	400	ML	Satisfactory	sampEventAsses =Non Compliant
BE-FASFC-17	Canned Tuna	3	Histamine	BE-FASFC-17/03/01	20	VAL	200	400	ML	Satisfactory	sampEventAsses =Non Compliant
BE-FASFC-17	Canned Tuna	4	Histamine	BE-FASFC-17/04/01	500	VAL	200	400	ML	Unsatisfactory.	sampEventAsses =Non Compliant
BE-FASFC-17	Canned Tuna	5	Histamine	BE-FASFC-17/03/01	300	VAL	200	400	ML	Acceptable	sampEventAsses =Non Compliant
BE-FASFC-17	Canned Tuna	6	Histamine	BE-FASFC-17/03/01	40	VAL	200	400	ML	Satisfactory	sampEventAsses =Non Compliant
BE-FASFC-17	Canned Tuna	7	Histamine	BE-FASFC-17/03/01	50	VAL	200	400	ML	Satisfactory	sampEventAsses =Non Compliant
BE-FASFC-17	Canned Tuna	8	Histamine	BE-FASFC-17/03/01	60	VAL	200	400	ML	Satisfactory	sampEventAsses =Non Compliant
BE-FASFC-17	Canned Tuna	9	Histamine	BE-FASFC-17/03/01	70	VAL	200	400	ML	Satisfactory	sampEventAsses =Non Compliant

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Table 16: Evaluation for calculated parameter (e.g. residue definitions)

SampId	SampMatText	anPortId	Param	resId	resVal	evalLowLimit	evalHighLimit	evalLimitType	evalCode	evalInfo
BE-FASFC-18	Apples		Pesticide 1	BE-FASFC-18/01	0.1				Result not evaluated	sampEventAsses=Compliant
BE-FASFC-18	Apples		Pesticide 2	BE-FASFC-18/02	4.3				Result not evaluated	sampEventAsses=Compliant
BE-FASFC-18	Apples		Pesticide 3	BE-FASFC-18/03	1.0				Result not evaluated	sampEventAsses=Compliant
BE-FASFC-18	Apples		Residue definition (sum of Pesticide 1, 2 and 3	BE-FASFC-18/04	5.4	10		MRL	≤ maximum permissible quantities	sampEventAsses=Compliant

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7. Preliminary attributes defined for SSD2 compound elements

Table 17 (Preliminary attributes of the compound elements) contains the definition of the attributes for each compound element reported in Table 4. Table 17 provides only a preliminary version of the attributes defined as compound elements, reflecting the currently defined needs. This table will be included in the published StandardSampleDescription controlled terminologies catalogues and will be subject to the same revision process of other SSD catalogues. The purpose is to make the structure of the SSD more flexible and adaptable to data collection specific requirements which will inevitably evolve over time.

Most of the attributes currently defined are related to the full implementation of the FoodEx2 facets for the elements sampMatCode (E.02) and anMatCode (G.01).

The compound elements are defined by the following information:

- **Attribute id**: unique identification of the attribute;
- **Attribute name**: the name of the attribute to be used in database systems;
- **Attribute label**: the label for the attribute to be displayed to users;
- **Type**: the XML type for the attribute;
- **Base /facet**;
- **S/R**: Simple or repeatable;
- **Controlled terminology**: the controlled terminology to be used for the catalogue;
- **Description**: a user level description of the catalogue attribute.

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Table 17: Preliminary attributes of the compound elements

Element Code	Element name	Attribute id	Attribute name	Attribute label	Type	Base/Facet	S/R	Controlled terminology	Description
A.03	localOrgInfo		Com	Comment	xs:string (250)	B	R		Comment field for the local organisation.
B.08	progInfo		Com	Comment	xs:string (250)	B	R		Comment field for the sampling program information.
B.08	progInfo		targetVerif	Verification of *Salmonella* target	xs:string (5)	F	S	YESNO	Information if the data are used for the verification of the *Salmonella* reduction target.
B.08	progInfo		contrFlocks	Number of flocks/herds under control program	xs:integer (10)	F	S		Number of flocks/herds under the control program.
B.08	progInfo		contrAnimals	Number of animals under control program	xs:integer (10)	F	S		Number of animals under the control program.
B.08	progInfo		totUnitsTested	Total number of samples tested	xs: integer (10)	F	S		This is a numerical field. It means the total number of biological samples investigated (e.g. carcass, fresh meat, faeces), within a specific reporting MS, for the presence of specific bacterial species, strains or serovars. Samples may test positive or negative to the specific bacterial species, strain or serovar. Any isolate will subsequently be submitted to antimicrobial susceptibility testing.
B.08	progInfo		totSampUnitsTested	Total number of sampling units tested	xs: integer (10)	F	S		This is a numerical field. It means the total number of epidemiological units of interest (e.g. flock, herd, slaughtered batch, ..) investigated, within a specific reporting MS, for the presence of specific bacterial species, strains or serovars (or other zoonotic agent).
C.05	sampUnitIds		animalId	Animal identification code	xs:string (250)	F	S		Identification code of the animal.
C.05	sampUnitIds		herdId	Flock/herd identification code	xs:string (250)	F	S		Identification code of the flock/ herd.
C.05	sampUnitIds		batchId	Batch/ slaughter batch identification code	xs:string (250)	F	S		Identification code of the batch/ slaughter batch.
C.05	sampUnitIds		sampHoldingId	Sampling Holding identification code	xs:string (250)	F	S		Identification code of the holding at the moment the sampling was performed.
C.05	sampUnitIds		slaughterHouseId	Slaughterhouse identification code	xs:string (250)	F	S		Identification code of the slaughterhouse where the animal was slaughtered.
C.05	sampUnitIds		sampPlantId	Sampling plant identification code	xs:string (250)	F	S		Identification code of the plant where the sample was taken.

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Element Code	Element name	Attribute id	Attribute name	Attribute label	Type	Base/Facet	S/R	Controlled terminology	Description
C.06	sampEventInfo		Com	Comment	xs:string (250)	B	R		Comment field for the sampling event.
C.06	sampEventInfo		statusHerd	Status of the holding/ herd	xs:string (5)	F	S	SUSTAT	Status of holding/herd regarding infection.
C.06	sampEventInfo		vaccStatus	Vaccination status of the animal or flock/ herd	xs:string (5)	F	S	YESNO	Status of animal or flock/ herd regarding vaccination.
C.06	sampEventInfo		medicStatus	Medication status of the animal or flock/ herd	xs:string (5)	F	S	PARAM	Medication status of the animal or flock/ herd with reference to the last two weeks.
C.06	sampEventInfo		birthCountry	Country of birth	xs:string (2)	F	S	COUNTRY	Country of birth of the animal.
C.06	sampEventInfo		slaughterY	Year of slaughtering	xs:integer (4)	F	S		Year of slaughtering in the format YYYY.
C.06	sampEventInfo		slaughterM	Month of slaughtering	xs:integer (2)	F	S		Month of slaughtering in the format MM.
C.06	sampEventInfo		slaughterD	Day of slaughtering	xs:integer (2)	F	S		Day of slaughtering in the format DD.
C.06	sampEventInfo		slaughterCountry	Country of slaughtering	xs:string (2)	F	S	COUNTRY	Country of slaughtering.
D.11	sampInfo		arrivalY	Arrival Year	xs:integer (4)	F	S		Year of arrival in the laboratory in the format YYYY.
D.11	sampInfo		arrivalM	Arrival Month	xs:integer (2)	F	S		Month of arrival in the lab in the format MM.
D.11	sampInfo		arrivalD	Arrival Day	xs:integer (2)	F	S		Day of arrival in the lab in the format DD.
E.02	sampMatCode	F00	building	Building hierarchy	xs:string (5)	B	S	building	Hierarchy serving as a master for the management of all the terms contributed by the different domains.
E.02	sampMatCode	F01	source	Source	xs:string (5)	F	R	source	This facet describes the plant, animal, other organism or other source from which a raw primary commodity (RPC) has been obtained. The information brought by this facet is very often implicitly included in the food list groups. In the case of plant and animals, the RPCs are parts separated from (e.g. meat, leaves, fruit…) or directly produced by (e.g. milk, eggs…) a source. In most cases a RPC has only one source. However, in some food groups, like milk or minced or diced meat, products of the same nature of the ones from one source, but from

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Element Code	Element name	Attribute id	Attribute name	Attribute label	Type	Base/Facet	S/R	Controlled terminology	Description
									mixed sources are encountered. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
E.02	sampMatCode	F02	part	Part-nature	xs:string (5)	F	R	part	This facet describes the nature of the food item or the part of plant or animal it represents. The information brought by this facet is very often implicitly included in the food list groups. The descriptors are hierarchically structured in levels reflecting the natural relationship of terms. The list of descriptors is not designed to systematically include all the possible parts or nature, but intends to cover all the parts of plant or animal or the different nature types of food referred to in the food list. More than one descriptor can be applied to each entry (e.g. mixed offals, mixed seafood), provided they are not contradicting each other.
E.02	sampMatCode	F03	state	Physical-state	xs:string (5)	F	S	state	This facet describes the form (physical aspect) of the food as reported by the consumer (as estimated during interview or as registered in the diary) (Consumption Data) or as expressed in the analysis results in the laboratory (Occurrence Data). Only one descriptor from this facet can be added to each entry, apart from the specification "with solid particles". This facet should only be used in case of raw foods and ingredients (not for composite foods).
E.02	sampMatCode	F04	ingred	Ingredient	xs:string (5)	F	R	ingred	This facet collects ingredients and/or flavour note. Regarding ingredients this facet serves the purpose of providing information on ingredients of a composite food being important from some point of view, like allergic reactions, hazards, but also aspect, taste. The descriptors for this facet are taken from a selected subset of the main list (actually a relevant part of the food list). Regarding flavour note, this facet allows providing information on flavour or taste

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Element Code	Element name	Attribute id	Attribute name	Attribute label	Type	Base/Facet	S/R	Controlled terminology	Description
									notes, when obtained by exclusive use of chemical or extracted flavours instead of using the named ingredient (e.g. banana flavour obtained by using commercial flavour instead of real banana). More than one descriptor can be applied to each entry, provided they are not contradicting each other.
E.02	sampMatCode	F06	medium	Surrounding-medium	xs:string (5)	F	R	medium	This facet is intended for food packed in any container, together with any additional (usually fluid) medium. Not to be used to define an aggregated composite. This facet is needed to allow understanding the chemically/microbiologically relevant condition of the food (intended as the food surrounded by the medium). More than one descriptor can be applied to each entry, provided they are not contradicting each other.
E.02	sampMatCode	F07	fat	Fat-content	xs:string (5)	F	S	fat	This is a facet with numerical descriptors, to allow providing the fat content (as percentage w/w) of a food item. Only one descriptor from this facet can be added to each entry.
E.02	sampMatCode	F08	sweet	Sweetening-agent	xs:string (5)	F	R	sweet	This facet allows providing information on the added ingredient(s) used to impart sweetness to a food item. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
E.02	sampMatCode	F09	fort	Fortification-agent	xs:string (5)	F	R	fort	This facet allows providing information on the added ingredient(s) used to fortify a food item. The descriptors of this facet are taken from a selected small subset of the food list. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
E.02	sampMatCode	F10	qual	Qualitative-info	xs:string (5)	F	R	qual	This facet provides some principal claims related to important nutrients-ingredients, like fat, sugar etc. It is not intended to include health claims or similar. The present

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Element Code	Element name	Attribute id	Attribute name	Attribute label	Type	Base/Facet	S/R	Controlled terminology	Description
									guidance provides a limited list, to be eventually improved during the evolution of the system. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
E.02	sampMatCode	F11	alcohol	Alcohol-content	xs:string (5)	F	S	alcohol	This is a facet containing information from the food label.This is a facet with numerical descriptors, to allow providing the alcohol (ethanol) content (as percentage v/v) of a food item. The European Union follows recommendations of the International Organization of Legal Metrology (OIML). OIML International Recommendation No. 22 (1973) provides standards for measuring alcohol strength by volume and by mass. A preference for one method over the other is not stated in the document, but in this case alcohol strength by volume is used, expressed as a percentage (%) of total volume, assuming that the water/alcohol mixture have a temperature of 20°C when measurement is performed. The descriptors of this facet are a positive list of numbers (approx. 200). The list proposes numbers from 0 to 10 at interval of 0.1 and from 11 to 100 at interval of 1. Only one descriptor from this facet can be added to each entry.
E.02	sampMatCode	F12	dough	Dough-Mass	xs:string (5)	F	R	dough	This facet is proposed to provide information on the original dough-mass, for bakery products. The descriptors for this facet are taken from a selected subset of the food list. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
E.02	sampMatCode	F13	cookmeth	Cooking-method[(a)]	xs:string (5)	F	S	cookmeth	This facet allows recording the way a food item has been heat treated before consumption. In many cases, one descriptor from this facet is added to each entry, though more than one descriptor can be applied to each entry in case of sequential treatments.
E.02	sampMatCode	F14	prep	Final-preparation[(b)]	xs:string (5)	F	R	prep	This facet is particularly needed for

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Element Code	Element name	Attribute id	Attribute name	Attribute label	Type	Base/Facet	S/R	Controlled terminology	Description
									consumption surveys and includes preparation (like battering or breading) as well as heat treatment steps. It allows recording the way a food item has been prepared for final consumption. In many cases, one descriptor from this facet is added to each entry, though applied to each entry d in case of sequential treatments.
E.02	sampMatCode	F15	preserv	Preservation-technique[c]	xs:string (5)	F	R	preserv	This facet allows recording different preservation treatments a food item underwent. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
E.02	sampMatCode	F16	treat	Treatment-modifying-structure-or-nature[d]	xs:string (5)	F	R	treat	This facet allows recording different technological steps or treatments applied while producing a food item. Preservation treatments are excluded, because collected separately in another facet. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
E.02	sampMatCode	F17	cookext	Extent-of-cooking	xs:string (5)	F	R	cookext	This facet describes the intensity of heat treatment having been applied to a food item in the categories meat, fish-seafood, vegetables, eggs, bread and similar. Only one descriptor from this facet can be added to each entry, apart from the specification "Meat/fish/bakery/vegetables: presence of burned spots-parts".
E.02	sampMatCode	F18	packformat	Packaging-format	xs:string (5)	F	R	packformat	This facet is used for packaged food and allows recording the container or wrapping form. Only one descriptor from this facet can usually be added to each entry.
E.02	sampMatCode	F19	packmat	Packaging-material	xs:string (5)	F	S	packmat	This facet is used for packaged food and allows recording the material constituting the packaging containing the food. In case of combined material, it describes all the material, not only the part in contact with food. Only one descriptor for this facet may be used for each entry.
E.02	sampMatCode	F20	partcon	Part-consumed-	xs:string (5)	F	R	partcon	When reporting food analysed or consumed,

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Element Code	Element name	Attribute id	Attribute name	Attribute label	Type	Base/Facet	S/R	Controlled terminology	Description
				analysed					this facet allows specifying in which form the food item was analysed or consumed. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
E.02	sampMatCode	F21	prod	Production-method	xs:string (5)	F	R	prod	The facet production method describes the method used to produce the food. It is mainly applicable for animals and for foods from plant and of animal origin. This facet should only be used in case of raw foods and ingredients (not for composite foods). More than one descriptor can be applied to each entry, provided they are not contradicting each other (e.g. domesticated and wild).
E.02	sampMatCode	F22	place	Preparation-production-place	xs:string (5)	F	S	place	This facet allows recording the place where the food was prepared for consumption. Only one descriptor from this facet can be added to each entry.
E.02	sampMatCode	F23	targcon	Target-consumer	xs:string (5)	F	R	targcon	This facet allows recording different consumer classes intended as target for the food item. For laboratory uses when reporting analyses for feed, a list or target animals is also included. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
E.02	sampMatCode	F24	use	Intended-use	xs:string (5)	F	S	use	This facet allows recording the intended use of a food item, in particular with respect to further treatment expected (or not expected) before consumption. Only one descriptor from this facet can be added to each entry.
E.02	sampMatCode	F25	riskingred	Risky-Ingredient	xs:string (5)	F	R	riskingred	This facet (of specific interest in the microbiological domain) allows recording the presence of microbiologically high-risk ingredients. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
E.02	sampMatCode	F26	gen	Generic-term	xs:string (5)	F	S	gen	This facet allows recording whether the food list code was chosen because of lack of information on the food item or because the proper entry in the food list was missing.

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Element Code	Element name	Attribute id	Attribute name	Attribute label	Type	Base/Facet	S/R	Controlled terminology	Description
									Only one descriptor from this facet can be added to each entry.
E.02	sampMatCode	F27	rawsource	RPC-source-of-derivatives	xs:string (5)	F	R	rawsource	This facet describes the RPC from which an ingredient or derivative has been obtained. The information brought by this facet is very often implicitly included in the food list groups. In most cases a RPC has only one raw source. However, in some food groups, like cheeses or fruit juices, products of the same nature of the ones from one raw source, but from mixed raw sources are encountered. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
E.02	sampMatCode	F28	techno	Process	xs:string (5)	F	R	Techno	This facet allows recording different characteristics of the food: preservation treatments a food item underwent, technological steps or treatments applied while producing a food item, the way a food item has been heat treated before consumption and the way a food item has been prepared for final consumption (particularly needed for consumption surveys and includes preparation (like battering or breading) as well as heat treatment steps). More than one descriptor can be applied to each entry, provided they are not contradicting each other.
E.02	sampMatCode	F29	purpose-of-raising	Purpose-of-raising	xs:string (5)	F	R	fpurpose	This facet allows recording the purpose of farming, keeping or breeding (e.g. milk production, egg production). More than one descriptor can be applied to each entry, provided they are not contradicting each other.
E.02	sampMatCode	F30	reproductive-level	Reproductive-level	xs:string (5)	F	R	replev	This facet allows recording classes of animals from the point of view of reproduction. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
E.02	sampMatCode	F31	animal-age-class	Animal-age-class	xs:string (5)	F	R	animage	This facet allows recording the classes of the animal used in legislation or in the practice,

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Element Code	Element name	Attribute id	Attribute name	Attribute label	Type	Base/Facet	S/R	Controlled terminology	Description
									based on age or development stage. More than one descriptor can be applied to each entry, provided they are not contradicting each other..
E.02	sampMatCode	F32	gender	Gender	xs:string (5)	F	R	gender	This facet allows recording the status of an animal or animal group, with respect to sex. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
E.02	sampMatCode	F33	food-additive legislative-class	Food-additive-legislative-class	xs:string (5)	F	S	addit	This facet allows recording the food additives classes as reported in the legislation in order. Only one descriptor from this facet can be added to each entry.
E.10	sampMatCode		brandName	Brand name	xs:string (250)	F	S		Brand name of the product under analysis.
E.10	sampMatCode		manuf	Manufacturer	xs:string (250)	F	S		Company manufacturer of the product.
E.10	sampMatInfo		Com	Comment	xs:string (250)	B	R		Comment field for the matrix sampled.
E.10	sampMatInfo		prodY	Year of production	xs:integer (4)	F	S		Year of production.
E.10	sampMatInfo		prodM	Month of production	xs:integer (2)	F	S		Month of production.
E.10	sampMatInfo		prodD	Day of production	xs:integer (2)	F	S		Day of production.
E.10	sampMatInfo		expiryY	Year of expiry	xs:integer (4)	F	S		Year of expiry.
E.10	sampMatInfo		expiryM	Month of expiry	xs:integer (2)	F	S		Month of expiry.
E.10	sampMatInfo		expiryD	Day of expiry	xs:integer (2)	F	S		Day of expiry.
F.06	sampAnInfo		compY	Completion year of analysis	xs:string (4)	B	R		If the analysis has been performed over a period of time the completion year of analysis should be stated in this field.
F.06	sampAnInfo		compM	Completion month of analysis	xs:string (2)	B	R		If the analysis has been performed over a period of time the completion month of analysis should be stated in this field.
F.06	sampAnInfo		compD	Completion day of analysis	xs:string (2)	B	R		If the analysis has been performed over a period of time the completion day of analysis should be stated in this field.
F.06	sampAnInfo		Com	Comment	xs:string (250)	B	R		Comment field for the matrix sampled.
G.01	AnMatCode	F00	building	Building hierarchy	xs:string (5)	B	S	building	Hierarchy serving as a master for the management of all the terms contributed by the different domains.
G.01	sampMatCode	F01	source	Source	xs:string (5)	F	R	source	This facet describes the plant, animal, other organism or other source from which a raw primary commodity (RPC) has been obtained. The information brought by this facet is very often implicitly included in the food list groups. In the case of plant and

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Element Code	Element name	Attribute id	Attribute name	Attribute label	Type	Base/Facet	S/R	Controlled terminology	Description
									animals, the RPCs are parts separated from (e.g. meat, leaves, fruit…) or directly produced by (e.g. milk, eggs…) a source. In most cases a RPC has only one source. However, in some food groups, like milk or minced or diced meat, products of the same nature of the ones from one source, but from mixed sources are encountered. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
G.01	sampMatCode	F02	part	Part-nature	xs:string (5)	F	R	part	This facet describes the nature of the food item or the part of plant or animal it represents. The information brought by this facet is very often implicitly included in the food list groups. The descriptors are hierarchically structured in levels reflecting the natural relationship of terms. The list of descriptors is not designed to systematically include all the possible parts or nature, but intends to cover all the parts of plant or animal or the different nature types of food referred to in the food list. More than one descriptor can be applied to each entry (e.g. mixed offals, mixed seafood), provided they are not contradicting each other.
G.01	sampMatCode	F03	state	Physical-state	xs:string (5)	F	S	state	This facet describes the form (physical aspect) of the food as reported by the consumer (as estimated during interview or as registered in the diary) (Consumption Data) or as expressed in the analysis results in the laboratory (Occurrence Data). Only one descriptor from this facet can be added to each entry, apart from the specification "with solid particles". This facet should only be used in case of raw foods and ingredients (not for composite foods).
G.01	sampMatCode	F04	ingred	Ingredient	xs:string (5)	F	R	ingred	This facet collects ingredients and/or flavour note. Regarding ingredients this facet serves the purpose of providing information on ingredients of a composite food being

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Element Code	Element name	Attribute id	Attribute name	Attribute label	Type	Base/Facet	S/R	Controlled terminology	Description
									important from some point of view, like allergic reactions, hazards, but also aspect, taste. The descriptors for this facet are taken from a selected subset of the main list (actually a relevant part of the food list). Regarding flavour note, this facet allows providing information on flavour or taste notes, when obtained by exclusive use of chemical or extracted flavours instead of using the named ingredient (e.g. banana flavour obtained by using commercial flavour instead of real banana). More than one descriptor can be applied to each entry, provided they are not contradicting each other.
G.01	sampMatCode	F06	medium	Surrounding-medium	xs:string (5)	F	R	medium	This facet is intended for food packed in any container, together with any additional (usually fluid) medium. Not to be used to define an aggregated composite. This facet is needed to allow understanding the chemically/microbiologically relevant condition of the food (intended as the food surrounded by the medium). More than one descriptor can be applied to each entry, provided they are not contradicting each other.
G.01	sampMatCode	F07	fat	Fat-content	xs:string (5)	F	S	fat	This is a facet with numerical descriptors, to allow providing the fat content (as percentage w/w) of a food item. Only one descriptor from this facet can be added to each entry.
G.01	sampMatCode	F08	sweet	Sweetening-agent	xs:string (5)	F	R	sweet	This facet allows providing information on the added ingredient(s) used to impart sweetness to a food item. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
G.01	sampMatCode	F09	fort	Fortification-agent	xs:string (5)	F	R	fort	This facet allows providing information on the added ingredient(s) used to fortify a food item. The descriptors of this facet are taken from a selected small subset of the food list.

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Element Code	Element name	Attribute id	Attribute name	Attribute label	Type	Base/Facet	S/R	Controlled terminology	Description
									More than one descriptor can be applied to each entry, provided they are not contradicting each other.
G.01	sampMatCode	F10	qual	Qualitative-info	xs:string (5)	F	R	qual	This facet provides some principal claims related to important nutrients-ingredients, like fat, sugar etc. It is not intended to include health claims or similar. The present guidance provides a limited list, to be eventually improved during the evolution of the system. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
G.01	sampMatCode	F11	alcohol	Alcohol-content	xs:string (5)	F	S	alcohol	This is a facet containing information from the food label.This is a facet with numerical descriptors, to allow providing the alcohol (ethanol) content (as percentage v/v) of a food item. The European Union follows recommendations of the International Organization of Legal Metrology (OIML). OIML International Recommendation No. 22 (1973) provides standards for measuring alcohol strength by volume and by mass. A preference for one method over the other is not stated in the document, but in this case alcohol strength by volume is used, expressed as a percentage (%) of total volume, assuming that the water/alcohol mixture have a temperature of 20°C when measurement is performed. The descriptors of this facet are a positive list of numbers (approx. 200). The list proposes numbers from 0 to 10 at interval of 0.1 and from 11 to 100 at interval of 1. Only one descriptor from this facet can be added to each entry.
G.01	sampMatCode	F12	dough	Dough-Mass	xs:string (5)	F	R	dough	This facet is proposed to provide information on the original dough-mass, for bakery products. The descriptors for this facet are taken from a selected subset of the food list. More than one descriptor can be applied to each entry, provided they are not contradicting each other.

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Element Code	Element name	Attribute id	Attribute name	Attribute label	Type	Base/Facet	S/R	Controlled terminology	Description
G.01	sampMatCode	F13	cookmeth	Cooking-method[(a)]	xs:string (5)	F	S	cookmeth	This facet allows recording the way a food item has been heat treated before consumption. In many cases, one descriptor from this facet is added to each entry, though more than one descriptor can be applied to each entry in case of sequential treatments.
G.01	sampMatCode	F14	prep	Final-preparation[(b)]	xs:string (5)	F	R	prep	This facet is particularly needed for consumption surveys and includes preparation (like battering or breading) as well as heat treatment steps. It allows recording the way a food item has been prepared for final consumption. In many cases, one descriptor from this facet is added to each entry, though applied to each entry d in case of sequential treatments.
G.01	sampMatCode	F15	preserv	Preservation-technique[(c)]	xs:string (5)	F	R	preserv	This facet allows recording different preservation treatments a food item underwent. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
G.01	sampMatCode	F16	treat	Treatment-modifying-structure-or-nature[(d)]	xs:string (5)	F	R	treat	This facet allows recording different technological steps or treatments applied while producing a food item. Preservation treatments are excluded, because collected separately in another facet. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
G.01	sampMatCode	F17	cookext	Extent-of-cooking	xs:string (5)	F	R	cookext	This facet describes the intensity of heat treatment having been applied to a food item in the categories meat, fish-seafood, vegetables, eggs, bread and similar. Only one descriptor from this facet can be added to each entry, apart from the specification "Meat/fish/bakery/vegetables: presence of burned spots-parts".
G.01	sampMatCode	F18	packformat	Packaging-format	xs:string (5)	F	R	packformat	This facet is used for packaged food and allows recording the container or wrapping form. Only one descriptor from this facet can usually be added to each entry.
G.01	sampMatCode	F19	packmat	Packaging-material	xs:string (5)	F	S	packmat	This facet is used for packaged food and

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Element Code	Element name	Attribute id	Attribute name	Attribute label	Type	Base/Facet	S/R	Controlled terminology	Description
									allows recording the material constituting the packaging containing the food. In case of combined material, it describes all the material, not only the part in contact with food. Only one descriptor for this facet may be used for each entry.
G.01	sampMatCode	F20	partcon	Part-consumed-analysed	xs:string (5)	F	R	partcon	When reporting food analysed or consumed, this facet allows specifying in which form the food item was analysed or consumed. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
G.01	sampMatCode	F21	prod	Production-method	xs:string (5)	F	R	prod	The facet production method describes the method used to produce the food. It is mainly applicable for animals and for foods from plant and of animal origin. This facet should only be used in case of raw foods and ingredients (not for composite foods). More than one descriptor can be applied to each entry, provided they are not contradicting each other (e.g. domesticated and wild).
G.01	sampMatCode	F22	place	Preparation-production-place	xs:string (5)	F	S	place	This facet allows recording the place where the food was prepared for consumption. Only one descriptor from this facet can be added to each entry.
G.01	sampMatCode	F23	targcon	Target-consumer	xs:string (5)	F	R	targcon	This facet allows recording different consumer classes intended as target for the food item. For laboratory uses when reporting analyses for feed, a list or target animals is also included. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
G.01	sampMatCode	F24	use	Intended-use	xs:string (5)	F	S	use	This facet allows recording the intended use of a food item, in particular with respect to further treatment expected (or not expected) before consumption. Only one descriptor from this facet can be added to each entry.
G.01	sampMatCode	F25	riskingred	Risky-Ingredient	xs:string (5)	F	R	riskingred	This facet (of specific interest in the microbiological domain) allows recording the presence of microbiologically high-risk

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Element Code	Element name	Attribute id	Attribute name	Attribute label	Type	Base/Facet	S/R	Controlled terminology	Description
									ingredients. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
G.01	sampMatCode	F26	gen	Generic-term	xs:string (5)	F	S	gen	This facet allows recording whether the food list code was chosen because of lack of information on the food item or because the proper entry in the food list was missing. Only one descriptor from this facet can be added to each entry.
G.01	sampMatCode	F27	rawsource	RPC-source-of-derivatives	xs:string (5)	F	R	rawsource	This facet describes the RPC from which an ingredient or derivative has been obtained. The information brought by this facet is very often implicitly included in the food list groups. In most cases a RPC has only one raw source. However, in some food groups, like cheeses or fruit juices, products of the same nature of the ones from one raw source, but from mixed raw sources are encountered. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
G.01	sampMatCode	F28	techno	Process	xs:string (5)	F	R	Techno	This facet allows recording different characteristics of the food: preservation treatments a food item underwent, technological steps or treatments applied while producing a food item, the way a food item has been heat treated before consumption and the way a food item has been prepared for final consumption (particularly needed for consumption surveys and includes preparation (like battering or breading) as well as heat treatment steps). More than one descriptor can be applied to each entry, provided they are not contradicting each other.
G.01	sampMatCode	F29	purpose-of-raising	Purpose-of-raising	xs:string (5)	F	R	fpurpose	This facet allows recording the purpose of farming, keeping or breeding (e.g. milk production, egg production). More than one descriptor can be applied to each entry, provided they are not contradicting each other.

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Element Code	Element name	Attribute id	Attribute name	Attribute label	Type	Base/Facet	S/R	Controlled terminology	Description
G.01	sampMatCode	F30	reproductive-level	Reproductive-level	xs:string (5)	F	R	replev	This facet allows recording classes of animals from the point of view of reproduction. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
G.01	sampMatCode	F31	animal-age-class	Animal-age-class	xs:string (5)	F	R	animage	This facet allows recording the classes of the animal used in legislation or in the practice, based on age or development stage. More than one descriptor can be applied to each entry, provided they are not contradicting each other..
G.01	sampMatCode	F32	gender	Gender	xs:string (5)	F	R	gender	This facet allows recording the status of an animal or animal group, with respect to sex. More than one descriptor can be applied to each entry, provided they are not contradicting each other.
G.01	sampMatCode	F33	food-additive legislative-class	Food–additive-legislative-class	xs:string (5)	F	S	addit	This facet allows recording the food additives classes as reported in the legislation in order. Only one descriptor from this facet can be added to each entry.
G.03	anMatInfo		com	Additional information on the matrix analysed	xs:string (250)	B	R		Additional information on the analysed matrix.
H.04	anPortInfo		com	Additional information on the sample analysed portion	xs:string (250)	B	R		Data collection specific information on the sample analysed portion.
I.02	isolParamCode		isolate	Isolate	xs:string (15)	B	S	PARAM	Encoding of the isolate parameter according to the PARAM catalogue. It is used to report the speciation or serotyping of the isolate.
I.04	isolInfo		isolY	Isolation year	xs:integer (4)	F	S		Year when the isolate was identified.
I.04	isolInfo		isolM	Isolation month	xs:integer (2)	F	S		Month when the isolate was identified.
I.04	IsolInfo		isolD	Isolation day	xs:integer (2)	F	S		Day when the isolate was identified.
I.04	isolInfo		com	Additional information on the isolate	xs:string (250)	B	R		Data collection specific additional information on the isolate.
J.04	labInfo		labTotIsol	Total number of isolates available in the laboratory	xs:integer (10)	F	S		Data collection specific additional information on the laboratory (e.g. total number of isolates available in the laboratory, total number of isolates available in the laboratory testing for AMR for the

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Element Code	Element name	Attribute id	Attribute name	Attribute label	Type	Base/Facet	S/R	Controlled terminology	Description
									specific bacterial species, strains or serovars).
J.04	labInfo		com	Comment	xs:string (250)	B	R		Data collection specific additional information on the laboratory.
K.02	paramCode		param	base parameter	xs:string (15)	B	S	PARAM	Encoding of the parameter/ analyte according to the PARAM catalogue.
K.02	paramCode		esbl	ESBL phenotype	xs:string (250)	F	S	PARAM	ESBL phenotypes in isolates of Salmonella and E. coli through enzyme formulas (e.g. TEM-52, SHV-2, CTX-M-1).
K.02	paramCode		ampC	AMPC phenotype	xs:string (250)	F	S	PARAM	AmpC phenotypes in isolates of Salmonella and E. coli through enzyme formulas (e.g. CMY-2, AAC-1).
K.02	paramCode		carbapenem	Carbapenemase phenotype	xs:string (250)	F	S	PARAM	Carbapenemase phenotypes in isolates of Salmonella and E. coli through enzyme formulas (e.g. KPC, OXA-48).
K.02	paramCode		synTest	Synergy test	xs:string (250)	F	S	PARAM	Outcome of a synergy test executed to detect certain -lactam resistant phenotypes.
L.06	anMethInfo		diskConc	Disk concentration	xs:double	F	S		Substance concentration in the disk used for the diffusion method (expressed in micrograms).
L.06	anMethInfo		diskDiam	Disk diameter	xs:double	F	S		Diameter of the disk used in the diffusion method (expressed in mm).
L.06	anMethInfo		methSensitivity	Method sensitivity	xs: decimal (5,2)	F	S		Sensitivity of the analytical method.
L.06	anMethInfo		methSpecificity	Method specificity	xs: decimal(5,2) (10,2)	F	S		Specificity of the analytical method.
L.06	anMethInfo		contactTime_min	Contact time	xs:double	F	S		The contact time represents the length of the exposition of the food packaging to a food stimulant (in minutes).
L.06	anMethInfo		contactTemp_C	Contact temperature	xs:double	F	S		The contact temperature represents the temperature of the bath in which both the packaging material and the simulant are contained (as °C).
L.06	anMethInfo		com	Comment	xs:string (250)	B	R		Additional information on the analytical method depending on specific requirements of the different data collection domains.
M.13	exprResPerc		fatPerc	Percentage of fat	xs:double	B	R		Percentage of fat in the analysed sample.
M.13	exprResPerc		moistPerc	Percentage of moisture	xs:double	B	R		Percentage of moisture in the analysed sample.
M.13	exprResPerc		alcoholPerc	Percentage of alcohol	xs:double	B	R		Percentage of alcohol in the analysed sample.

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Element Code	Element name	Attribute id	Attribute name	Attribute label	Type	Base/Facet	S/R	Controlled terminology	Description
M.20	resInfo		perCC	Performed CC for MRSA characterisation	xs:string (5)	F	S	YESNO	Binary list to record whether determination of clonal complex for MRSA isolate has been performed through laboratory experiments (YES) or has been inferred from e.g. online databases (NO).
M.20	resInfo		perMLST	Performed MLST MRSA characterisation	xs:string (5)	F	S	YESNO	Binary list to record whether determination of MLST type for MRSA isolate has been performed through laboratory experiments (YES) or has been inferred from e.g. online databases (NO).
M.20	resInfo		com	Comment	xs:string (250)	B	R		Additional information on the results.
N.06	evalInfo		sampAnAsses	Sample analysed assessment	xs:string (5)	F	S	RESEVAL	Sample analysed assessment.
N.06	evalInfo		sampTkAsses	Sample taken assessment	xs:string (5)	F	S	RESEVAL	Sample taken assessment.
N.06	evalInfo		sampEventAsses	Sampling event evaluation	xs:string (5)	F	S	RESEVAL	Sampling event assessment.
N.06	evalInfo		com	Comment	xs:string (250)	B	R		Additional information on the evaluation section.

(a): The elements listed in the facet 'Cooking-method' are now all included in the facet F28 'Process' and their use is deprecated since September the 1st, 2013.
(b): The elements listed in the facet 'Final-preparation' are now all included in the facet F28 'Process' and their use is deprecated since September the 1st, 2013.
(c): The elements listed in the facet 'Preservation-technique' are now all included in the facet F28 'Process' and their use is deprecated since September the 1st, 2013.
(d): The elements listed in the facet 'Treatment-modifying-structure-or-nature' are now all included in the facet F28 'Process' and their use is deprecated since September the 1st, 2013.

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8. Description for the Controlled terminologies, maintenance and versioning

The use of controlled terminologies facilitates the aggregation of data during analysis and ensures comparability between datasets. Controlled terminologies are language independent (only the code needs to be returned) and the description of the code in any language can be linked to the code. To ensure that the data are of sufficient quality to allow analysis at EU level, controlled terminologies have been applied extensively in the SSD. Although every effort has been made to ensure controlled terminologies are comprehensive to ensure full description of the sample, the ongoing evolution of the catalogues over time, in line with reality, may result in temporary catalogue deficiencies. For some elements with controlled terminologies, an optional additional companion free text element is included. This allows the provision of relevant information when the controlled terminology is insufficient to fully characterise the item described.

8.1. Maintenance and versioning of SSD2 controlled terminology catalogues

As already pointed out in the previous chapters, controlled terminologies are paramount to the SSD. In a similar way to the SSD structure, terminologies need to have a standard set of data elements describing the terms, so that they can be easily imported into data provider databases.

These data elements include not only the ‘term code’, ‘term name’ and the ‘term definition’ but also additional information used to describe the relationships between the terms, the terms’ version control and the data domain validity of terms.

The maintenance of the controlled terminologies is one of the major challenges to ensure a fully working SSD. For this reason, provisions are made to add some data elements to the controlled terminology intended to describe the life cycle of the terms (described in Table 22) and the logical model (described in Table 4) .

The life cycle of the term can be summarised as follows:

1. After the term is published and distributed, the ‘term code’ cannot be changed;
2. Correction for spelling mistakes in the ‘term name’, ‘term definition’ or additional elements can be considered minor changes and will be amended in the next published version of the catalogues;
3. When terminology changes are required, new terms can be added to the controlled terminology, amended or logically removed (deprecated). Terms are logically removed from the terminology rather than deleted in order to avoid backward compatibility issues with data already submitted. A flagging mechanism is defined to highlight these logically removed terms as ‘deprecated’;
4. It is possible that terms may be added to the system between two official releases, bypassing the standard approval process presented below. These terms will be referred to as ‘preliminary terms’ and a flagging mechanism is defined to highlight them. These terms will be validated through the standard approval process in the next official release of the controlled terminologies. ‘Preliminary terms’ may be flagged ‘deprecated’ in the next official release if they do not pass the standard approval process;
5. A notification system to the users of the SSD should be set up in order to inform them of any update operation of the SSD catalogues.

In the validation of an SSD data transmission, the terms reported will be checked against the relevant controlled terminology active at the time of the data transmission (‘Valid from’ <= transmission date < ‘Valid to’ or transmission date > ‘Valid from’ if the ‘Valid to’ date is missing).

It should be noted that in the first version of the controlled terminology ‘valid from’ date will be set to ‘1 Dec 2009’. For SSD2, the new catalogues included will have the ‘valid from’ date set to ‘1 Dec 2012’.

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The amount of information stored in the catalogues due to history management, although very useful for IT system management, can be misleading for users. As policy and to avoid the inadvertent use of deprecated terms by data providers, terms listed as 'deprecated' for a time period exceeding six years will also be removed from the published catalogues. For example, terms 'logically deleted' on 01/12/2010 will be actually removed from the full list of terms on the first release of the controlled terminology after 02/12/2016.

The SSD2 also allows the possibility of adding new terminology catalogues to support data reporting in the new compound elements when needed. For these catalogues, the validity data will be set from the date of creation of the new catalogue itself. In order to support a better user understanding of the SSD2, terminologies will also be distributed in a user friendly format, which will report only the terms active at the time of catalogue publication and only the data element subset which is applicable for users. The user friendly format will be a Microsoft Excel workbook containing all the terminologies used in the SSD2.

The full list of terms and data elements will then be distributed in machine friendly format, the procedure for the distribution and the format will be described in the GDE2.

8.2. Frequency of review

The new structure for SSD and its amended catalogues cater for the new requirements described in the mandate of this WG-SSD2.

A maintenance process is needed to allow:

- addition and/or deprecation of terms within the catalogues over time;
- addition and/or deprecation of compound elements;
- addition and/or deprecation of catalogues used in compound elements.

The frequency and timing of such updates are of particular concern to MSs since it is not always feasible to collect or code data to new SSD catalogue versions once the data have already been created or captured. Ideally, changes will be relatively infrequent, enabling correct controlled terminologies to be in use at the time of capture. In principle, before including the changes in the SSD catalogue a review of the terms inserted/deprecated should be performed by the relevant networks.

It is recommended that the implementation schedule for catalogue changes should be mainly annual (*major release*) and take into account of the ability of MSs to implement such changes. In addition it is anticipated that each data collection domain will need to make changes at a time of year which best coincides with their data collection schedules. In several instances, these annual schedules are well established within the relevant networks and a draft annual schedule is proposed below.

Each network will collate their comments and consider the requests which impact their specific domain.

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Table 18: Proposed annual plan for the update of the controlled terminologies

Data Collection Domain	Annual reporting deadline for data submission	Deadline for terms to include in next release	Deadline for discussion within the network	Submission of updates from the network	Submission of updated version for final comments	Deadline for SSD Revision Publication
Biological Monitoring	31st May	15 October	Up to the networks	No later than 30 November	31 January	28 February
Pesticides	31st August					
Chemical Contaminants	1st October					
Additives	Ad-hoc basis					
Food contact materials	Ad-hoc basis					

EFSA should set up an internal controlled terminology group to discuss the catalogue amendments proposed by the networks so that decisions are always taken in an harmonised way taking into account all the requirements and implications in the different data collection domains. When necessary this group may involve additional experts from the MSs or data providers.

Before the new annual version of the controlled terminologies is released, the collated suggestions should be circulated to the relevant networks for their comments and approval. In order to streamline the process, the networks could establish a specific subgroup to deal with these requests. Once comments are gathered, the EFSA controlled terminology group could make the final decision and publish the updated catalogues.

In addition, it is anticipated that there may be instances where unplanned calls for data are launched which require catalogue revisions or new terms need to be added with urgency to the catalogue to enable the transmission of the data of an existent data collection. In this case, preliminary terms can be added. This type of updates should be kept to a minimum due to the impact that they may cause on the data providers. The addition of preliminary terms will require publication of additional 'minor releases' of the catalogues during the year. This process would also be controlled by the EFSA controlled terminology group.

8.3. Mechanism for revision suggestions

SSD users should be able to make suggestions regarding controlled terminology changes and other issues regarding transmission of their data. The WG-SSD2 suggests the following as a possible format for this type of requests as an example:

Item:	**Example:**
Suggestion #	1
Suggested Date	1 Feb 2012
Suggested by (Name)	Eileen O'Dea
Email Address	eodea@fsai.ie
Member State	IE
Organisation	FSAI
SSD catalogue	PARAM
SSD Element Code	K.02
SSD Term Code	RF-00000489-TOX
SSD Term Description	New Parameter
New/Change/Deprecate	New
Value	'New Parameter'
Urgency	'Urgent' or 'Standard'
Comments	'Please add this item for which we now analyse samples'

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Suggestion Outcome	'Approved as preliminary term' 'To be included in the annual release' 'Rejected - use instead xxx'
Closed date	1st June 2012.

All the 'standard' requests will be dealt in the annual release with the mechanisms presented. The 'urgent' requests could be:

- dealt with as preliminary term additions;
- included then in the annual release after the consideration of the networks.

All the requests received in the defined format by EFSA during the data collection period will be submitted to the relevant networks on 15 October each year. The requests received after 15 October will be considered in the following review period.

The data providers suggesting amendments should be informed of the result of their requests within 15 working days.

8.4. Synchronisation with international standard terminologies

Wherever possible, existing terminologies have been adopted to control the terms which will be transmitted for the elements. These may be from international or national sources and have been specified in this document either exactly as available in the source terminology or in an adapted form. Where the terminology has been adapted, this has been noted.

The EFSA controlled terminology group should ensure an annual check for amendments to the source terminologies in these cases. The following table (Table 19) details the terminologies within SSD2 which should be verified annually for changes.

Table 19: Table of Source Controlled Terminologies

SSD Controlled Terminology	Source Terminology	Source Terminology Organisation
COUNTRY	ISO 3166-1-alpha-2	ISO
SAMPSTR	Doc. ESTAT/F5/ES/201 "Typology of sampling strategy"	Eurostat
SAMPNT	Doc. ESTAT/F5/ES/155 "Data dictionary of activities of the establishments"	Eurostat
NUTS	Nomenclature of territorial units for statistics - NUTS	Eurostat
FAREA	FAO Fisheries areas	FAO/WHO

8.5. Publication process

In order to support the reporting of data to EFSA, the WG-SSD2 believes that an effective publication and maintenance system of the EFSA's controlled terminology is crucial. Controlled Terminologies are to be made available to data providers on the Internet (i.e. the EFSA Website and through web-services). It is essential that the published version, in any format available, will always report the version of the SSD they refer to (as release date or in the format <major release>.<minor release>).

An alternative possibility for publishing the SSD catalogues to data providers could be to have a dedicated web-service tool. This functionality will be further discussed in the GDE2. In this case, a mechanism of notification of publication to the SSD users should be defined. For example, MSs could subscribe to a notification tool informing them when a new version of the SSD catalogues becomes available.

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A file with all controlled terminology terms and a file with the list of changes since the previous version should be made available for download or synchronisation to facilitate MSs who implement mapping in their system from national terms to EFSA terms. Formats for these files will be described in the GDE2.

There are a number of options for the format of publication and these include at least XML and Microsoft Excel. The eventual support of additional formats will also be discussed in the GDE2.

Controlled terminologies will include for each term the 'Valid from (date)', 'Valid to (date)', 'Last update (date)' and 'status (Preliminary (P)/Approved (A))' information to facilitate identification of the changes, the dates of changes and the period for which terms were valid. No term should be physically deleted from the Controlled Terminology for a period of six years, a 'logical deletion' of term will result in flagging the term as 'deprecated' and its 'Valid to (date)' filled in. In any case, as policy, terms listed as 'deprecated' for a time period exceeding six years will also be removed from the file. Therefore terms 'logically deleted' on 01/12/2010 will be actually removed from the full list of terms on the first release of the controlled terminology after 02/12/2016.

8.6. Controlled terminologies database

A controlled terminology is in principle a list of terms identified by a code that can be used in data transmission. These lists of terms can be organised in one or more hierarchies.

In principle, there is always one hierarchy, the master hierarchy, which includes all the terms of the controlled terminology (a list can be seen as a flattened hierarchy). The master hierarchy will be identified by the same identifier as the controlled terminology. The additional hierarchies, that will be defined as 'analysis hierarchies' can define a subset of the terms available in the master hierarchy or can even change the parent-child relationship of the terms as available in the master hierarchy.

The master hierarchy can allow many-to-many relationships between parent terms and child terms, while in the analysis hierarchies only one-to-many relationships are allowed between a parent term and its child terms. Each hierarchy will define an applicability scope to one-or-more data domains, which will be listed in the SSD as an additional catalogue (DMN). The list included in Table 20 is preliminary and reflects the list of domains at the time the SSD2 was published.

Table 20: Domain (DMN catalogue)[36]

code	Name	parentCode	validFrom	validTo	lastUpdate
ALL	All domains	ROOT	01/09/2012		
ADD	Additives	ALL	01/09/2012		
CHEM	Chemical contaminants	ALL	01/09/2012		
FCM	Food contact materials	ALL	01/09/2012		
BIOMO	Biological monitoring	ALL	01/09/2012		
PEST	Pesticide residues	ALL	01/09/2012		

The SSD will provide an additional standard terminology called 'Dictionary' (DICT), listing all the standard terminologies, with master hierarchies and analysis hierarchies available.

For example, the extract of the catalogue UNIT, at the time this guidance is published, is reported in Table 21, including its related hierarchies as available in the dictionary catalogue.

In Table 20 you can distinguish the catalogues from the hierarchies since catalogues are reporting as parentCode the code 'ROOT', while hierarchies are reporting as parentCode the associated catalogue code e.g. 'Unit'.

[36] The column 'sortingCode' is not reported in this table

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The SSD allows the defining of more than one hierarchy for each catalogue. These additional hierarchies are used to populate compound fields such as for food classification using the FoodEx2 system.

Table 21: Preliminary list of catalogues and hierarchies as available in the dictionary catalogue (Only the most relevant columns are included).

Controlled terminology	Catalogue name	parentCode
UNIT	Unit of measurement	ROOT
microUnit	Units allowed in biological monitoring	UNIT
pestUnit	Units allowed in pesticide monitoring	UNIT
fcmUnit	Units allowed in FCM data collections	UNIT

Catalogues have some data elements that are common to all the catalogues, and other elements that are catalogue specific.

The common data elements are described in the Table 22.

Table 22: Common Catalogue Structure Data Elements

Element Code	Element Name	Element Label	Type	R	Description	Mandatory
T.01	code	Term code	xs:string (20)		Unique code for the term. This is the only code that should be reported in the Standard Description.	**M**
T.02	name	Term name	xs:string (50)		Term name	**M**
T.03	parentCode	Parent code	xs:string (20)	R	Link to the parent term for the master hierarchy	
T.04	sortingCode	Hierarchy code	xs:string (20)		Code representing the hierarchy, for displaying and sorting purposes. This code should not be used for reporting.	
T.NN	<dictionary catalogue code>ParentCode	Parent code	xs:string (20)		Link to the parent term for the specific analysis hierarchy	**M**
T.NN	<dictionary catalogue code>SortingCode	Hierarchy code	xs:string (20)		Code representing the hierarchy, for displaying and sorting purposes. This code should not be used for reporting.	**M**
T.96	Status		xs:string (1)		Status of the term: preliminary (P), approved (A)	**M**
T.97	validFrom	Valid from date	xs:date		Start date of the validity interval	**M**
T.98	validTo	Valid to date	xs:date		End date of the validity interval. If null the term is valid	
T.99	lastUpdate	Last update date	xs:date		Date when this record was updated	

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A specific table will define which additional catalogue structure elements are needed in the different catalogues. The catalogue structure elements required are the following:

- **Catalogue**: the code of the catalogue for which the elements are specified;
- **Attribute name**: the name of the attribute to be used in database systems;
- **Attribute label**: the label for the attribute to be displayed to users;
- **Type**: the XML type for the attribute;
- **S/R/C**: simple, repeatable, compound fields;
- **Controlled terminology**: the controlled terminology to be used for the catalogue;
- **Description**: a user level description of the catalogue attribute.

An example on how this metadata can be filled-in for the FoodEx2 catalogue is presented in Table 23.

The usage of validFrom and validTo elements as presented above is capable of supporting version control for the terms but it cannot provide a version control for the association of the term to the different hierarchies and of the parent-child relationship in the hierarchies. This support would result in a MS Excel file too complex. The version control of the hierarchies is essential and should be included in the XML version of the catalogue proposed in the GDE2.

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Table 23: Example of definition of additional catalogue structure elements for the FoodEx2 catalogue

Catalogue	Attribute name	Attribute label	Type	S/R/C	Controlled terminology	Description
MTX	foodexOldCode	FoodEx1 code	xs:string (20)	R	FOODEX	Former FoodEx1 Code.
MTX	matrixCode	Pesticide MATRIX catalogue code	xs:string (20)	R	MATRIX	MATRIX classification catalogue.
MTX	GEMSCode	GEMS code	xs:string (20)	R		
MTX	langualCode	Langual code	xs:string (20)	R		
MTX	statef	State flag	xs:string (1)	S	STATEF	Define the type of term..
MTX	corex	Core - extended flag	xs:string (1)	S	COREX	Define if the term is core or extended.
MTX	scientificNames	Scientific names	xs:string (1000)	R		List of scientific names for the term.
MTX	commonNames	Common names	xs:string (1000)	R		List of common names for the term.
MTX	facetSchema	Applicable facets	xs:string (1000)	R		List of applicable facets.
MTX	facetSchemaCard	Cardinality of the applicable facets	xs:string (1000)	R	CARD	Cardinality of the applied facets (mandatory single, optional single, mandatory repeatable, optional repeatable).
MTX	facetApplicability	Applicable descriptors	xs:String (1000)	C		Compound field describing the applicable facets.
MTX	ImplicitFacets	Applied descriptors	xs:string (1000)	C		Compound field describing the applied facets.

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CONCLUSIONS AND RECOMMENDATIONS

CONCLUSIONS

This document is intended as a guidance for data providers, in particular MSs, to use in transmitting sample data to EFSA and planning future developments and evolution of local, regional and national systems with the objective of harmonising EU-wide data transmissions according to SSD2.

The SSD2, defining data elements and controlled terminologies, is a mature effort designed to harmonise the transmission of sample level data from data providers to EFSA in several data collection domains.

The SSD2 has been developed specifically to address transmission at sample level of:

- food additive occurrence data;
- chemical contaminants occurrence data;
- pesticide residues occurrence data;
- animal/flock/herd level prevalence data on zoonotic agents in animals;
- isolate-based data on antimicrobial resistance;
- microbiological contaminants occurrence data.

The WG-SSD2 anticipates that the SSD2 structure is sufficiently flexible to accommodate additional data domains, if required in the future, through additions to the controlled terminologies and the compound elements.

In the case of data collection domains for which sample-level data transmission is not agreed or appropriate, the SSD2 still provides a harmonised basis for aggregation of data prior to transmission. The successful use of this data collection standard is linked to its tailoring to the specific reporting requirements in the different data collection domains.

In addition, being a standard in current and ongoing use, it is essential to put in place a process for further maintenance of the controlled terminologies and the compound elements.

Harmonisation of data collections by establishing a common data model is recognised as a fundamental step to the development of an effective EFSA Data Warehouse containing data from a range of data collection domains. The establishment of the EFSA Data Warehouse is seen as a key milestone for achieving a shared and transparent access to data to support Europe-wide risk assessment by EFSA and - with appropriate access policies - for MS and other stakeholders.

The WG-SSD2 recognises that the ability of each MS to transmit data to EFSA according to the SSD2 data model will vary.

In countries where SSD1 has already been implemented for one or more data domains, this will require adjustments to comply with SSD2; the WG-SSD2 has identified where adjustments are needed and have listed them in detail in Appendix B.

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RECOMMENDATIONS

The WG-SSD2, taking into account the characteristics of SSD data elements and the controlled terminologies, makes the following recommendations which are aimed to facilitate the format, harmonisation and mechanism of transmission of data to EFSA, for all the data collection domains specified within its mandate.

The following recommendations are addressed to EFSA as the leading organisation but they require a cooperative effort involving EFSA, the EU Member States and the European Commission.

Recommendation 1: Implementation of SSD2 and parallel running of SSD1

For each data collection domain, the implementation of SSD2 will depend on different circumstances:

- in cases where the SSD1 is not yet in use, the SSD2 should be directly adopted;
- in cases where SSD1 is already in use for data transmission, data providers should agree on a plan in stages to replace SSD1 with SSD2. Each domain network should define a time frame by which only SSD2 will be used by all data providers. In the case of the Pesticides data collection domain, the use and the timeline for implementation of SSD2 must additionally be approved by the Standing Committee on the Food Chain and Animal Health (SCoFCAH)[37] and the legislation prescribing the use of SSD1 should be updated before SSD2 can be officially implemented.

The general recommendation of the WG-SSD2 is that the period of parallel use of SSD1 and SSD2 should be as short as possible and in any case, should not exceed five years i.e. all data relating to sampling year 2018 should be transmitted to EFSA in SSD2 format.

Recommendation 2: Implementation of SSD2 at national level by the launch of pilot projects

The WG-SSD2 recommends that a pilot project on the use of SSD2, supported by EFSA, is initiated within each data domain. This is already in progress for FoodEx2 in some data collection domains (e.g. chemical contaminants and consumption data). Additional pilot projects for the other domains are recommended (e.g. piloting the usage of FoodEx2 in the pesticide residues domain).

Recommendation 3: Harmonising reporting requirements and usage of SSD2

Different reporting templates are currently used and defined in the legislation for the reporting of data in different domains. Efforts should be made towards rationalising the reporting requirements for these data and towards endorsing the use of the SSD2 throughout the different data collection domains.

Recommendation 4: Usage and maintenance of the SSD2 and its controlled terminologies

A maintenance process should be set up to allow agreed changes to be made to the controlled terminologies of the SSD2 according to the proposal presented in the section 8 of this guidance. The same maintenance process should also support the addition of new data elements to be reported through the compound elements.

[37] Report to the Standing Committee on the Food Chain and Animal Health (SCoFCAH) on the relationship between analytical results, the measurement uncertainty, recovery factors and the provisions in EU food and feed legislation with particular focus on the community legislation concerning contaminants in food and undesirable substances in feed, 2004. Available online: http://ec.europa.eu/comm/food/food/chemicalsafety/contaminants/report-sampling_analysis_2004_en.pdf

Recommendation 5: Harmonised matrix classification system

The WG-SSD2 recommends that the FoodEx2 catalogue is used as a harmonised matrix classification system throughout all the data domains within the remit of EFSA. For this reason FoodEx2 should be extended to include non-food items as proposed in the controlled terminology revisions.

The SSD2 relies on the availability of a stable and tested version of FoodEx2. By this, the WG-SSD2 means a version in which:

- the terms in the building hierarchy cover the predictable needs of the data domains under the remit of the Working Group on SSD Extension (WG-SSD2) (chemical contaminants, pesticides residues, biological monitoring and additives);
- the relevant implicit facets are defined and inappropriate combinations of base terms and facets are prevented;
- the applicability and mandatory status of facets to the appropriate terms are defined for each data collection domain;
- FoodEx2 is thoroughly tested in its usage for data reporting in all the data collection domains through ad-hoc pilot projects.

Therefore the WG-SSD2 recommends that the implementation of the SSD2 for official use in data reporting is subject to the availability of a stable and tested version of FoodEx2. Due to the complexity of FoodEx2, the WG-SSD2 recommends EFSA support for Member States in the implementation and roll out of this new catalogue. This support could include:

- practical user guidance document for FoodEx2 implementation (how to add facets, which hierarchy to choose as starting points, how to manage cases where there is no clear correspondence between FoodEx2 codes and the LIMS system catalogue);
- financial support;
- mapping tools defining the conversion between FoodEx1, MATRIX to FoodEx2 classification terms.

Recommendation 6: Policies for access and use of data

The WG-SSD2 recognises that data transmitted to EFSA using the SSD2 contains a much deeper level of detail than the previous SSD. As such, the data will provide a greatly improved data analysis and risk assessment capability but also a correspondingly increased responsibility for maintaining secure access controls and preventing breaches of data confidentiality.

It is recommended that policies for access and use of the data are defined and published to establish clarity regarding what are appropriate uses of the transmitted data.

Recommendation 7: Domain specific data collection requirements and business rules

In the SSD2 and GDE2 guidance documents, the WG-SSD2 included only the general reporting requirements and business rules applicable to all data domains specified in the working group's mandate. Specific requirements and business rules can be defined at each data domain level and are already available for some data collection domains (e.g. pesticide residues and chemical contaminants).

In order to enhance the communication of these domain specific requirements and business rules, the WG-SSD2 recommends that EFSA collects, in a single section of the EFSA web site, all these requirements. Additionally, the WG-SSD2 recommends that the specific requirements and business rules are subject to the same maintenance approval process as defined for the controlled terminologies and compound elements.

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Recommendation 8: Languages

At present, the elements and controlled terminologies have been defined in English only.

It is recommended that translations of the elements and terminologies be accommodated as synonyms of the English values defined and that the development of such translations be co-ordinated with the Member States. It is also recommended that, wherever possible, data in free text fields be transmitted in English.

It should be noted that throughout the data model, the use of free text fields has been minimised both to reduce the amount of non-standard data and to reduce the difficulty of usage of multiple languages.

Recommendation 9: Multiple Character sets

Due to language variations across Member States, a number of characters are in common usage in one or more languages which are not standard in all languages e.g. Greek and Cyrillic characters.

It is recommended that the file formats and the EFSA Data Warehouse system support the UNICODE character set, so as to support all the character sets currently in use in Europe.

Recommendation 11: Consider the extension of the SSD2 to other data collection areas

The SSD2 was built to support the data collection domains described in the mandate for the present WG-SSD2 on SSD Extension. The WG-SSD2 notes that it also provides a very generic and flexible model which can be applied to a variety of data collection domains based on the use of the compound data elements and extensions of the available controlled terminologies. Therefore, this standard should be taken into account, when new data collections are launched in the future, as a potential reference model.

REFERENCES

Codex Alimentarius Commission, FAO, WHO, 2009. Report of the thirtieth session of the Codex Committee on methods of analysis and sampling. Balatonalmádi, Hungary, 9 - 13 March 2009, 1-69. Available online: http://www.codexalimentarius.org/input/download/report/720/al32_23e.pdf

EFSA (European Food Safety Authority), 2010a. EFSA Guidance on Standard sample description for food and feed. EFSA Journal 2010;8(1):1457, 54 pp., doi:10.2903/j.efsa.2010.1457

EFSA (European Food Safety Authority), 2010b. EFSA Guidance on Data Exchange. EFSA Journal 2010;8(11):1895, 50 pp., doi:10.2903/j.efsa.2010.1895

EFSA (European Food Safety Authority), 2010c. EFSA Report on Data Collection: Future Directions. EFSA Journal 2010;8(5):1533, 35 pp., doi:10.2903/j.efsa.2010.1533

EFSA (European Food Safety Authority), 2011a. Report on the development of a food classification and description system for exposure assessment and guidance on its implementation and use. EFSA Journal 2011;9(12):2489, 84 pp., doi:10.2903/j.efsa.2011.2489

EFSA (European Food Safety Authority), 2011b. The food classification and description system FoodEx2 (draft-revision1). Supporting Publications 2011:EN-215, 438 pp.

EFSA (European Food Safety Authority), 2012a. A pilot with volunteering reporting countries on submission of antimicrobial resistance isolate-based data in Excel and XML formats using the Data Collection Framework. Supporting Publications 2012:EN-220, 16 pp.

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Nicolau A, Tudorie C and Craciun C, 2012. Technical report submitted to EFSA. Electronic transmission of chemical occurrence data. CFP/EFSA/DATEX/2010/01. Available online: http://www.efsa.europa.eu/en/supporting/doc/237e.pdf

Eurostat, 2007. Data dictionary of activities of the establishments. Doc. ESTAT/F5/ES/155, Technical Group "Food and feed control and monitoring activities". Meeting of 1-2 October 2007, Luxembourg. Available online: https://circabc.europa.eu/faces/jsp/extension/wai/navigation/container.jsp

Eurostat, 2010. Typology of sampling strategies. Doc. ESTAT/F5/ES/201. Working group "Food Safety Statistics". Meeting of 24-25 June 2010, Luxembourg. Available online: https://circabc.europa.eu/faces/jsp/extension/wai/navigation/container.jsp

Eurostat, 2013. Nomenclature of territorial units for statistics - NUTS. Available online: http://epp.eurostat.ec.europa.eu/portal/page/portal/nuts_nomenclature/introduction

FAO (Food and Agriculture Organization of the United Nations), 2013. FAO Major Fishing Areas, FAO. Available online: http://www.fao.org/fishery/area/search/en

IOLM (International Organisation for Legal Metrology), 1973. International Recommendation No 22 Alcoholometry – International Alcoholometric tables. Fourth International Conference of Legal Metrology, Paris, France, pp. 71. Available online: http://www.itecref.com/pdf/OIML_Alcoholometric_Tables.pdf

ISO (International Organisation for Standardisation), 2004. Information technology -- Metadata registries (MDR) -- Part 1: Framework. ISO/IEC 11179-1:2004, ISO.

ISO (International Organisation for Standardisation), 2005. Microbiology of food and animal feeding stuffs -- Horizontal method for the detection of *Salmonella* spp. ISO 6579:2002, ISO.

ISO (International Organisation for Standardisation), 2005. General Requirements for the Competence and Testing of Calibration Laboratories. EN ISO/IEC 17025:2005, ISO.

ISO (International Organisation for Standardisation), 2006. Codes for the representation of names of countries and their subdivisions -- Part 1: Country code. ISO 3166-1:2001, ISO.

ISO (International Organisation for Standardisation), 2007. Codes for the representation of names of countries and their subdivisions -- Part 2: Country subdivision code. ISO 3166-2:2007, ISO.

IUPAC, 2013. Compendium of Chemical Terminology - the Gold Book. Available online: http://goldbook.iupac.org/

Keith L H, Crummett W, Deegan J Jr, Libby R A, Taylor J K, Wentler G, 1983. Principles of environmental analysis. Analytical Chemistry, 55 (14), 2210-2218.

United Nations, 2000. Handbook on Geographic Information Systems and Digital Mapping, Studies in Methods, Series F, No. 79. ST/ESA/STAT/SER.F/79. United Nations Department of Economic and Social Affairs, Statistics Division, New York, Annex VI – Glossary. 197 pp.

W3C, online. XML Schema Part 2: Datatypes Second Edition. W3C recommendation 28 October 2004. Available online:,http://www.w3.org/TR/xmlschema-2/

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APPENDICES

APPENDIX A. LIST OF CONTROLLED TERMINOLOGIES

The list of all controlled terminologies is available in the Microsoft Excel® file "StandardSampleDescription.xls".

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APPENDIX B. COMPATIBILITY BETWEEN SSD VER. 1.0 AND SSD VER. 2.0

The WG-SSD2 highlights that the SSD ver. 2.0 is compatible, from a content perspective, with the previous version (SSD1). Of course, the contrary is not possible due to the introduction of repeatable and compound data elements and the implementation of the 'FoodEx2'.

In order to give sufficient time to all stakeholders to adapt their system to the new structure, the WG-SSD2 suggests that both the standards (SSD2 and SSD1) should be supported in 'parallel running' for a certain period of time. Each competent network should then agree a plan and a timeframe to fully implement the SSD2. In the case of data domains where SSD1 (e.g. Zoonoses) is not yet used, the WG-SSD2 recommends implementation of SSD2 directly. This should also be the case for those data providers who have not yet started implementation of SSD1 format data transmission.

B.1 Resolution of overlapping between SSD Elements and FoodEx2 Facets

Since the introduction of FoodEx2 catalogue with the structure of facets, there now exist some areas of overlap between data elements which exist with SSD1 and the information to be recorded in the FoodEx2 facets. The WG-SSD2 remarks that these data elements ('Product Code', 'Product Treatment', 'Method of Production', 'Packaging and Ingredients') were introduced preliminarily in SSD1 since a complete, harmonised and systematic food classification system was, at that time, not yet available and a detailed recommendation was reported in the guidance for the Food Classification by the working group to transform these data elements into facets of the FoodEx2 (Recommendation 3 – Guidance on Standard Sample Description for Food and Feed (EFSA, 2010a)).

The WG-SSD2, in line with the above listed recommendation, is therefore proposing in this guidance to allow the recording of this information in the FoodEx2 facets. In addition, this method, introducing a compound element, allows a more flexible and extensible approach for describing food characteristics.

For the period of parallel running, the SSD1 will still maintain all the overlapping catalogues ('FOODEX1', 'MATRIX', 'PRODMD', 'PRODPACK' and 'PRODTREAT'). The 'FoodEx2 building hierarchy' will be added to the SSD1 catalogues, replacing 'FoodEx1'. For this reason the 'FoodEx'[38] catalogue will not be modified (i.e. addition of new terms) any longer in the future and data providers will need to report terms from the FoodEx2 'building hierarchy'. This process will help data providers to begin the transition to the usage of a fully facetted FoodEx2 catalogue. FoodEx1 terms will be still maintained active in the SSD1 catalogues, for the period of the parallel running of the two versions SSD1 and SSD2, to support backward compatibility.

The Table 24 shows areas in which overlaps between SSD1 and SSD2 data elements were identified and in which all this information is now reported using the 'FoodEx2 code' (i.e. the FoodEx2 base terms plus the relevant facets and facet descriptors).

[38] FoodEx catalogue refers here to the entire content of the FOODEX tab present in the StandardSampleDescription.xls file. This catalogue includes not only FoodEx1 terms but also food simulants - Commission Regulation (EU) No 10/2011 (OJ L 12, 15.01.2011, p.1-89) and feed terms - Commission Regulation (EU) No 575/2011 (OJ L 159, 17.06.2011, p. 25-65).

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Table 24: Overlaps between SSD1 elements and SSD2 FoodEx2 facets

Element Name	SSD1 Catalogue[(a)] or Controlled terminology	Element Name FoodEx2[(a)]	SSD2 Facet[(b)]
EFSAProdCode	FoodEx (Classification without facets)	anMatCode: FoodEx2 (Classification with facets)	
prodCode	MATRIX		
prodProdMeth	PRODMD		Production Method
prodPack	PRODPAC		Packaging Format Packaging Material
prodTreat	PRODTR		Process Extent of Cooking (Doneness) Part consumed/analysed
Ingredients	Free text (with $ delimiter) switch to a coded list		Ingredients[+]

(a) Single value choice only (except Ingredients which is free text)
(b) Multiple value choice possible

It should be noted that some of the SSD1 elements listed in Table 24 are mandatory in some data collection domains (e.g. the SSD1 element 'prodTreat' is mandatory for pesticide residues data collection). These conditions should be reflected in SSD2, imposing the corresponding facets to be mandatory. Data providers should be aware that 'FoodEx2' defines default values for some facets (implicit facets) for those terms where the facets characteristics can be unambiguously derived by term definition (i.e. 'scope note'). For example for orange juice, the 'process-technology facet' can be assumed to be 'juicing'. In principle, the implicit facets do not need to be reported by the data providers but they will be automatically populated by the system, before the check of the mandatory facets is performed.

Data providers, in this case, should be aware that only the 'source facet' and the 'part-nature facet' are systematically populated for all terms flagged as 'core' or 'extended'. The data providers should verify before transmission if mandatory facets are present in a term definition, because otherwise they risk rejection of the transmission.

B.2 Catalogue updates to support SSD1 and SSD2 in parallel

EFSA will make sure that the catalogues used both in SSD1 and SSD2 will be kept consistent. The WG-SSD2 recognises that EFSA will have to maintain both catalogues for SSD1 and SSD2 during the period of parallel running. In fact, it is expected that the relevant networks will ask to include new terms in the catalogues during this period (e.g. new substance such as new pesticides). If new terms were to be entered only in the SSD2 catalogues, the SSD1 structure could not be used for the data transmission of information requiring the new term.

Details of the linking between SSD2 catalogues and SSD1 catalogues are included in Appendix B, chapter B.4 ('Correspondence of SSD1 and SSD2 Data Element Codes') and Table 25 ('Mapping from SSD1 to SSD2 in the contaminants area') and in the same appendix, chapter B5 ('Correspondence of AMR isolate base structure and SSD2 Data Element Codes') in Table 26 ('Mapping from SSD1 to SSD2 in the zoonoses area').

The maintenance of SSD1 catalogues will be performed for all catalogues with the exception of the FOODEX catalogue, where only the new FoodEx2 terms will be maintained as presented in Appendix B, chapter B.1 (Resolution of overlapping SSD Elements and FoodEx2 Facets) and in Table 24.

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B.3 Conditions to support SSD1 and SSD2 in parallel

During the period of parallel running, data providers should be able to use either the SSD1 structure or SSD2 structure. It will not be possible to send some SSD2 elements in an SSD1 transmission. The two data structures will be completely separated.

The WG-SSD2 is aware that the SSD2 introduces new sample characteristics (e.g. Target Species for Feed) that could be of benefit for some data collections currently using SSD1. These sample characteristics will be not reflected in SSD1 since the SSD1 model will not be amended.

In these cases, where the features introduced by the SSD2 are paramount for a data collection, the relevant networks should decide the most appropriate time frame for adoption of the SSD2 data model.

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B.4 Correspondence of SSD1 and SSD2 Data Element Codes

In order to evaluate the impact of the new structure a full mapping from the SSD1 structure and SSD2 structure is presented in Table 25. Only the relevant SSD2 columns for this mapping are reported.

Table 25: Mapping from SSD1 to SSD2 in the contaminants and pesticide area

Standard Sample Description version 1 (SSD1)				Standard Sample Description version 2 (SSD2)				
Element Code	**Element Name**	**Element Label**	**Controlled terminology**	**Element Code**	**Element Name**	**Element Label**	**Controlled terminology**	**Changes between SSD1 and SSD2**
S.01	labSampCode	Laboratory sample code		D.01	SampId	Sample taken identification code		This is not an exact matching. Depending on the data domain it should be re-considered.
S.02	labSubSampCode	Laboratory sub-sample code		H.01	AnPortSeq	Sample analysed portion sequence		This is not an exact matching. Depending on the data domain it should be re-considered.
S.03	Lang	Language	LANG					This element has been removed
				D.02	repCountry	Reporting Country	COUNTRY	This element has been added and has to be populated to be correctly integrated with the transmission context
				D.05	repYear	Reporting Year		This element has been added and has to be populated to be correctly integrated with the transmission context
S.04	sampCountry	Country of sampling	COUNTRY	D.03	sampCountry	Country of sampling	COUNTRY	The name of this element is unchanged
S.05	sampArea	Area of sampling	NUTS	D.04	sampArea	Area of sampling	NUTS	The name of this element is unchanged
S.06	origCountry	Country of origin of the product	COUNTRY	E.04	origCountry	Country of origin of the sample taken	COUNTRY	The name of this element is unchanged
S.07	origArea	Area of origin of the product	NUTS	E.05	origArea	Area of origin of the sample taken	NUTS	The name of this element is unchanged
S.08	origFishAreaCode	Area of origin for fisheries or aquaculture activities code	FAREA	E.06	origFishAreaCode	Area of origin for fisheries or aquaculture activities code of the sample taken	FAREA	The name of this element is unchanged

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Standard Sample Description version 1 (SSD1)				Standard Sample Description version 2 (SSD2)				
Element Code	**Element Name**	**Element Label**	**Controlled terminology**	**Element Code**	**Element Name**	**Element Label**	**Controlled terminology**	**Changes between SSD1 and SSD2**
S.09	origFishAreaText	Area of origin for fisheries or aquaculture activities text		E.07	origFishAreaText	Area of origin for fisheries or aquaculture activities text of the sample taken		The name of this element is unchanged
S.10	procCountry	Country of processing	COUNTRY	E.08	procCountry	Country of processing of the sample taken	COUNTRY	The name of this element is unchanged
S.11	procArea	Area of processing	NUTS	E.09	procArea	Area of processing of the sample taken	NUTS	The name of this element is unchanged
				E.01	sampMatType	Type of matrix	MTXTYP	This element is necessary to restrict the sub-domain in use (food, feed, etc.), it should be calculated from the anMatCode
S.12	EFSAProdCode	EFSA Product Code	FOODEX	G.01	anMatCode	Coded description of the matrix of the sample analysed	MTX	This element is now managed by FoodEx2 food classification
S.13	prodCode	Product code	MATRIX	--	--	--	--	This element has been removed
S.14	prodText	Product full text description		G.02	anMatText	Text description of the matrix analysed		The name of this element has been changed
S.15	prodProdMeth	Method of production	PRODMD	G.01	anMatCode	Coded description of the matrix analysed	MTX	This element is now managed by FoodEx2 food classification
S.16	prodPack	Packaging	PRODPAC	G.01	anMatCode	Coded description of the matrix analysed	MTX	This element is now managed by FoodEx2 food classification
S.17	prodTreat	Product treatment	PRODTR	G.01	anMatCode	Coded description of the matrix analysed	MTX	This element is now managed by FoodEx2 food classification
S.18	prodBrandName	Brand name		E.10	brandName	Brand name		This element has been included as compound element of sampMatInfo
S.19	prodManuf	Manufacturer		E.10	Manuf	Manufacturer		This element has been included as compound element of sampMatInfo

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Standard Sample Description version 1 (SSD1)				Standard Sample Description version 2 (SSD2)				
Element Code	**Element Name**	**Element Label**	**Controlled terminology**	**Element Code**	**Element Name**	**Element Label**	**Controlled terminology**	**Changes between SSD1 and SSD2**
S.20	prodIngred	Ingredients		E.02	F04	Coded description of the matrix analysed	MTX	This element is now managed by FoodEx2 food classification
S.21	prodCom	Product comment		E.10	sampMatInfo	Additional information on the analysed matrix		The name of this element has been changed
S.22	prodY	Year of production		F.06	prodY	Production year		This element has mapped to a compound element
S.23	prodM	Month of production		F.06	prodM	Production month		This element has mapped to a compound element
S.24	prodD	Day of production		F.06	prodD	Production day		This element has mapped to a compound element
S.25	expiryY	Year of expiry		F.06	expiryY	Expiry year		This element has mapped to a compound element
S.26	expiryM	Month of expiry		F.06	expiryM	Expiry month		This element has mapped to a compound element
S.27	expiryD	Day of expiry		F.06	expiryD	Expiry day		This element has mapped to a compound element
S.28	sampY	Year of sampling		D.06	sampY	Year of sampling		The name of this element is unchanged
S.29	sampM	Month of sampling		D.07	sampM	Month of sampling		The name of this element is unchanged
S.30	sampD	Day of sampling		D.08	sampD	Day of sampling		The name of this element is unchanged
S.31	progCode	Sampling programme code		B.01	progId	Sampling programme identification code		The name of this element has been changed
S.32	progLegalRef	Programme legal reference		B.02	progLegalRef	Programme legal reference	LEGREF	A catalogue has been linked to this element
S.33	progSampStrategy	Sampling strategy	SAMPSTR	B.03	sampStrategy	Sampling strategy	SAMPSTR	The name of this element has been changed
S.34	progType	Type of sampling programme	SRCTYP	B.04	progType	Programme type	PRGTYP	The name of this element is unchanged
S.35	sampMethod	Sampling method	SAMPMD	B.05	sampMethod	Sampling method	SAMPMD	The name of this element is unchanged
S.36	sampleNum	Number of samples						This element has been removed

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Standard Sample Description version 1 (SSD1)				Standard Sample Description version 2 (SSD2)				
Element Code	**Element Name**	**Element Label**	**Controlled terminology**	**Element Code**	**Element Name**	**Element Label**	**Controlled terminology**	**Changes between SSD1 and SSD2**
S.37	lotSize	Lot size		C.03	sampUnitSize	Sampling unit size		The name of this element has been changed
S.38	lotSizeUnit	Lot size unit	UNIT	C.04	sampUnitSizeUnit	Sampling unit size unit	UNIT	The name of this element has been changed
S.39	sampPoint	Sampling point	SMPNT	B.07	sampPoint	Sampling point	SAMPNT	The name of this element has been changed
L.1	labCode	Laboratory		J.01	labId	Laboratory identification code		The name of this element has been changed
L.2	labAccred	Laboratory accreditation	LABACC	J.02	labAccred	Laboratory accreditation	LABACC	The name of this element is unchanged
L.3	labCountry	Laboratory country	COUNTRY	J.03	labCountry	Laboratory country	COUNTRY	The name of this element is unchanged
O.1	localOrg	Local organisation		A.01	localOrgId	Local organisation identification code		The name of this element has been changed
O.2	localOrgCountry	Local organisation country	COUNTRY	A.02	localOrgCountry	Local organisation country	COUNTRY	The name of this element is unchanged
R.01	resultCode	Result code		M.01	resId	Result identification code		The name of this element has been changed
R.02	analysisY	Year of analysis		F.03	analysisY	Year of analysis		The name of this element is unchanged
R.03	analysisM	Month of analysis		F.04	analysisM	Month of analysis		The name of this element is unchanged
R.04	analysisD	Day of analysis		F.05	analysisD	Day of analysis		The name of this element is unchanged
R.05	EFSAParamCode	EFSA Parameter Code						This element has been removed
R.06	paramCode	Parameter code	PARAM	K.02	paramCode	Coded description of the parameter	PARAM	The name of this element is unchanged
R.07	paramText	Parameter text		K.03	paramText	Parameter text		The name of this element is unchanged
R.08	paramType	Type of parameter	PARTYP	K.01	paramType	Type of parameter	PARAMTYP	The name of this element is unchanged
R.09	anMethRefCode	Analytical method reference code		L.01	anMethRefId	Analytical method reference identification		The name of this element has been changed

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Standard Sample Description version 1 (SSD1)				Standard Sample Description version 2 (SSD2)				
Element Code	**Element Name**	**Element Label**	**Controlled terminology**	**Element Code**	**Element Name**	**Element Label**	**Controlled terminology**	**Changes between SSD1 and SSD2**
R.10	anMethCode	Analytical method code	ANLYMD	L.04	anMethCode	Analytical method code	ANLYMD	The name of this element is unchanged
R.11	anMethText	Analytical method text		L.05	anMethText	Analytical method text		The name of this element is unchanged
R.12	accredProc	Accreditation procedure for the analytical method	MDSTAT	M.02	accredProc	Accreditation procedure for the analytical method	MDACC	The name of this element has been changed
R.13	resUnit	Result unit	UNIT	M.03	resUnit	Result unit	UNIT	The name of this element is unchanged
R.14	resLOD	Result LOD		M.04	resLOD	Result LOD		The name of this element is unchanged
R.15	resLOQ	Result LOQ		M.05	resLOQ	Result LOQ		The name of this element is unchanged
R.16	CCalpha	CC alpha		M.08	CCalpha	CC alpha		The name of this element is unchanged
R.17	CCbeta	CC beta		M.09	CCbeta	CC beta		The name of this element is unchanged
R.18	resVal	Result value		M.10	resVal	Result value		The name of this element is unchanged
R.19	resValRec	Result value recovery		M.11	resValRec	Result value recovery rate		The name of this element is unchanged
R20	resValRecCorr	Result value corrected for recovery	YESNO	M.12	resValRecCorr	Result value corrected for recovery	YESNO	The name of this element is unchanged
R.21	resValUncertSD	Result value uncertainty Standard deviation		M.18	resValUncertSD	Result value uncertainty standard deviation		The name of this element is unchanged
R.22	resValUncert	Result value uncertainty		M.17	resValUncert	Result value uncertainty		The name of this element is unchanged
R.23	moistPerc	Percentage of moisture in the original sample		M.13	exprResPerc	Percentage of component in which the result is expressed		This element has mapped to a compound element
R.24	fatPerc	Percentage of fat in the original sample		M.13	exprResPerc	Percentage of component in which the result is expressed		This element has mapped to a compound element

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Standard Sample Description version 1 (SSD1)				Standard Sample Description version 2 (SSD2)				
Element Code	**Element Name**	**Element Label**	**Controlled terminology**	**Element Code**	**Element Name**	**Element Label**	**Controlled terminology**	**Changes between SSD1 and SSD2**
R.25	exprRes	Expression of result	EXRES	M.14	exprResType	Expression of result	EXPRRES	The name of this element has been changed
R.26	resQualValue	Result qualitative value	POSNEG	M.15	resQualValue	Result qualitative value	POSNEG	The name of this element is unchanged
R.27	resType	Type of result	VALTYP	M.16	resType	Type of result	VALTYP	The name of this element is unchanged
R.28	resLegalLimit	Legal Limit for the result		N.02	evalHighLimit	Limit for the result evaluation (High limit)		The name of this element has been changed
R.29	resLegalLimit Type	Type of legal limit	LMTTYP	N.03	evalLimitType	Type of limit for the result evaluation	LMTTYP	The name of this element has been changed
R.30	resEvaluation	Evaluation of the result	RESEVAL	N.04	evalCode	Evaluation of the result	RESEVAL	The name of this element has been changed
R.31	actTakenCode	Action taken	ACTION	N.05	actTakenCode	Action Taken	ACTION	The name of this element is unchanged
R.32	resComm	Comment of the result		M.20	resInfo	Additional information on the result		The name of this element has been changed

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The SSD2 introduces for the contaminants and pesticide area three major changes:

- SSD2 uses a more complete sample identification approach based on four levels (sampling event identification, sample taken identification, sample analysed identification, sample analysed portion sequence). The mapping suggested in Table 24 will be correct in the greatest majority of the cases, but users are recommended to amend it where applicable particularly for food additives, food contact materials and process contaminants. For the pesticide area the mapping suggested in Table 3 shows that only one level of sample identification will be needed.

- FoodEx2 provides a greater level of detail, which was not available in the former FoodEx or MATRIX classification systems. Although mapping tables to transform both FoodEx and MATRIX codes in FoodEx2 codes exist, users are recommended to re-check their mapping procedures to ensure the mappings are providing accurate transformations.

- FoodEx2 includes as facets some fields from the SSD1 which are mandatory in some areas such as pesticide e.g. 'prodProdMeth' (Method of production and PROMD catalogue in SSD1) and 'prod' (F21 Production-method facet and 'prod' catalogue in FoodEx2) and 'prodTreat' (Product treatment and PRODTR catalogue in SSD1) and 'techno' (F28 Process and 'techno' catalogue in FoodEx2) (Table 25). While mapping from the PRODMD catalogue to the corresponding FoodEx2 facet (F21, Production-method facet) is simple, mapping from the PRODTR catalogue to the corresponding FoodEx2 facet (F28 'Process') is not trivial.

In general, most variables relevant for the contaminants and pesticide area have been transferred without change from SSD1 to SSD2. In some cases variables and catalogues have been given new names to enhance the clarity and consistency of the data model especially where these fields interact with the new needs for SSD2 (e.g. extension to other domains like zoonoses). This will create a need for existing systems to revise some names and the order of elements in the XML-file. In some cases catalogues (e.g. methods, parameters) and variables have been set up to handle compound elements; handling of such changes will also have to be implemented.

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B.5 Correspondence of AMR isolate based data model and SSD2 Data Element Codes

The following table (Table 26) represents the mapping from the data model used to collect AMR Quantitative data at Isolate-based level and the SSD2 structure.

Table 26: Mapping from AMR isolate-based data model to SSD2

AMR Quantitative isolate-based level data model				Standard Sample Description version 2 (SSD2)				Changes between AMR model and SSD2 system
Element Code	**Element Name**	**Short Element name for XML/Excel transfer**	**Current picklists in zoonoses**	**Element Code**	**Element name**	**Element label**	**Controlled terminology**	
AMR.01	ResultCode	resultCode		M.01	resId	Result identification code		The name of this element has been changed
AMR.02	Reporting Year	repYear		D.05	repYear	Reporting year		The name of this element is unchanged
AMR.03	Reporting Country	repCountry	ZOO_CAT_COUNTRY	D.02	repCountry	Reporting country	COUNTRY	The name of this element is unchanged
AMR.04	Language	lang	ZOO_CAT_LANG					This element has been removed
AMR.05	Zoonotic Agent	zoonosis	ZOO_CAT_PARAM_ZOO	I.02	isolParamCode	Coded description of the isolate	PARAM	This element has been mapped to a compound element
AMR.06	Matrix	matrix	ZOO_CAT_MATRIX	E.02	sampMatCode	Coded description of the matrix of the sample taken	MTX	This element is now managed by FoodEx2 food classification
AMR.07	Total number of samples tested	totUnitsTested		B.08	totUnitsTested	Total number of sampling units tested		This element has been mapped to a compound element
AMR.08	Sampling Stage	sampStage	ZOO_CAT_SMPNT_ST	B.07	sampPoint	Sampling point	SAMPNT	The name of this element has been changed

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AMR Quantitative isolate-based level data model				Standard Sample Description version 2 (SSD2)				Changes between AMR model and SSD2 system
Element Code	**Element Name**	**Short Element name for XML/Excel transfer**	**Current picklists in zoonoses**	**Element Code**	**Element name**	**Element label**	**Controlled terminology**	
AMR.09	Sample Type	sampType	ZOO_CAT_SMPTYP	E.01	sampMatType	Type of matrix	MTXTYP[(a)]	The name of this element has been changed
AMR.10	Sampling Context	sampContext	ZOO_CAT_SRCTYP	B.04	progType	Programme type	PRGTYP	The name of this element has been changed
AMR.11	Sampler	sampler	ZOO_CAT_SMPLR	B.06	sampler	Sampler	SAMPLR[(a)]	The name of this element is unchanged
AMR.12	Program Code	progCode		B.01	progId	Sampling programme identification code		The name of this element has been changed
AMR.13	Sampling Strategy	progSampStrategy	ZOO_CAT_SAMPSTR	B.03	sampStrategy	Sampling strategy	SAMPSTR	The name of this element has been changed
AMR.14	Sampling Details	sampDetails		B.08	progInfo	Comment		This element has been mapped to a compound element
AMR.15	Area of Sampling	sampArea	ZOO_CAT_NUTS	D.04	sampArea	Area of sampling	NUTS	The name of this element is unchanged
AMR.16	Laboratory Identification Code	labCode		J.01	labId	Laboratory identification code		The name of this element has been changed
AMR.17	Laboratory Isolate Code	labIsolCode		I.01	isolId	Isolate identification		The name of this element has been changed
AMR.18	Total number of isolates available in	labTotIsol		J.04	labTotIsol	Total number of isolates available in the laboratory		This element has been mapped to a compound element

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AMR Quantitative isolate-based level data model				Standard Sample Description version 2 (SSD2)				Changes between AMR model and SSD2 system
Element Code	**Element Name**	**Short Element name for XML/Excel transfer**	**Current picklists in zoonoses**	**Element Code**	**Element name**	**Element label**	**Controlled terminology**	
	the laboratory							
AMR.19	Sampling Year	sampY		D.06	sampY	Year of sampling		The name of this element is unchanged
AMR.20	Sampling Month	sampM		D.07	sampM	Month of sampling		The name of this element is unchanged
AMR.21	Sampling Day	sampD		D.08	sampD	Day of sampling		The name of this element is unchanged
AMR.22	Isolation Year	isolY		I.04	isolY	Isolation year		This element has been mapped to a compound element
AMR.23	Isolation Month	isolM		I.04	isolM	Isolation month		This element has been mapped to a compound element
AMR.24	Isolation Day	isolD		I.04	isolD	Isolation day		This element has been mapped to a compound element
AMR.25	Susceptibility Test Year	analysisY		F.03	analysisY	Year of analysis		The name of this element is unchanged
AMR.26	Susceptibility Test Month	analysisM		F.04	analysisM	Month of analysis		The name of this element is unchanged
AMR.27	Susceptibility Test Day	analysisD		F.05	analysisD	Day of analysis		The name of this element is unchanged
AMR.28	Method	anMethCode	ZOO_CAT_ANLYMD	L.04	anMethCode	Analytical method code	ANLYMD	This element has been mapped to a compound element
AMR.29	Antimicrobial Substance	substance	ZOO_CAT_PARAM_SUB	K.02	paramCode	Coded description of the parameter	PARAM	This element has been mapped to a compound element
AMR.30	Cut-off value	cutoffValue		N.01	evalLowLimit	Limit for the result		The name of this

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AMR Quantitative isolate-based level data model				Standard Sample Description version 2 (SSD2)				Changes between AMR model and SSD2 system
Element Code	**Element Name**	**Short Element name for XML/Excel transfer**	**Current picklists in zoonoses**	**Element Code**	**Element name**	**Element label**	**Controlled terminology**	
						evaluation		element and the catalogue have been changed
AMR.31	Lowest (limit)	lowest	ZOO_CAT_FIXMEAS	M.06	resLLWR	Result lower limit of the working range		Changed from catalogue to a numeric element and the name of this element has been changed
AMR.32	Highest (limit)	highest	ZOO_CAT_FIXMEAS	M.07	resULWR	Result upper limit of the working range		Changed from catalogue to a numeric element and the name of this element has been changed
AMR.33	MIC values (mg/L)	MIC	ZOO_CAT_FIXMEAS	M.10	resVal	Result value		Changed from catalogue to a numeric element and the name of this element has been changed
AMR.34	Disk concentration (microg)	diskConc		L.06	diskConc	Disk concentration		This element has been mapped to a compound element
AMR.35	Disk diameter (mm)	diskDiam		L.06	diskDiam	Disk diameter		This element has been mapped to a compound element
AMR.36	IZD values (mm)	IZD	ZOO_CAT_FIXMEAS	M.10	resVal	Result value		Changed from catalogue to a numeric element and the name of this element has been

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AMR Quantitative isolate-based level data model				Standard Sample Description version 2 (SSD2)				Changes between AMR model and SSD2 system
Element Code	**Element Name**	**Short Element name for XML/Excel transfer**	**Current picklists in zoonoses**	**Element Code**	**Element name**	**Element label**	**Controlled terminology**	
								changed
AMR.37	Comment	resComm		M.20	resInfo	Comment		This element has been mapped to a compound element
AMR.38	ESBL phenotype	esbl	ZOO_CAT_ESBL	K.02	ParamCode	Coded description of the isolate	PARAM	This element has been mapped to a compound element
AMR.39	AMPC phenotype	ampC	ZOO_CAT_AMPC	K.02	ParamCode	Coded description of the isolate	PARAM	This element has been mapped to a compound element
AMR.40	Carbapenemase phenotype	carbapenem	ZOO_CAT_CARBAPENEM	K.02	ParamCode	Coded description of the isolate	PARAM	This element has been mapped to a compound element
AMR.41	Ceftazidime synergy test	synTestCAZ	ZOO_CAT_POSNEG	K.02	ParamCode	Coded description of the isolate	PARAM	This element has been mapped to a compound element
AMR.42	Cefotaxime synergy test	synTestCTX	ZOO_CAT_POSNEG	K.02	ParamCode	Coded description of the isolate	PARAM	This element has been mapped to a compound element
AMR.43	Cefepime synergy test	synTestFEP	ZOO_CAT_POSNEG	K.02	ParamCode	Coded description of the isolate	PARAM	This element has been mapped to a compound element
AMR.44	Total number of sampling units tested	totSampUnitsTested		B.08	tot SampUnitsTested	Total number of sampling units tested		This element has been mapped to a compound element
AMR.45	Sampling unit type	sampUnitType	ZOO_CAT_UNIT	C.02	sampUnitType	Sampling unit type	SAMPUNTYP	The name of this element is unchanged

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AMR Quantitative isolate-based level data model				Standard Sample Description version 2 (SSD2)				Changes between AMR model and SSD2 system
Element Code	**Element Name**	**Short Element name for XML/Excel transfer**	**Current picklists in zoonoses**	**Element Code**	**Element name**	**Element label**	**Controlled terminology**	
AMR.46	Sample origin	sampOrig	ZOO_CAT_COUNTRY	E.04	origCountry	Country of origin of the sample taken	COUNTRY	The name of this element has been changed
AMR.47	Performed CC MRSA characterisation	perCC	ZOO_CAT_YESNO	M.20	perCC	Performed CC MRSA characterisation	YESNO	This element has been mapped to a compound element
AMR.48	Performed MLST MRSA characterisation	per MLST	ZOO_CAT_YESNO	M.20	per MLST	Performed MLST MRSA characterisation	YESNO	This element has been mapped to a compound element

(a): New pick list

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The SSD2 introduces for the zoonoses domain and AMR quantitative isolate-based level data model users only one major change: the use of the FoodEx2 which was not available in the former system.

A few additional changes impact mainly the data model's clarity and consistency, such as:

- some existing element names and controlled terminology names have been changed;
- some elements have been added;
- some elements have been removed or changed to compound elements.

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GLOSSARY AND ABBREVIATIONS

GLOSSARY

Analytical method: the element contains the information about the technical procedure for detection of an analyte and generation of a laboratory result.

Applicable domains: the list of data domains for which the term is applicable.

Attribute: see Simple data elements.

Attribute name: the name of the attribute to be used in database systems.

Attribute label: the label for the attribute to be displayed to users.

Base term: any of the terms listed in the 'Food' section of the FoodEx2 building hierarchy. These terms implicitly include some facet descriptors and may be further specified with additional facet descriptors. Base term is a synonym for food list term.

Building hierarchy: is a food hierarchy serving as a master for the management of all the terms contributed by the different domains, including the facet descriptors.

Business rules: the rules to verify the validity of a value reported in an individual data element.

Catalogue value/term: one defined value of a catalogue.

Catalogue: list of terms, linked hierarchically or not, which are valid values within the same 'Element name'. It is possible for one catalogue to be applicable to more than one element**.**

Compound data element: element containing one or many 'simple elements' which are named 'attributes' of the 'compound element'. Each attribute can have values which can be 'text values', 'numeric values' or 'catalogue values'. An example could be: '*element1=value1$element2=value2$...$ elementN=valueN'*.

Controlled terminology: finite and enumerated set of terms intended to convey information unambiguously.

Data collection domain or domain: this is an entity grouping together all the data collections on a specific risk assessment area and can be considered as a synonym of a risk assessment area.

Data collection: Entity linking all file transmissions included in a single collection of data on specific risk assessment areas over time. In general terms, the data receiver will define data collections on an ad-hoc basis: e.g. Heavy metal data collection which is within the domain of Chemical Contaminants.

Data element: it is a unique element name, referenced by a sequential alphanumeric code, this is to be used for column names, field names and tags depending on the software programs, files or databases implementing the SSD.

Data interchange protocol: logical protocol independent from the specific application protocol used for the physical delivery of the message itself.

Data model: lexical representation of data, specifying their properties, structure and interrelationships (ISO/IEC FDIS 11179-1:2004) built to comply with the need of data submission.

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Data structure: Implementation of a data model consisting of file structures used to represent various features (Handbook on Geographic Information Systems and Digital Mapping, Studies in Methods, Series F, No. 79, United Nations Department of Economic and Social Affairs, Statistics Division, New York, 2000, Annex VI – Glossary) and built to comply with the needs of data submission.

Denormalised: in databases**,** the approach to merge and store in one table separate data entities that would, in normalised format, be stored in separate tables.

Deprecation: process of removing logically from SSD2 system terms by adding a 'Valid to date' after which they should not be used in data transmissions.

Description: this provides a short description on what the data element should contain.

Element code: this is an alphanumeric code providing a unique identifier for the data element. The element code is made of the section identifier code plus a progressive number.

Element label: this provides a descriptive label for each data element to be used in reports, print outs or in the graphical interfaces of the software programs that will manage the SSD.

Element name: this element is provided to be used for column names, field names and tags depending on the software programs, files or databases implementing the SSD.

Element value: this is the specific term applicable to the SSD record for the element.

Evaluation: this element contains the assessment of one or more results, evaluating its compliance with a defined limit or an assessment.

Explicit facet: facet descriptor which has been added to a base term of the food list (specifically to the core list and extended list elements). These facet descriptors are not implicitly included in the definition of the food list term and serve the purpose of better specifying it, narrowing its scope.

Exposure hierarchy: food hierarchy arbitrarily defined based on experiences in exposure assessment of chemical contaminants. It particularly focuses on the needs of data analysis and exposure calculation in the domain of chemical contaminants;

Facet: collections of terms (facet descriptors) defining characteristics of food/ feed/ animal items/groups according to specific points of view. These should be clearly defined, mutually exclusive, and collectively exhaustive characteristics of a class or specific item.

File transmission: the element contains all the information linking all data submitted in a single file transmission. The entity should be described by some additional attributes such as the transmission date, the receipt date and other additional logging dates that may be needed by the transmission or receiver systems.

FoodEx2: facet-based food classification and description system, where the most commonly needed combinations of facet descriptors are condensed into list terms whose scope is narrow enough to be recognised in different food safety domains. These terms are aggregated into hierarchies accordingly to the needs of the different domains.

Hierarchy code: code representing the hierarchy, for displaying and sorting purposes. This code should not be used for reporting.

Implicit facet: facet descriptor which is included in the definition of a FoodEx2 food list term (base term). Even if not explicitly referenced (as for explicit facets defined above), they are always implied if a food list item is chosen.

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Isolate: the elements identify a culture of biological agent, isolated from a specific 'sample taken' and typically sent for further laboratory analysis e.g. speciation or anti-microbial resistance testing.

Laboratory: this element contains the identification of the laboratory in which the analysis for detection of an analyte was performed.

Local organisation: this element contains the identification of the organisation (Local/regional national Competent Authority) that initially requested or performed the sampling and is responsible for the implementation of the sampling programme.

Major release: published version of the SSD2 catalogues according to the scheduled implementation of catalogue changes.

Master hierarchy: is a generic food hierarchy proposed by the WG on Food Classification. It was used as starting point to produce domain specific hierarchies.

Matrix analysed: this element contains the description of the matrix analysed and its relevant characteristics. This will often be the same as the 'Matrix sampled'.

Matrix sampled: This element contains the description of the item sampled and of its relevant characteristics available before the analysis.

Metadata: data that defines and describes other data (ISO/IEC FDIS 11179-1).

Minor release: published interim version of the SSD2 catalogues provided to make necessary terms immediately available for use on a preliminary basis. Changes made in a minor release will then be ratified (or not) via the scheduled procedure and confirmed or rejected in the next major release.

Monitoring: the term means the intermittent performance and analysis of routine measurements and observations, aimed at detecting changes in the object of analysis, environment or health status of a population.

Parameter: this element contains the specific analyte for which the laboratory analysis was performed on the matrix analysed.

Parent code: link to the parent code in the master hierarchy**.**

Pesticide residues hierarchy: a specific food hierarchy for pesticide residues data reporting and analysis. It reflects and further details the lists defined in the MRL Regulations in the pesticide domain.

Preliminary term: term added to the terminology system between two scheduled (major) releases. Preliminary terms would be made available to data senders and receivers in Minor Releases of the SSD2 catalogues.

Receiver organisation: this element contains the description of the organisation receiving the data. In the current situation it is typically EFSA, but the WG-SSD2 defined the model in a way that it could be used by other organisations to receive data.

Reference period: this element contains the reporting period for the data collection, for example in years e.g. Pesticide residues 2010.

Repeatable data element: element containing one or many instances of an 'element value' for the specified data type. An example could be: '*value1$value2$...$valueN*'.

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Result: this element contains the result of the laboratory test, a quantitative or qualitative outcome value.

Sample analysed: this elements represents the material analysed in a laboratory (the sample taken as it is analysed by the laboratory).

Sample analysed portion: this element (often called 'test portion') contains the identification of a replicate, achieved by subdividing a sample analysed. It is a repetition of the sample analysed and is required under defined testing procedures for some analytes.

Sample level data: individual results from the measurement of the concentration of an analyte.

Sample taken: this element contains the material taken from the sampling unit at a certain time by the sampling officer.

Sampling event: this element represents the sampling unit or units extracted at certain time from the sampled population.

Sampling programme: this element describes criteria, purpose, method or legislative reference used to generate a sampling event.

Sampling unit: the unit which the specimens taken represent and which is considered either infected (contaminated) or not, based on the analyses result.

Section: the element describes the key entity of the SSD data model.

Sender country: the element contains the country of the organisation transmitting the data.

Sender organisation: the element contains the description of the organisation transmitting the data.

Simple data element: the element represents only one instance of an 'element value' which may be either a 'text value' or a 'numeric value' or a 'catalogue value'.

Surveillance: the term means the systematic ongoing collection, collation, and analysis of information related to food safety and the timely dissemination of information to those who need to know so that action can be taken.

Term code: unique code for the term. This is the only code that should be reported in the SSD transmissions.

Term definition: all the characteristics of a term (i.e. as described in the scope-note).

Type: A data type is associated to each data element and it defines the values that it can contain. Data types will be defined using the W3C XML schemas data type specification.

Zoonoses hierarchy: a specific food hierarchy within FoodEx2 addressing the needs for zoonoses and microbiological data reporting and analysis, particularly in the section on animal products.

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ABBREVIATIONS

AMR: Antimicrobial resistance

BIOMO: Biological Monitoring Unit of EFSA

CAS number: Chemical Abstracts Service number

CSV: Comma Separated Values

DCM: Dietary and Chemical Monitoring Unit of EFSA

DICT: Dictionary of SSD

EEA: European Economic Area

EFSA: European Food Safety Authority

EFSA Data Warehouse: Database designed specifically for providing access to the data collected by EFSA.

EU: European Union

EUROSTAT: European statistics organisation

FAO: Food and Agriculture Organisation of the United Nations

FoodEx2: Food classification and description system ver. 2

GDE/GDE2: Guidance on Data Exchange

ID: Identification code

IEC: International Electrotechnical Commission

ISO: International Organization for Standardization

IUPAC: International Union of Pure and Applied Chemistry

LOD: Limit of detection

LOQ: Limit of quantification

ML: Maximum limit

MRL: Maximum residue limit

MRPL: Minimum required performance limits of analytical methods

MS: Member State of European Union

NUTS: Nomenclature of territorial units for statistics

PARAM: Parameter catalogues of SSD

RPC: Raw Primary Commodities

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SCoFCAH: Standing Committee on the Food Chain and Animal Health

SSD: Standard Sample Description

SSD1: Standard Sample Description ver.1.0

SSD2: Standard Sample Description ver.2.0

TWG: Technical Working Group on SSD1

W3C XML: W3C Extensible Markup Language

WG-SSD2: Working Group on SSD Extension

WHO: World Health Organisation

XML: Extensible Markup Language

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